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**DEVELOPMENT OF A FOUR-WHEEL QUASI-STEADY STATE NUMERICAL
MODEL FOR VEHICLE DYNAMICS SIMULATION**

Florianópolis
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Advisor: Prof. Rodrigo de Souza Vieira, Dr.

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This Bachelor's Thesis has been considered adequate for the conferral of the degree of Bachelor in Mechanical Engineering and approved in its final form by the Undergraduate Program in Mechanical Engineering.

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Dedico este trabalho aos meus
pais, Marcus e Vera.

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It's what non-car people don't get. They see all cars as just two tons of wires, glass, metal and rubber. That's all they see. People like you or I know, we have an unshakeable belief that cars are living entities, ... you can develop a relationship with a car. And that's what non-car people just don't get.

Jeremy Clarkson

RESUMO

As simulações computacionais em dinâmica veicular apresentam diversas vantagens significativas em relação aos ensaios físicos, como a redução de tempo, esforço e custos, além de maior liberdade na definição dos cenários simulados. Por esses motivos, a modelagem veicular numérica tornou-se um elemento fundamental na indústria automotiva, ao permitir aprimorar o projeto de veículos em termos de segurança, eficiência e desempenho. Entretanto, muitas soluções comercialmente disponíveis em dinâmica veicular consistem em modelos transientes com múltiplos graus de liberdade, que são frequentemente demasiadamente complexos e custosos. Isso limita severamente as possibilidades de uso por equipes menores de automobilismo ou por empresas automotivas de pequeno porte. Por outro lado, modelos mais simples muitas vezes não representam adequadamente os fenômenos físicos que influenciam o projeto veicular. Este trabalho apresenta um modelo quase estático em quatro rodas que abrange os principais parâmetros relevantes para o projeto em dinâmica veicular, tais como características de suspensão e direção, comportamento de pneus, influência aerodinâmica e comandos do motorista. O modelo é implementado em C++ com um *loop* iterativo para verificar restrições físicas ao torque nas rodas, ao escorregamento dos pneus e à transferência de carga. A validação analítica do modelo é realizada, bem como sua comparação com uma solução comercial consolidada em simulação veicular, apresentando resultados satisfatórios em ambos os casos. De forma geral, o modelo propõe uma abordagem mais didática para um tema complexo, como a dinâmica veicular, podendo contribuir para o desenvolvimento de estudantes de engenharia e de equipes menores na indústria automotiva.

Palavras-chave: dinâmica veicular; modelo de pneu; transferência de carga; cinemática de suspensão; simulação numérica.

ABSTRACT

Computational simulation in vehicle dynamics has many substantial advantages over physical testing, such as reduced time, effort and expenses, in addition to more freedom in simulated scenarios. For these reasons, numerical vehicular modelling has become a cornerstone of the automotive industry, as it enhances vehicle design in terms of safety, efficiency and performance. However, many commercially available vehicle dynamics solutions consist of transient models with multiple degrees of freedom, which are often too expensive and complex. This severely limits the possibilities for smaller motorsport teams or automotive companies to use them. In contrast, simpler models are often not representative of the physical phenomena that affect vehicle design. This work presents a four-wheel quasi-steady-state model which covers the most important parameters for design in vehicle dynamics, such as suspension and steering parameters, tire behavior, aerodynamic influence and driver input. The model is implemented in C++ with an iterative loop to check for physical constraints on wheel torque, tire slip and load transfer. The model is analytically validated and compared to a well-established commercial solution in vehicle simulation, granting satisfactory results on both fronts. Ultimately, the model presents a more didactic approach to a complicated matter, such as vehicle dynamics, which may help engineering students and smaller teams to thrive in the automotive industry.

Keywords: vehicle dynamics; tire model; load transfer; suspension kinematics; numerical simulation.

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LIST OF SYMBOLS

κ	Slip ratio
V_x	Wheel longitudinal velocity [m/s]
V_t	Wheel tangential velocity [m/s]
α	Slip angle [rad]
V	Vehicle longitudinal velocity [m/s]
μ	Average tire friction coefficient
g	Acceleration due to gravity [m/s ²]
R	Cornering radius [m]
F_y	Tire pure lateral friction force [N]
α_y	Equivalent slip angle [rad]
C_y	Lateral shape factor
D_y	Peak lateral friction force [N]
μ_y	Lateral friction coefficient
E_y	Lateral curvature factor
$K_{y\alpha}$	Lateral cornering stiffness [N/rad]
B_y	Lateral stiffness factor
S_{Hy}	Lateral horizontal shift [rad]
$S_{Vy\gamma}$	Lateral vertical shift due to camber [N]
S_{Vy}	Lateral vertical shift [N]
F_x	Tire pure longitudinal friction force [N]
κ_x	Equivalent slip ratio
C_x	Longitudinal shape factor
D_x	Peak longitudinal friction force [N]
μ_x	Longitudinal friction coefficient
E_x	Longitudinal curvature factor
$K_{x\kappa}$	Longitudinal cornering stiffness [N]
B_x	Longitudinal stiffness factor
S_{Hx}	Longitudinal horizontal shift
S_{Vx}	Longitudinal vertical shift [N]
γ	Camber angle [rad]
F_z	Vertical load on the tire [N]
F_{zo}	Tire reference vertical load [N]
df_z	Normalized change in tire vertical load

$F_{x,comb}$	Tire combined longitudinal friction force [N]
$F_{y,comb}$	Tire combined lateral friction force of each tire [N]
F_{rr}	Tire rolling resistance force [N]
f	Tire rolling resistance force coefficient
MR	Suspension motion ratio
K	Generic wheel rate [N/m]
k	Generic suspension spring rate [N/m]
W	Total weight of the vehicle [N]
h_{CG}	Height of vehicle center of gravity [m]
t	Average track [m]
t_f	Front track [m]
t_r	Rear track [m]
L	Wheelbase [m]
$bias$	Ratio between tractive or braking forces
B_f	Front braking forces [N]
B_r	Rear braking forces [N]
A_f	Front tractive forces [N]
A_r	Rear tractive forces [N]
r_{ab}	Loaded radius of each tire [m]
φ	“anti” geometry angle [rad]
Ψ	Roll angle [rad]
Φ	Pitch angle [rad]
ΔW_{lat}	Lateral load transfer [N]
ΔW_{lon}	Longitudinal load transfer [N]
ΔW_k	Elastic load transfer [N]
$\Delta W_{lat,k,ab}$	Elastic lateral load transfer [N]
$\Delta W_{lon,k,ab}$	Elastic longitudinal load transfer [N]
ΔW_s	Sprung masses load transfer [N]
$\Delta W_{lat,s,ab}$	Sprung masses lateral load transfer [N]
$\Delta W_{lon,s,ab}$	Sprung masses longitudinal load transfer [N]
ΔW_{arb}	Elastic ARB lateral load transfer [N]
ΔW_u	Unsprung masses load transfer [N]
$\Delta W_{lat,u,ab}$	Unsprung masses lateral load transfer [N]
$\Delta W_{lon,u,ab}$	Unsprung masses longitudinal load transfer [N]
$\Delta W_{lat,g,ab}$	Geometric lateral load transfer [N]

$\Delta W_{lon,g,ab}$	Geometric longitudinal load transfer [N]
$\Delta W_{lon,drag}$	Aerodynamic drag longitudinal load transfer [N]
m	Total mass of the vehicle [kg]
m_s	Sprung masses [kg]
m_u	Unsprung masses [kg]
$m_{u,ab}$	Unsprung masses on each wheel [kg]
$h_{CG,s}$	Height of the sprung masses center of gravity [m]
$h_{CG,u}$	Height of the unsprung masses center of gravity [m]
$h_{CG,u,ab}$	Height of the unsprung masses center of gravity of each wheel [m]
h_o	Vertical distance from the sprung masses center of gravity to the roll axis [m]
M_u	Unsprung masses moment [Nm]
M_s	Sprung masses moment [Nm]
M_k	Suspension springs moment [Nm]
M_{arb}	ARB moment [Nm]
δ_{ab}	Steering angle of each wheel [rad]
δ_d	Driver input steering wheel angle [rad]
R_a	Actual cornering radius [m]
R_{ab}	Cornering radius of each wheel [m]
a	Distance from center of gravity to the front axle [m]
b	Distance from center of gravity to the rear axle [m]
x	Longitudinal weight distribution
y	Lateral weight distribution
S_f	Longitudinal distance from the cornering center to the front axle [m]
S_r	Longitudinal distance from the cornering center to the rear axle [m]
θ_{ab}	Effective steering angle of each wheel [rad]
T_{ab}	Torque generated by each tire [Nm]
T	Total torque [Nm]
κ_{des}	Desired slip ratio
$F_{x,comb,ab,tar}$	Target for combined longitudinal friction forces of each tire [N]
F_{lat}	Total chassis lateral forces [N]
$F_{lat,ab}$	Chassis lateral forces of each tire [N]
F_{lon}	Total chassis longitudinal forces [N]
$F_{lon,ab}$	Chassis longitudinal forces of each tire [N]

F_{rad}	Total chassis radial forces to cornering center forces [N]
$F_{rad,ab}$	Chassis radial forces to cornering center forces of each tire [N]
a_{lat}	Lateral acceleration [g]
a_{lon}	Longitudinal acceleration [g]
a_{rad}	Radial acceleration to cornering center [g]
V_{skid}	Maximum vehicle longitudinal velocity [m/s]
F_{down}	Aerodynamic downforce loads [N]
$F_{down,x}$	Longitudinal aerodynamic downforce loads distribution
$F_{down,y}$	Lateral aerodynamic downforce loads distribution
$F_{down,ab}$	Aerodynamic downforce load on each wheel [N]
F_{drag}	Aerodynamic drag loads [N]
$F_{drag,z}$	Height of aerodynamic drag center of pressure [m]
$F_{drag,y}$	Lateral aerodynamic drag loads distribution
h_{drag}	Perpendicular distance from aerodynamic drag center of pressure to pitch axis [m]
$n_{roll,ab}$	Lateral distance from roll center to wheel midplane of each wheel [m]
$n_{pitch,ab}$	Longitudinal distance from pitch center to axle of each wheel [m]
$p_{roll,ab}$	Height of roll instantaneous center of each wheel [m]
$p_{pitch,ab}$	Height of pitch instantaneous center of each wheel [m]
$q_{roll,ab}$	Lateral distance from roll instantaneous center to wheel midplane of each wheel [m]
$q_{pitch,ab}$	Longitudinal distance from pitch instantaneous center to axle of each wheel [m]
$h_{roll,a}$	Height of roll center of each axle [m]
$h_{pitch,b}$	Height of pitch center of each side [m]
w_{ab}	Total vertical displacement of each wheel [m]
$w_{roll,ab}$	Vertical displacement of each wheel caused by roll [m]
$w_{roll,k,ab}$	Vertical elastic displacement of each wheel caused by roll [m]
$w_{roll,t,ab}$	Deformation of each tire caused by roll [m]
$w_{roll,c,ab}$	Vertical chassis displacement of each wheel caused by roll [m]
$w_{roll,n,a}$	Negated vertical displacement of each axle caused by roll [m]
$w_{pitch,n,b}$	Negated vertical displacement of each side caused by pitch [m]
$K_{susp,ab}$	Suspension springs wheel rate of each wheel [N/m]
$K_{arb,ab}$	ARB wheel rate of each wheel [N/m]

$K_{t,ab}$	Tire rate [N/m]
$K_{roll,tot,a}$	Total roll stiffness of each axle [N/m]
$K_{pitch,tot,b}$	Total pitch stiffness of each side [N/m]
$k_{c,a}$	Torsional chassis stiffness to each axle [N/m]
$K_{c,a}$	Linear chassis stiffness to each axle [N/m]
$K_{c,b}$	Linear chassis stiffness to each side [N/m]
$M_{roll,k}$	Total roll elastic moment [Nm]
$M_{roll,k,ab}$	Roll elastic moment of each wheel [Nm]
$M_{roll,s}$	Total roll sprung masses moment [Nm]
$M_{roll,s,ab}$	Roll sprung masses moment of each wheel [Nm]
$M_{roll,u}$	Total roll unsprung masses moment [Nm]
$M_{roll,u,ab}$	Roll unsprung masses moment of each wheel [Nm]
$M_{roll,g}$	Total roll geometric moment [Nm]
$M_{roll,g,ab}$	Roll geometric moment of each wheel
$M_{pitch,k}$	Total pitch elastic moment [Nm]
$M_{pitch,k,ab}$	Pitch elastic moment of each wheel [Nm]
$M_{pitch,s}$	Total pitch sprung masses moment [Nm]
$M_{pitch,s,ab}$	Pitch sprung masses moment of each wheel [Nm]
$M_{pitch,u}$	Total pitch unsprung masses moment [Nm]
$M_{pitch,u,ab}$	Pitch unsprung masses moment of each wheel [Nm]
$M_{pitch,g}$	Total pitch geometric moment [Nm]
$M_{pitch,g,ab}$	Pitch geometric moment of each wheel [Nm]
$M_{pitch,drag}$	Pitch aerodynamic drag moment [Nm]
$F_{z,past,ab}$	Vertical load of each wheel on previous iteration [N]
$F_{z,tol}$	Vertical load tolerance for convergence [N]
M_{yaw}	Total yaw moment [Nm]
$M_{yaw,ab}$	Yaw moment of each wheel [Nm]

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1 INTRODUCTION

Vehicle dynamics is a discipline of indispensable industrial importance. It studies the physical phenomena applied to ground vehicles, which allows for the optimization of vehicular design and for the thriving of the modern automotive industry.

1.1 HISTORY OF THE AUTOMOTIVE INDUSTRY

Ever since the invention of the automobile by Carl Benz in 1886, with the Benz Patent Motorwagen (shown in Figure 1), industrial efforts aimed at expanding the usage of these new machines and making them more affordable, whereas engineers focused on developing techniques to optimize safety, efficiency and performance. Vehicles went from being a far-fetched concept to a mass-produced industrial product, becoming a consumer good in modern days (ECKERMAN, 2001).

Figure 1 – The Benz Patent Motorwagen, patented by Carl Benz in 1886



Source: ECKERMAN, 2001.

Motorsport has had a great impact on the development of new technologies for the automotive industry and, consequently, has driven the evolution of the automobile throughout the decades. Advanced aerodynamic design, hybrid powertrain units, data acquisition and analysis systems, fabrication with lightweight

composite materials and turbocharging technology are some examples of how racing had a substantial role in improving vehicle design over the years (EKANEM *et al.*, 2024).

However, vehicle design and development may come with considerable expenses for both motorsport teams and automotive companies. This cost is usually related to the necessity of on-track or bench testing with actual prototypes, as a means to acquire indispensable data, which is used to properly optimize the design of the vehicle. In light of this problem, vehicle mathematical and computational models are introduced as a solution. Many companies heavily invest in the development of said models, as they present several advantages over conventional testing. Some examples of said benefits are that computational models:

- allow an investigation on how different parameters interfere with vehicle behavior without having to drive the car;
- determine a theoretical performance potential for the vehicle, which allows the engineer to understand if the driver could deliver better results;
- facilitate the definition of a setup that manages to perform in certain conditions, being a specific race track or on the public roads of a region or country.

However, it is important to understand that simulation models are merely tools, and may not fully represent real scenarios as expected. For this reason, good engineering practices are required to create valid and reliable models and to critically analyze their output results (SEGERS, 2014).

The study of vehicle dynamics and automotive engineering is a complex matter, as there are several aspects to a proper holistic vehicle design, such as a multitude of antagonistic parameters to optimize and hard to understand physical phenomena to dominate. Those difficulties tend to worsen as new technologies are developed and vehicles become more complicated machines. This requires engineers to develop multidisciplinary technical skills and knowledge over a vast range of fields.

Created in 1981 in the United States of America by the Society of Automotive Engineers (SAE), Formula SAE is an event that aims precisely at helping engineering students to meet those requirements, as they are encouraged to participate in the

design, fabrication and validation stages of real racing vehicle prototypes and engage in an actual engineering competition (CASE, 1996).

Currently, this competition takes place in multiple countries, such as Australia, the United Kingdom, Italy, Germany and Brazil. The latter hosted its first version of the event in 2004, known as Fórmula SAE Brasil. In 2010, the Universidade Federal de Santa Catarina created its own team to participate for the first time in the 2011 edition. The team was born as Fórmula UFSC and rebranded to Protto UFSC Motorsport in 2023. Figures 2 and 3 show the first and latest prototypes developed by the team, respectively, demonstrating the evolution in design and fabrication capabilities of its members throughout the years.

Figure 2 – The first ever prototype of the Protto UFSC Motorsport team, at the time known as Fórmula UFSC



Source: the author.

Figure 3 – TYPE 25/C, the Protto UFSC Motorsport team's contender prototype for the 2025 Fórmula SAE Brasil competition



Source: the author.

1.2 JUSTIFICATION

As previously explained, vehicle dynamics simulation has become a cornerstone of modern automotive engineering, offering substantial advantages over conventional physical testing.

In spite of these benefits, the adoption of high-fidelity vehicle dynamics models remains a challenge for smaller motorsport teams and emerging automotive companies. Many commercially available vehicle dynamics software implementations rely on complex transient multi-degree-of-freedom models, which often makes it difficult for inexperienced engineers to fully exploit their potential. Additionally, licensing to these models is usually prohibitively expensive, fact that may completely negate access for budget-limited teams. These problems are further aggravated when considering engineering students and competition groups such as Formula SAE participants.

Conversely, while simpler models are easier to implement without cost, they may not offer the fidelity needed by an engineer regarding suspension tuning, tire

behavior and aerodynamic influence. They may omit critical phenomena such as lateral and longitudinal load transfer, tire slip and its non-linearities, or driver input realism.

The development of a model that fills the gap in between would be substantially beneficial. Both engineering students that need to acquire experience in vehicle simulation in order to launch their careers, and smaller motorsport teams or automotive companies with limited resources would be positively impacted. Such a model should have a didactic approach and be reasonably accessible, while complete enough to provide substantial realism to the simulation, allowing the user to develop a proper engineering design in vehicle dynamics.

1.3 OBJECTIVES

The main objective of this work is **to develop a four-wheel quasi-steady state numerical model for vehicle dynamics simulation**. As a means to achieve the global objective, minor steps have been defined in the form of specific objectives:

- a) Review existing tire and vehicle mathematical models used in simulation;
- b) Propose an algorithm that calculates the tire slip and other conditions subjecting each wheel for any driving circumstance;
- c) Propose a general load transfer model capable of representing geometric and dynamic asymmetry of the vehicle both laterally and longitudinally;
- d) Analytically evaluate the physical results of the model and benchmark it against a commercial solution.

1.4 WORK OUTLINE

This work is divided into five chapters.

Chapter 1 brings a brief contextualization about the automotive industry, vehicle dynamics modelling and Formula SAE, along with the work's justification and objectives.

Chapter 2 presents the literature review, introducing fundamental concepts of vehicle dynamics, tire and vehicle models, basics of suspension geometries and a description of the load transfer phenomenon.

Chapter 3 details the proposed vehicle model. It presents the geometric analysis of a vehicle, which allows for the definition of slip inputs for the tire models. Following, the loads imposed on the vehicle are determined and load transfer is calculated, through an asymmetrical proposed model. Finally, after achieving numerical convergence, the yaw of the vehicle is analyzed.

Chapter 4, the vehicle's parameters sources are presented and the load transfer model is analytically tested for physical inconsistencies. Additionally, the results of the vehicle model as a whole are compared to a commercial solution.

Finally, chapter 5 brings the author's conclusions to the work's objectives, along with proposals of future developments.

2 LITERATURE REVIEW

This chapter introduces fundamental concepts of vehicle dynamics, followed by description of vehicle and tire mathematical modelling. In the end suspension geometries are presented along with the definition of load transfer.

2.1 VEHICLE DYNAMICS

This subsection brings basic concepts and important nomenclature to the study in vehicle dynamics, especially for the development described in this work.

2.1.1 Sprung and unsprung masses

The sprung masses of a vehicle belong to all the components supported by the suspension mechanism. They may include, but are not limited to the frame, body, engine, transmission, passengers and a portion of the suspension mechanism itself. Unsprung masses are not carried by the suspension, but rather are directly supported by the tires. They include the masses of tires, wheels, all the upright and hub assembly and a percentage of the suspension system (SAE INTERNATIONAL, 2008).

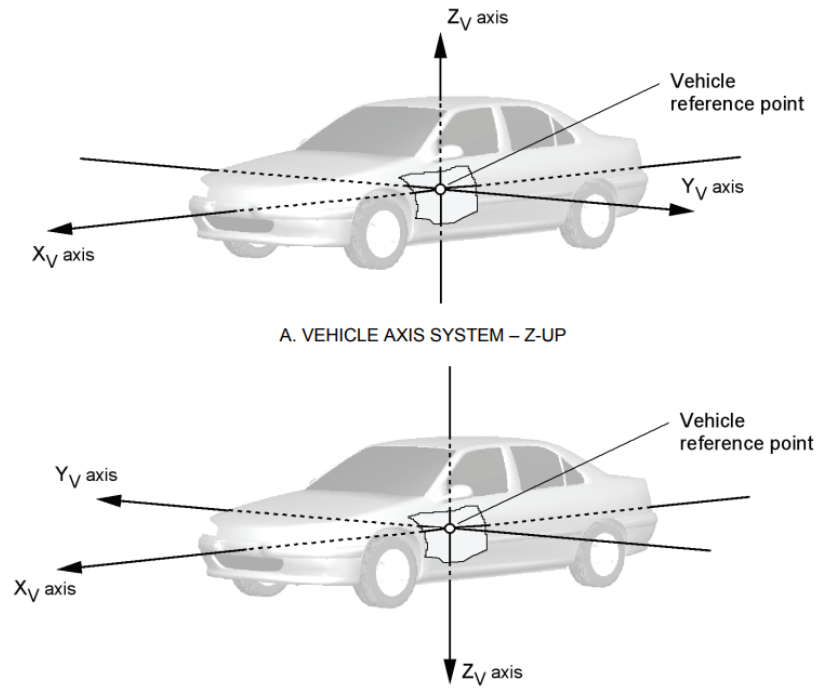
Each portion of mass has its own center of gravity. Usual engineering efforts aim at lowering the height of those centers as a means to reduce load transfer effects. Lesser values of unsprung masses tend to favor damping actuation, further improving performance. The sum of all sprung and unsprung masses is equal to the total vehicle mass.

2.1.2 Coordinate systems

It is usual in vehicle dynamics to use a coordinate system fixed to the vehicle's sprung masses reference frame. It is set so that the X axis is horizontal pointing forwards (with the vehicle at rest), parallel to its lateral plane of symmetry. The Y axis points to the left and the Z axis upwards in the Z-UP variation, both of which are inverted in the Z-DOWN variation of the system (SAE INTERNATIONAL,

2008). Figure 4 shows the axis system described above. This work will follow the Z-DOWN variation.

Figure 4 – Orientation of vehicle axis systems

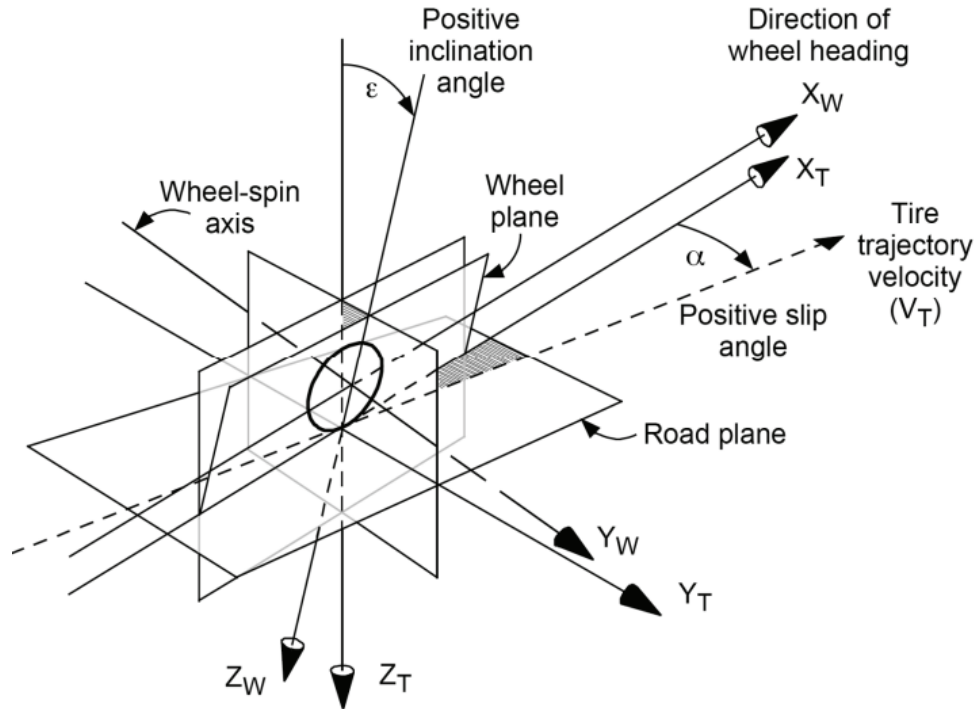


Source: SAE INTERNATIONAL, 2008

Following, the main movements to which the chassis is subjected are described in terms of the coordinate system presented above. The rotation about the three axis each receive a name, being roll a rotation about the longitudinal X axis, pitch a rotation about the lateral Y axis and yaw a rotation about the vertical Z axis. Additionally, a linear vertical movement of the chassis is called heave, whereas longitudinal and lateral linear movements are known as surge and sway, respectively.

Each tire also has its own coordinate system, with its origin fixed at the contact center point (a wheel coordinate system has its origin at the wheel center), with a longitudinal X axis, a lateral Y axis and a vertical Z axis, as shown in Figure 5.

Figure 5 – Z-DOWN tire and wheel axis systems



Source: SAE INTERNATIONAL, 2008.

2.1.3 Tire slip

The phenomenon known as slip occurs when the tires of a vehicle generate horizontal friction forces greater than the necessary for a free rolling movement (PACEJKA, 2012).

When a torque is applied to the tire, a difference between the tire's contact patch tangential velocity and its translational velocity appears, which allows for tractive/braking forces to be generated (NICOLAZZI, 2023). This difference may be represented as a slip ratio (or longitudinal slip) κ . Under traction conditions it is determined as in Equation 1.

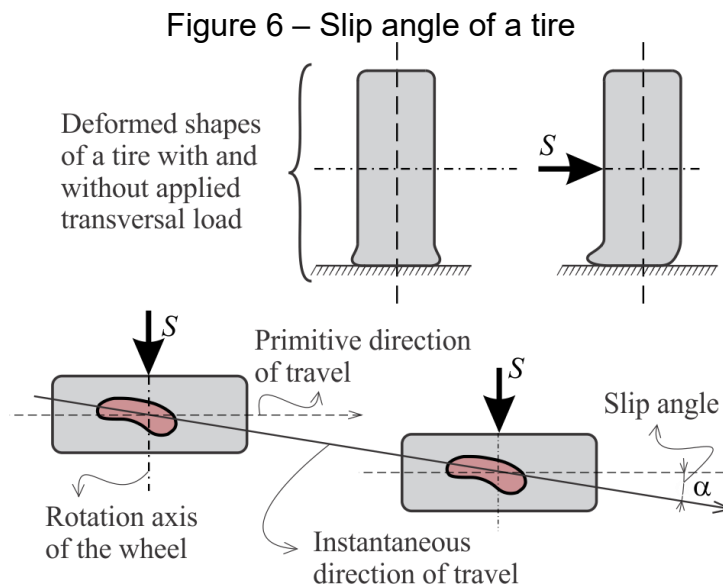
$$\kappa = \frac{V_x - V_t}{V_x} \quad (1)$$

And for braking conditions it is expressed as in Equation 2.

$$\kappa = \frac{V_t - V_x}{V_t} \quad (2)$$

Where V_x is the tire's longitudinal tangential velocity and V_t is its translational velocity.

Slip angle (or lateral slip) α is defined as the angle between the tire's longitudinal X axis and its trajectory direction. The slip angle occurs due to the deflection of the tire's contact patch as it is subjected to lateral loads, as shown in Figure 6.



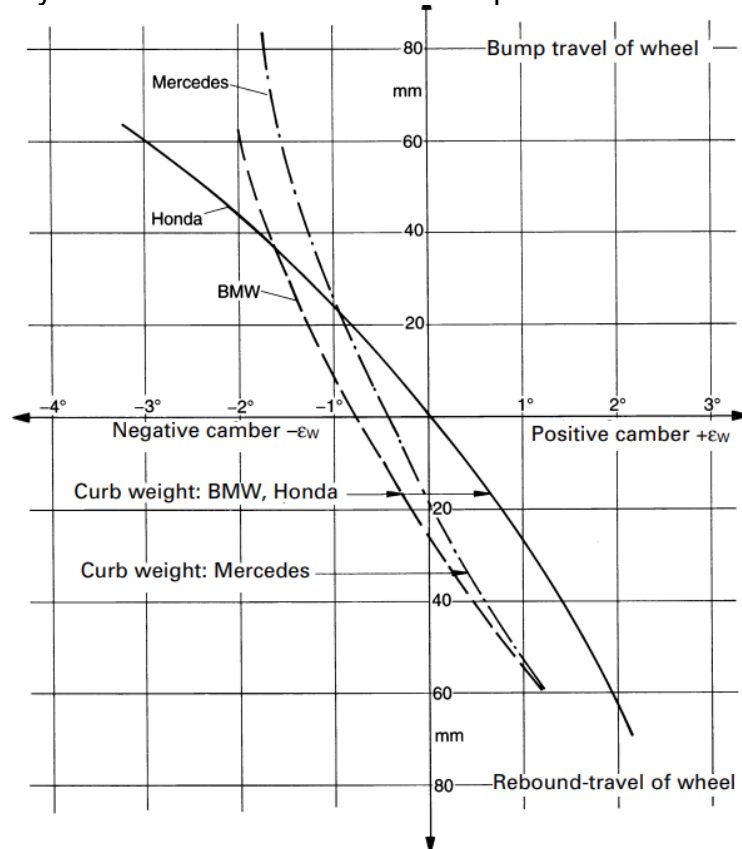
2.1.4 Wheel angles

Out of many concepts regarding steering and suspension geometry, camber and toe are two of the most influential and addressed dynamic parameters with respect to both design and setup tuning.

Camber (or inclination angle) is the lateral angle between the wheel plane and a vertical plane intercepting the tire's contact patch center point. By convention it is considered positive when the superior portion of the wheels point outwards from the vehicle and negative when the opposite is true. The camber angle may significantly influence the performance of a vehicle due to a difference in tire behavior induced by it. The displacement of the suspension mechanism and chassis roll may cause a variation on the inclination angles of each wheel, commonly referred to as dynamic camber. Figure 7 shows examples of dynamic camber in real vehicles,

where camber angle is induced on the wheels as the wheels travel vertically (bump and rebound). For this reason, the wheels of racing cars tend to be set with a negative camber configuration, as a means to mitigate those effects and improve overall tire performance.

Figure 7 – Dynamic camber due to wheel displacement on different vehicles

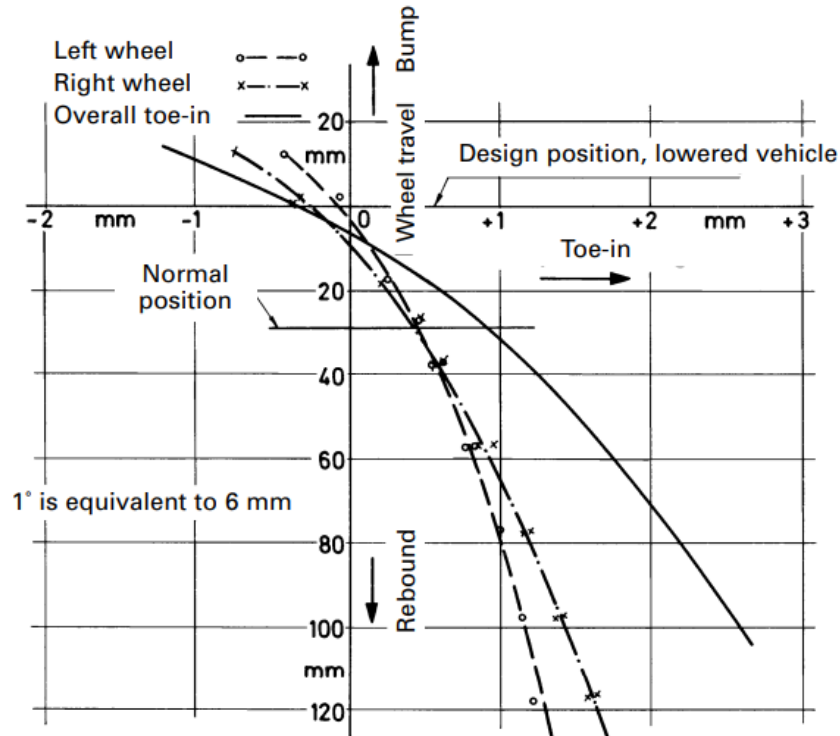


Source: REIMPELL, 2001.

Toe may be interpreted as a static steering angle setup on each wheel. It is called toe-in when the wheels are steered inward to the vehicle and toe-out when the steering angle points outwards. This setup is used as a means to modify the slip angles of each tire as the vehicle manages a corner and the wheels are steered. The downside to this method is that a slip angle is induced on the wheels even when the vehicle follows a straight trajectory, which substantially increases tire wear. Similarly to camber, toe is a dynamic parameter and may vary with wheel travel, a phenomenon known as bump steer for a single wheel or roll steer when the displacements of the suspension mechanisms are induced by chassis roll. Figure 8 shows an example of bump steer, where toe angle is induced as the wheel travels

vertically (bump and rebound). The Figure shows the difference in left and right wheels' bump steer on a real vehicle.

Figure 8 – Toe variation of a VW Golf GTi



Source: REIMPELL, 2001.

The graph shows the kinematics of a vehicle lowered by 30 mm. The normal position is the one specified by the manufacturer, which presents less variation in toe between wheels when compared to the lowered position. The behavior of the modified suspension may profoundly impact vehicle control and tire wear.

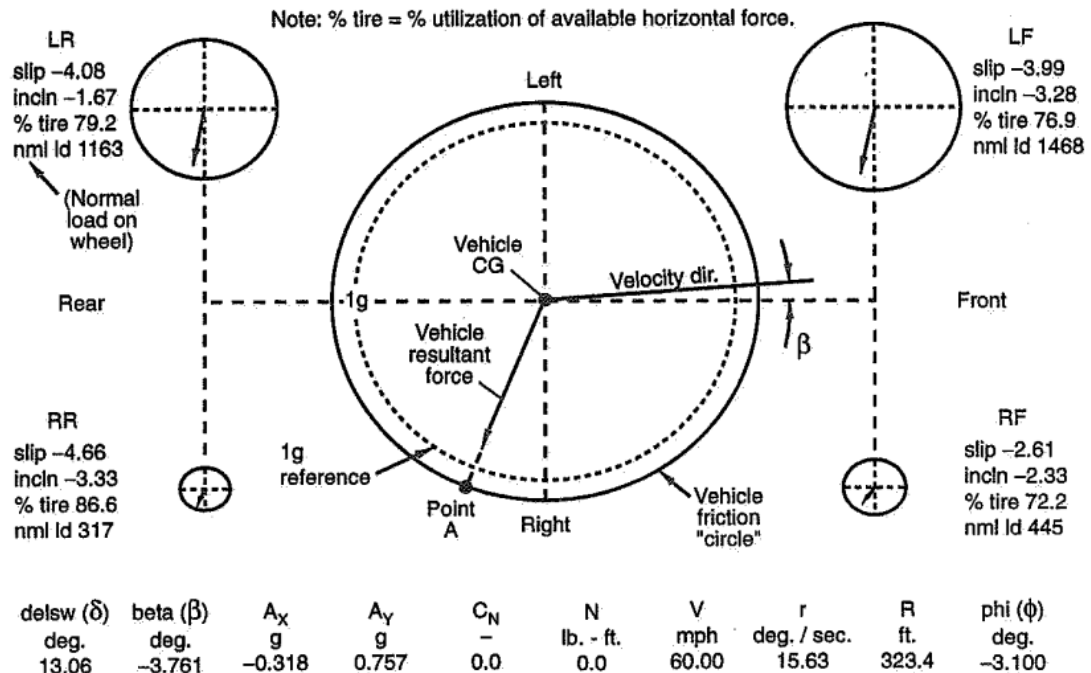
2.1.5 Driver inputs

The design of a race car requires that the driver is provided with enough control and stability over the vehicle to operate it following a “g-g” diagram. This diagram basically represents the limits of acceleration which the car is capable of achieving. Those limits are imposed mainly by tire behavior, load transfer and aerodynamic influence (MILLIKEN; MILLIKEN, 1995). Figure 9 shows a generic representation of a “g-g” diagram and the influence of each tire. The vertical and

longitudinal axis represent lateral and longitudinal accelerations, respectively, both in terms of gravitational acceleration g .

Figure 9 – “g-g” diagram of a rear-wheel drive vehicle under braking and cornering

Moderate braking dropped throttle



Source: MILLIKEN; MILLIKEN, 1995.

The shape of the presented diagram impels the driver to deliver his/her inputs in such a way to create combined accelerations that follow the outer limits of the ellipse, in order to achieve maximum performance of the vehicle during cornering. For this reason, each corner is usually divided into five segments, each related to the driver's input in that section, being them:

- pure braking
- combined braking and steering (trail braking)
- pure steering (corner apex)
- combined throttle and steering (unwinding)
- pure throttle

A standard turn in racing will include all the segments presented above, and the driver follows the order of inputs as listed. As the vehicle achieves the desired speed after pure braking, the driver smoothly releases the brakes as he/she starts to steer, applying proper trail braking technique. After fully releasing the brakes, throttle

is carefully applied to control wheel spin and the steering wheel is simultaneously straightened, ending the corner unwinding with full throttle input.

2.2 VEHICLE MODELS

There are many objectives in mathematically representing a vehicle, which may vary over a spectrum of complexity and realism compromises. This subsection presents different approaches to vehicle modelling and their main purposes and advantages.

2.2.1 Point mass model

The simplest vehicle model is obtained by collapsing all the characteristics of a vehicle to a single point of mass, which has a single velocity state V and may be accelerated laterally and/or longitudinally by an average friction force, determined with the average friction coefficient μ from all four tires. Equation 3 exemplifies the velocity of a point mass vehicle negotiating a corner with radius R .

$$V = \sqrt{\mu g R} \quad (3)$$

Where V is the velocity which the vehicle carries through the corner and g is the vertical acceleration due to gravity.

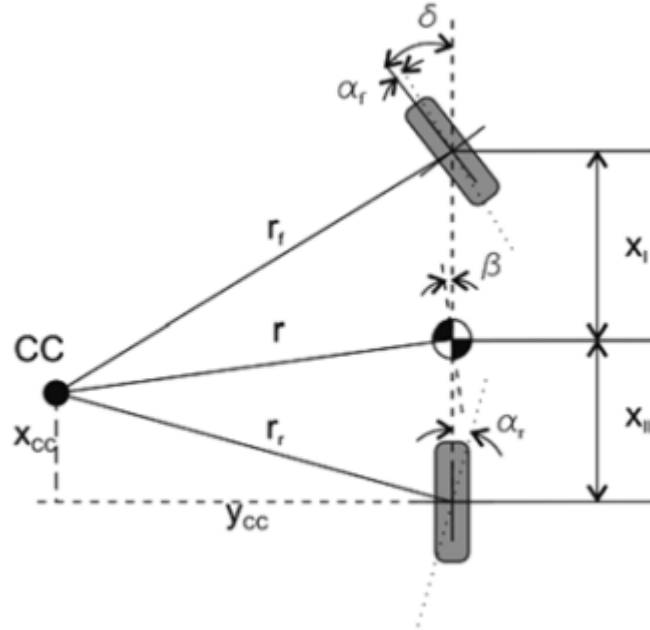
Other phenomena may be included in this model to increase fidelity to real vehicular conditions, such as aerodynamic influences and more complex tire behaviors.

The point mass model is often used for long-term trajectory definitions such as in lap time simulators (NORONHA, 2025). This type of software focuses more on the overall vehicle behavior and capabilities throughout a described route, such as a track circuit lap. However, they do not consider internal physical phenomena regarding the vehicle's suspension and steering, for example, as it is usually not needed. Different models are presented to satisfy those necessities.

2.2.2 Steady-state bicycle and four-wheel models

The bicycle model (or single-track model) consists of a simplified representation of a four-wheel car, where both wheels of either axle are collapsed onto a single wheel, as illustrated in Figure 10.

Figure 10 – Bicycle model and tire slip angles



Source: VIEIRA, *et al.*, 2012.

This model introduces the concept of tire slip angles α , steering angles δ and chassis slip angle β .

The bicycle model can be expanded to a four-wheel model, which considers that each wheel has its own steering and slip angles. This allows for a more complete and realistic analysis of vehicle behavior. Figure 11 represents the relation between a bicycle model and the four-wheel model, known as an Ackermann diagram.

The benefits to this type of model relate to mathematical simplicity and the need for little computational power to implement them. This is true because of their non-dependency on differential equations of motion. Quasi-steady state models have a much greater capability of representing physical phenomena that profoundly impact vehicle behavior and design when compared to steady-state models and are significantly easier to understand and implement than transient models. However, if there is a need to simulate oscillatory and time-dependent concepts such as damping and complex tire behavior, the use of a transient model is necessary.

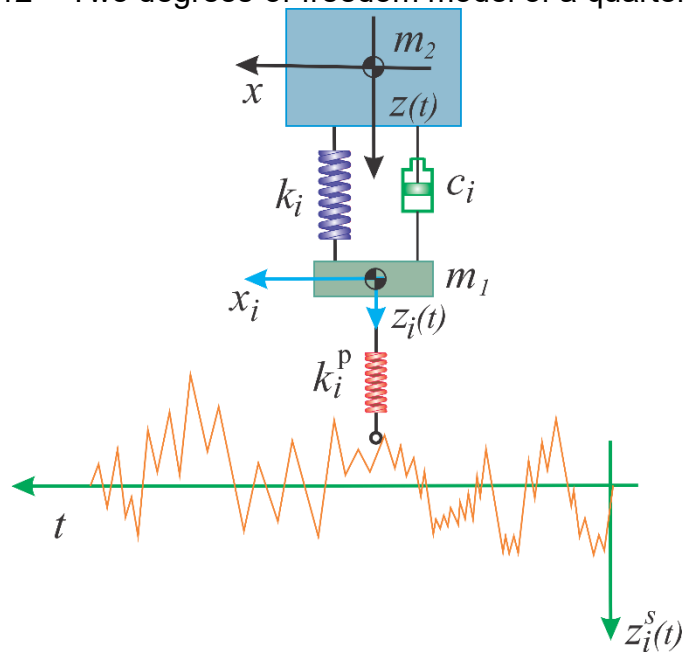
2.2.4 Transient models

Linear mechanical systems may exhibit time-varying responses due to dynamic excitations. The models that allow analysis of said responses are called transient, which are expressed as equations of motion following Newton's laws (MILLIKEN; MILLIKEN, 1995).

This concept may be applied to simpler models as a means to determine different parameters, such as yaw velocity and acceleration, acquired from a transient bicycle representation, for example.

A common use of transient models is in vertical dynamics, which consists of the analysis of dynamic responses of the vehicle masses to external excitation forces, usually caused by road-profile disturbances. Figure 12 presents a two degree-of-freedom transient model, which collapses the vehicle onto a single wheel, but represents the influence of tire and suspension stiffnesses, suspension damping and detachment of sprung and unsprung masses.

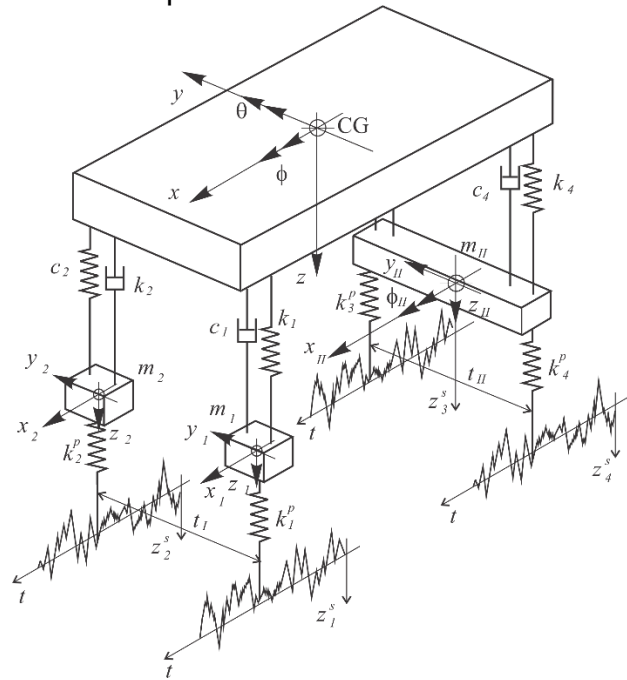
Figure 12 – Two degrees-of-freedom model of a quarter vehicle



Source: NICOLAZZI, 2023.

Figure 13 shows an expansion of the previous model to a seven degree-of-freedom model for a vehicle with front independent suspension and rear solid axle suspension. This model considers separate sprung masses to each front wheel, the inclination angle and vertical displacement of the rear axle and the three motions of the chassis, being roll, pitch and heave.

Figure 13 – Seven degrees-of-freedom model for a vehicle with independent front suspension and rear solid axle



Source: NICOLAZZI, 2023.

2.3 TIRE MODELS

Tires are the main source of forces and moments that act upon a vehicle. As so, the car designer is responsible for understanding their characteristics to aim at optimizing vehicle control and stability. However, the behavior of tires is often too complex to be analyzed as a whole, which makes it usual to approach each of their aspects separately (MILLIKEN; MILLIKEN, 1995). Similar to vehicle models, tires can be mathematically described over a spectrum of complexity, depending on its application. They may vary from a simple friction ellipse that defines the static limits of grip to transient/vibratory models used to determine complex characteristics such as tire dynamic damping (PACEJKA, 2012).

2.3.1 The *Magic Formula* model

The so-called *Magic Formula* is one of the most broadly used models in the automotive industry. It is considered a semi-empirical tire model which contemplates steady-state forces, moments and non-linearities associated with tire behavior. This

model is based on the similarity method, which states that the curves of tire forces and moments caused by pure slip maintain a similar shape, even when the tire is subjected to conditions different to their reference (PACEJKA, 2012).

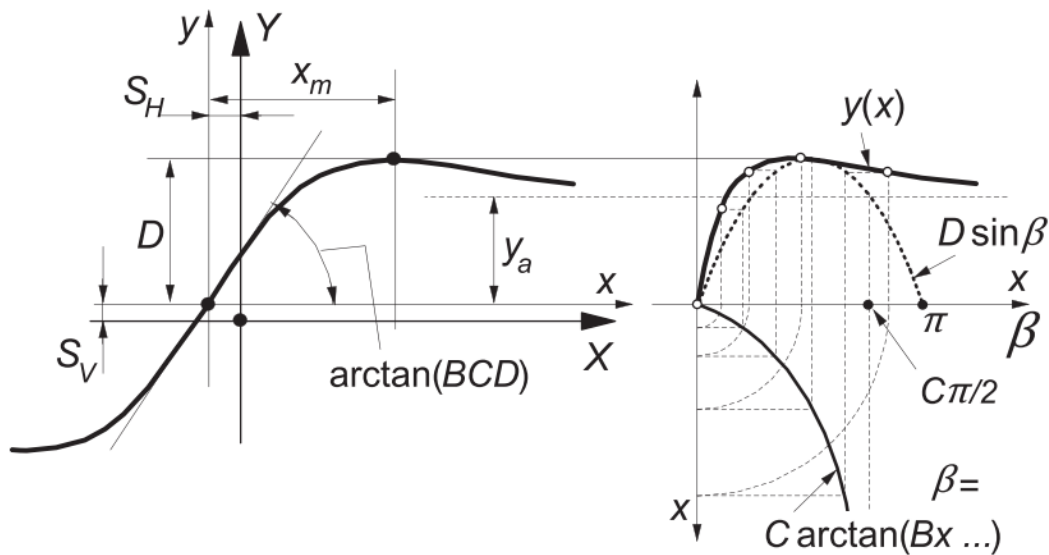
In essence, the *Magic Formula* is a curve fitting method that emulates tire behaviors observed in empirical testing. Its generic version is given below by Equations 4 and 5.

$$Y(x) = S_V + D \sin\{C \arctan\{B x - E[B x - \arctan(B x)]\}\} \quad (4)$$

$$x = X + S_H \quad (5)$$

Figure 14 shows the generic shape of a curve given by the *Magic Formula*.

Figure 14 – Curve produced by the original sine version of the *Magic Formula*



Source: PACEJKA, 2012.

Here, Y is the output variable (longitudinal force F_x or lateral force F_y) and X the input (lateral slip angle α or longitudinal slip ratio κ). The formula also contains factors that influence the results, being B the stiffness factor, C the shape factor, D the peak value, E the curvature factor, S_H a horizontal shift and S_V a vertical shift on the plot. Additionally, a cornering stiffness factor K is determined, which represents the curve inclination near the origin. It is given by Equation 6.

$$K = B C D \quad (6)$$

Finally, each parameter is defined separately for lateral and longitudinal forces and are determined by several empirical parameters p .

The full set of Equations for lateral forces is given in Equations 7 – 17.

$$F_y = S_{Vy} + D_y \sin\{C_y \operatorname{atan}\{B_y \alpha_y - E_y [B_y \alpha_y - \operatorname{atan}(B_y \alpha_y)]\}\} \quad (7)$$

$$\alpha_y = \alpha + S_{Hy} \quad (8)$$

$$C_y = p_{Cy1} \quad (9)$$

$$D_y = \mu_y F_z \quad (10)$$

$$\mu_y = (p_{Dy1} + p_{Dy2} df_z)(1 - p_{Dy3} \gamma^2) \quad (11)$$

$$E_y = (p_{Ey1} + p_{Ey2} df_z)[1 - (p_{Ey3} + p_{Ey4} \gamma) \operatorname{sgn}(\alpha_y)] \quad (12)$$

$$K_{y\alpha} = p_{Ky1} F_{zo} (1 - p_{Ky3} |\gamma|) \sin \left[2 \operatorname{atan} \left(\frac{F_z}{p_{Ky2} F_{zo}} \right) \right] \quad (13)$$

$$B_y = \frac{K_{y\alpha}}{C_y D_y} \quad (14)$$

$$S_{Hy} = (p_{Hy1} + p_{Hy2} df_z) + (p_{Hy3} \gamma) \quad (15)$$

$$S_{Vyy} = F_z \gamma (p_{Vy3} + p_{Vy4} df_z) \quad (16)$$

$$S_{Vy} = F_z (p_{Vy1} + p_{Vy2} df_z) + S_{Vyy} \quad (17)$$

Here, B_y , C_y , D_y , E_y , F_y , $K_{y\alpha}$, S_{Hy} and S_{Vy} are the lateral *Magic Formula* factors derived from the generic ones described above. Additionally, α_y is an equivalent slip angle influenced by the horizontal shift, μ_y is the lateral friction coefficient and S_{Vyy} is a vertical shift caused by camber variation.

Following, the Equations for longitudinal forces are described below in Equations 18 – 27.

$$F_x = S_{Vx} + D_x \sin\{C_x \operatorname{atan}\{B_x \kappa_x - E_x [B_x \kappa_x - \operatorname{atan}(B_x \kappa_x)]\}\} \quad (18)$$

$$\kappa_x = \kappa + S_{Hx} \quad (19)$$

$$C_x = p_{Cx1} \quad (20)$$

$$D_x = \mu_x F_z \quad (21)$$

$$\mu_x = (p_{Dx1} + p_{Dx2} df_z)(1 - p_{Dx3} \gamma^2) \quad (22)$$

$$E_x = (p_{Ex1} + p_{Ex2} df_z)[1 - p_{Ex4} \operatorname{sgn}(\kappa_x)] \quad (23)$$

$$K_{x\kappa} = F_z (p_{Kx1} + p_{Kx2} df_z) \exp(p_{Kx3} df_z) \quad (24)$$

$$B_x = \frac{K_{x\kappa}}{C_x D_x} \quad (25)$$

$$S_{Hx} = p_{Hx1} + p_{Hx2} df_z \quad (26)$$

$$S_{Vx} = F_z (p_{Vx1} + p_{Vx2} df_z) \quad (27)$$

Here, B_x , C_x , D_x , E_x , F_x , $K_{x\kappa}$, S_{Hx} and S_{Vx} are the longitudinal *Magic Formula* factors derived from the generic ones described above. Additionally, κ_x is an equivalent slip ratio influenced by the horizontal shift and μ_x is the longitudinal friction coefficient.

Moreover, γ is the camber angle, F_z is the vertical load imposed on the tire, F_{zo} is the reference vertical load and df_z is a normalized change in vertical load, expressed as in Equation 28.

$$df_z = \frac{F_z - F_{zo}}{F_{zo}} \quad (28)$$

It is interesting to note that for both lateral and longitudinal sets of equations, most parameters are heavily influenced by the variation of vertical load on the tire. This means that vehicle weight, aerodynamic downforce and load transfer profoundly impacts tire behavior, which in turn may alter vehicle drivability. Some parameters are also function of camber. Because of this, it is necessary for an engineer to understand how each parameter influences tire behavior, and how they change depending on conditions imposed for the tires. This understanding allows for the correct choice of tire for a vehicle, or for the correct optimization of load distribution and suspension kinematics to better suit a certain tire.

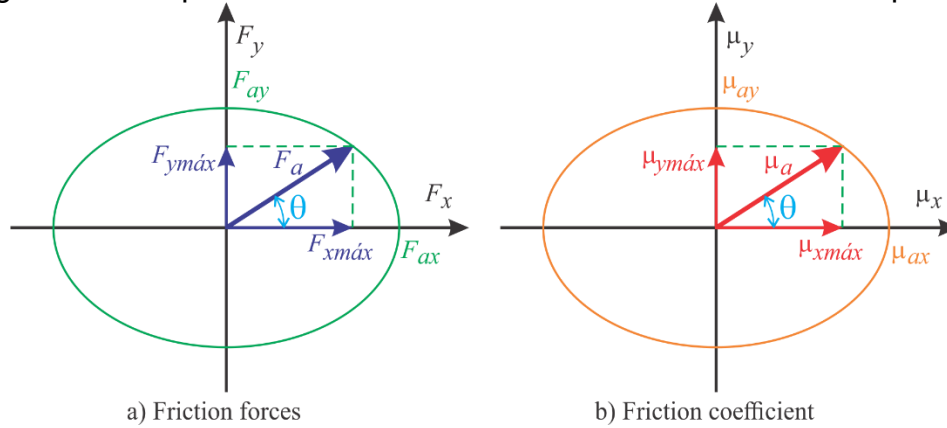
The *Magic Formula* also contemplates many other tire behavior phenomena, such as aligning torque, overturning moments and inflation pressure influence, but these concepts will not be discussed here.

2.3.2 Combined slip model

When subjected to both lateral and longitudinal slips simultaneously, tires tend to generate forces following a significantly different behavior. This adds another layer of non-linearity to tire modelling. Many models are elaborated as efforts to emulate these changes.

The simplest of these models is called the friction ellipse or friction circle. It states that the total friction force is generated at an angle relative to the tire plane and its magnitude is restricted by an ellipse, as shown in Figure 15.

Figure 15 – Ellipses for friction forces and coefficients for orthotropic tires



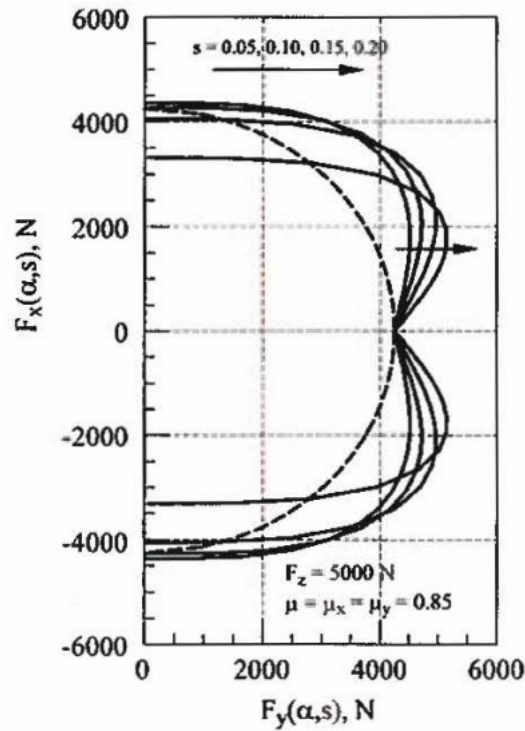
Source: adapted from NICOLAZZI, 2023.

However, the friction ellipse is an idealized and simplified model that cannot fully characterize the real behavior of tires (BRACH, 2011). Because of this, it is more adequate to use different models when seeking to achieve greater realism.

A combined slip model for the *Magic Formula* is presented by PACEJKA (2012). However, this work will discuss an alternative model also based on the *Magic Formula*, the Modified Nicolas-Comstock model (MNC).

The MNC model receives the pure slip forces $F_x(\kappa_x)$ and $F_y(\alpha_y)$ given by the *Magic Formula* to determine the combined slip forces $F_{x,comb}(\alpha_y, \kappa_x)$ and $F_{y,comb}(\alpha_y, \kappa_x)$. If a plot similar to the friction ellipse is generated using the MNC, it is possible to observe that as combined slip is introduced, the shape of the curve is deformed, ceasing to have an actual ellipse shape, as shown in Figure 16 (in the Figure, the symbol s is used for slip ratio instead of κ). In the Figure, a friction ellipse is plotted with pure lateral slip (dashed line). The full lines represent the effect of adding slip ratio to the curves, from left to right, but maintaining a constant slip angle, culminating in combined slip forces.

Figure 16 – Tire force curves generated by the MNC model compared to the friction ellipse model (dashed line)



Source: BRACH, 2011.

The expressions to determine the combined slip forces following the MNC model are given by Equations 29 and 30, below:

$$F_{x,comb} = \frac{F_x F_y}{\sqrt{\kappa_x^2 F_y^2 + F_x^2 \tan^2(\alpha_y)}} \frac{\sqrt{\kappa_x^2 K_{y\alpha}^2 + (1 - |\kappa_x|)^2 \cos^2(\alpha_y) F_x^2}}{K_{y\alpha}} \quad (29)$$

$$F_{y,comb} = \frac{F_x F_y}{\sqrt{\kappa_x^2 F_y^2 + F_x^2 \tan^2(\alpha_y)}} \frac{\sqrt{\sin^2(\alpha_y) K_{x\kappa}^2 + (1 - |\kappa_x|)^2 \cos^2(\alpha_y) F_y^2}}{K_{x\kappa} \cos(\alpha_y)} \quad (30)$$

The first quotient of the Equations is the original Nicolas-Comstock model, and the second is a correction factor (BRACH, 2011). When lateral load is exclusively being generated, the introduction of longitudinal forces causes the first to be reduced and vice-versa, causing the total friction force to follow the deformed “ellipse” shape presented.

2.3.3 Rolling resistance model

As a vehicle is in motion, some energy is lost to keep the tires rolling. This happens due to deformation of the soil, contact of the tire with the ground and hysteresis in tire compression caused by its damping characteristics (NICOLAZZI, 2023). These effects cause resistances to motion for the vehicle, called tire rolling resistance forces F_{rr} .

These forces can be interpreted as additional longitudinal friction forces oriented following the tire's coordinate system. They are function of the vertical load imposed on the tire F_z and its rolling resistance coefficient f , as given in Equation 31.

$$F_{rr} = f F_z \quad (31)$$

However, the coefficient f is influenced by the type of tire and soil, inflation pressure, vertical load and longitudinal velocity. For simplicity, the rolling resistance coefficient will be given as a function of the tire's longitudinal velocity V_x , as presented in Equation 32.

$$f = 0,0258 + 0,052 \left(\frac{V_x}{100} \right)^2 \quad (32)$$

This expression, as presented by NICOLAZZI (2023), considers parameters for high hysteresis tires, for racing and high-performance purposes.

2.4 SUSPENSION DESIGN

The suspension system of a vehicle is responsible for providing a comfortable ride for its passengers and greater durability to its structure. Additionally, it may be used as a means to manage loads and the handling characteristics of the vehicle under a design perspective (NICOLAZZI, 2023).

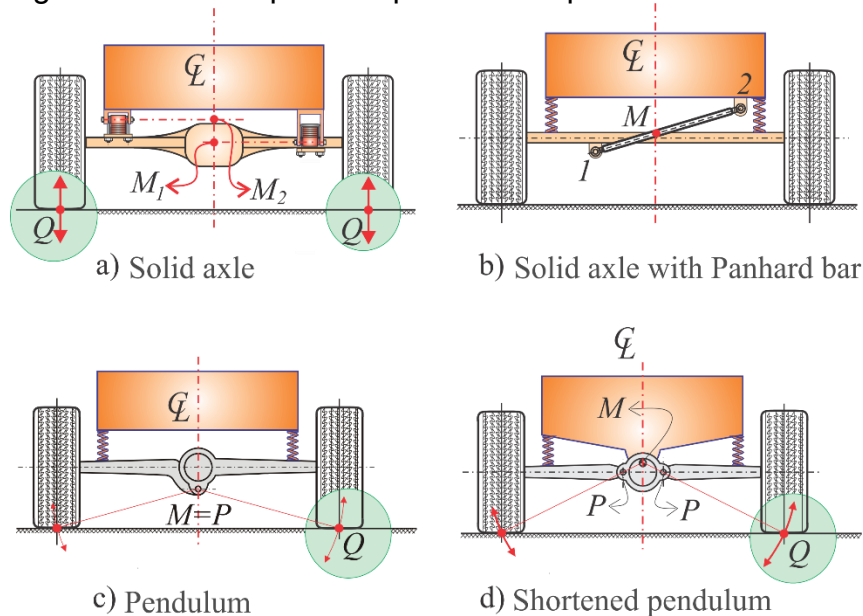
2.4.1 Types of suspension mechanisms

There are many different concepts of suspension mechanisms, each offering advantages for particular purposes and vehicle applications. Most of these concepts fall within two major groups, being:

- a) Dependent
- b) Independent

Dependent suspension systems are usually of simpler design, being found on older muscle cars, specialized all-terrain, utility and heavy-duty load vehicles, such as SUVs, pickup and semi-trucks, applications where they excel at. However, this type of mechanism presents some shortcomings, specially involving performance factors, due to its usual greater mass, mutual influence on wheels of the same axle and limited possibility for kinematic optimization (REIMPELL, 2001). Figure 17 shows some variations to this concept.

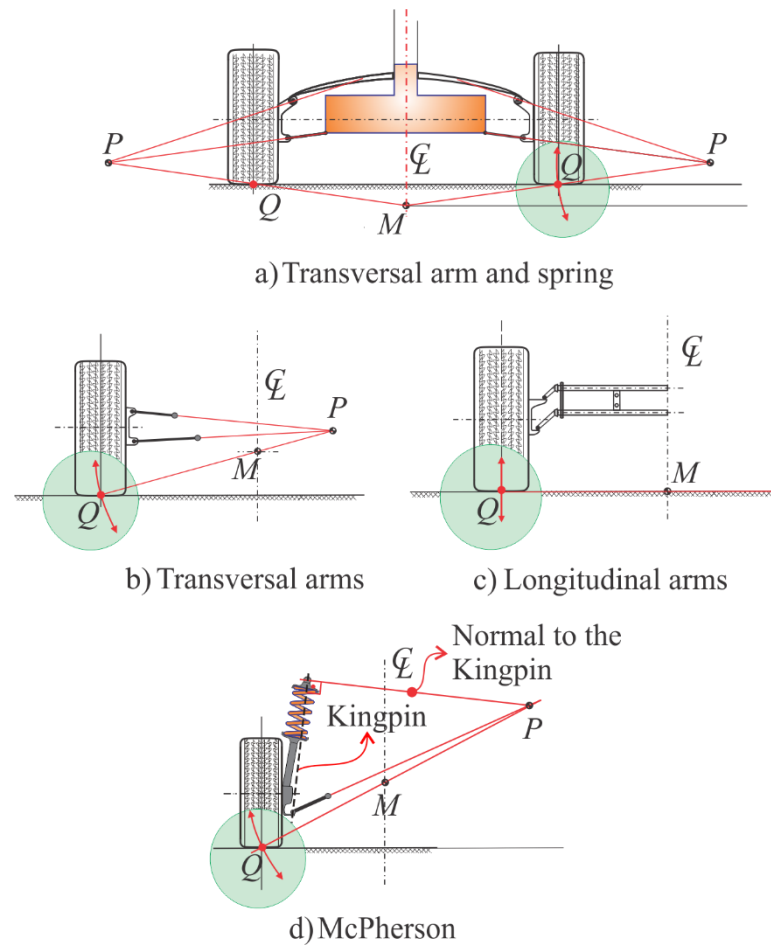
Figure 17 – Concepts of dependent suspension mechanisms



Source: adapted from NICOLAZZI, 2023.

Conversely, independent suspension mechanisms tend to offer greater control over geometry and kinematics, better road-holding characteristics and lower overall mass, granting both superior performance and improved freedom on its design. Trailing, semi-trailing and McPherson are amongst the most widespread concepts, being the usual choice for standard passenger vehicles, whereas multi-link and double wishbone suspensions are associated with high-end luxury and performance cars, the latter being more exclusively used for sports and racing purposes. Figure 18 shows different concepts of independent suspension mechanisms.

Figure 18 – Concepts of independent suspension mechanisms



Source: adapted from NICOLAZZI, 2023.

2.4.2 Double wishbone suspension kinematics

As previously mentioned, double wishbone suspension mechanisms are the main choice to equip sports and racing cars due to their significant performance advantages. Figure 19 shows an example of a racing vehicle equipped with this concept of suspension.

Figure 19 – McLaren MP 4/4 driven by Ayrton Senna during the 1988 F1 World Championship, at which he was consecrated champion

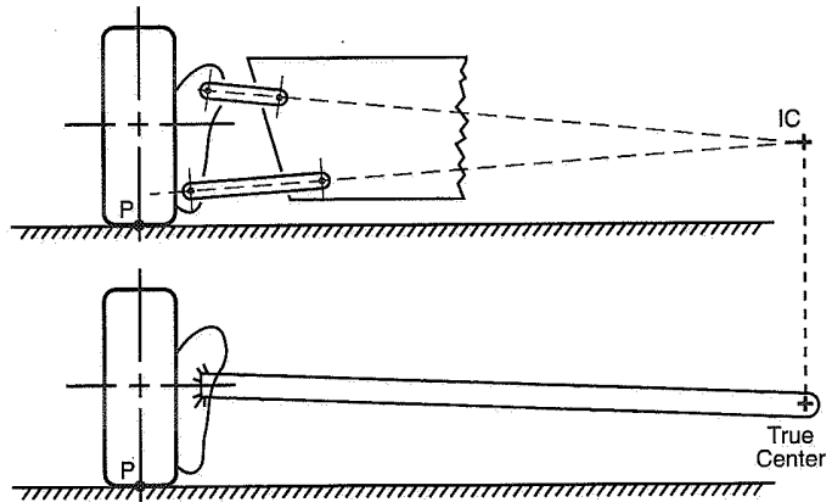


Source: KOIKE, 1988.

The double wishbone mechanism consists of two pivoted links (control arms or A-arms) that connect the sprung masses to the upright assembly. The spring/damper actuation is usually accomplished via push-rod or pull-rod and a rocker on high performance vehicles; however, it may also be directly assembled on the suspension arms.

The roll instantaneous center is a virtual point that acts like an effective pivot to the unsprung masses. It is determined by extending the control arms geometry on a front view plane, as shown in Figure 20. It is important to note that this pivot does not have a fixed position in space. Its movement is dictated by the suspension geometry itself as it is displaced, and the direction and velocity at which it moves may significantly influence the vehicle behavior as a whole.

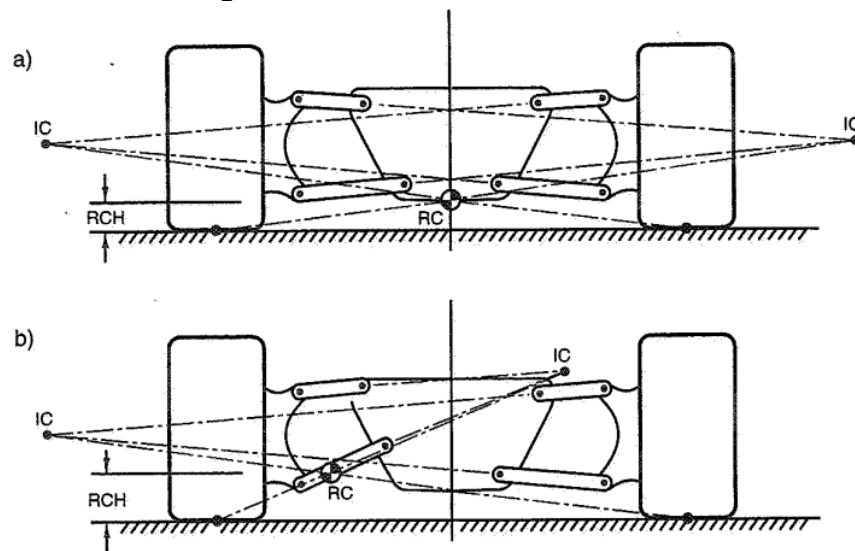
Figure 20 – Roll instantaneous center concept



Source: MILLIKEN; MILLIKEN, 1995.

Connecting the roll instantaneous center to the tire's contact patch center point is a virtual line called the front view swing arm. The point at which the left and right wheels' swing arms of the same axle cross defines the roll center, a virtual point that dictates most of the suspension behavior under roll motions. Similarly to the instantaneous center, the roll center moves as the vehicle rolls and heaves. These movements should be understood by the suspension designer to better optimize its operation. The geometric definition of the roll center is shown in Figure 21.

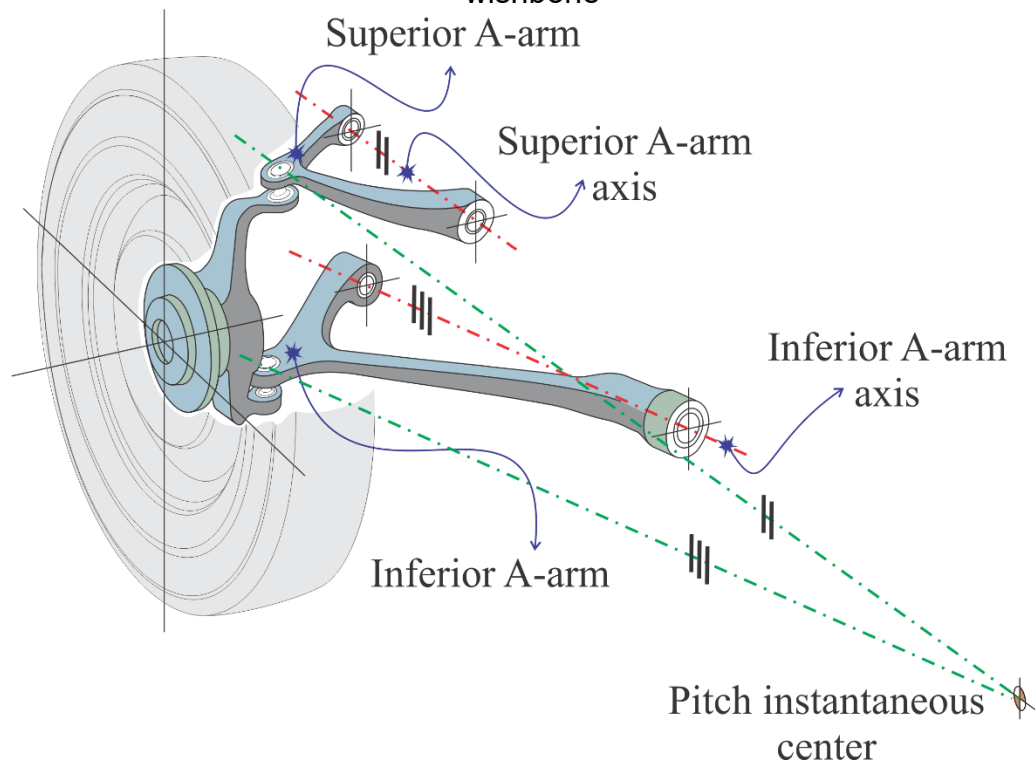
Figure 21 – Roll center construction



Source: MILLIKEN; MILLIKEN, 1995.

Similarly, a pitch instantaneous center may be defined on a side view plane as the convergence point of virtual lines parallel to the control arms axis that intercept the upright joints, as shown in Figure 22. Additionally, the side view swing arms connect the instantaneous pitch center to the wheels' contact patch center point. The front and rear swing arms of the same side of the vehicle converge to a virtual point called the pitch center. These geometric points may also move due to wheel vertical displacement.

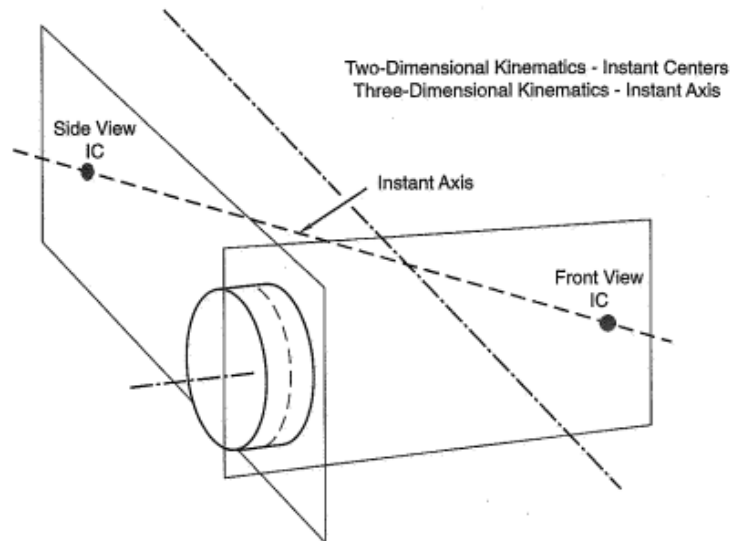
Figure 22 – Geometric definition of the pitch instantaneous center of a double wishbone



Source: adapted from NICOLAZZI, 2023.

Connecting the roll and pitch instantaneous centers is the instantaneous axis, around which the unsprung masses of each wheel instantaneously rotate as dictated by the suspension geometry. Figure 23 represents the instantaneous axis as described. Alternatively, linking the front and rear roll centers defines a roll axis, as for the union of left and right pitch centers create a pitch axis, around which the sprung masses of the vehicle rotate under roll and pitch movements, respectively.

Figure 23 – Instantaneous axis concept



Source: MILLIKEN; MILLIKEN, 1995.

2.4.3 Anti-roll bars

Anti-roll bars or ARBs are mechanical components applied to suspension mechanisms with two main goals:

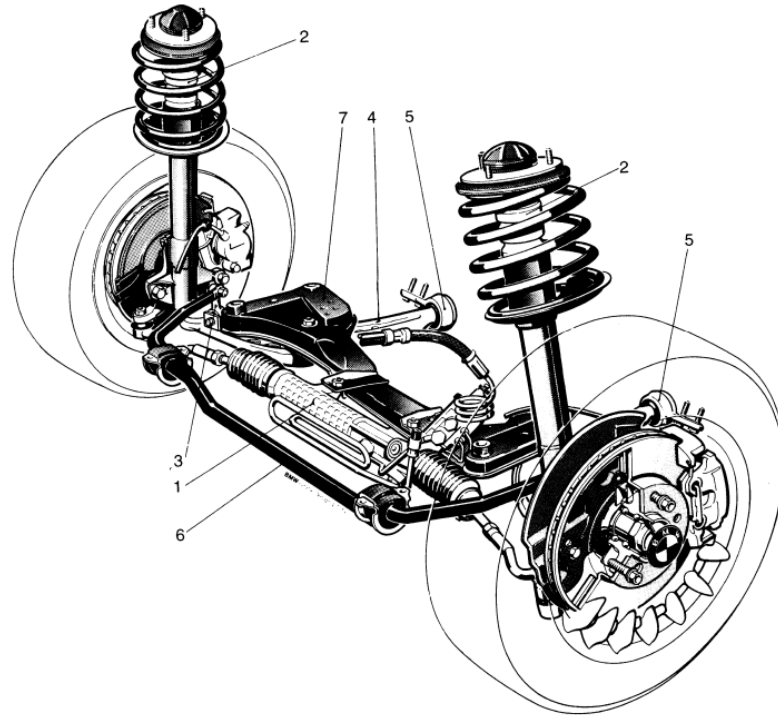
- a) decouple roll stiffnesses during design stages;
- b) fine tune roll distribution for specific track conditions.

For the first goal, ARBs are simply considered an extra set of springs that connects the left and right wheels of a same axle, altering the stiffness of the suspension; however, they act only under roll conditions. This property facilitates the achievement of optimal roll stiffness for the vehicle without interfering with the ride comfort and pitch behaviors.

The second objective is usually associated with racing vehicles, and demand a set of adjustable ARBs. They are used as tools to finely control the percentage of roll moments absorbed by each axle, which influences the handling characteristics of the car without altering the overall roll stiffness and lateral load transfer. This helps racing engineers to find the best vehicle setup for a specific track, driver and/or environmental conditions.

Figure 24 gives an example of an anti-roll bar, represented by the number 6.

Figure 24 – Suspension system equipped with anti-roll bar (represented in 6)



Source: REIMPELL, 2001.

2.4.4 Wheel rates and motion ratio

The motion ratio MR is the mechanical advantage given to the suspension springs by its mechanism. It is as a ratio between the wheel displacement and the spring deflection and is presented in Equation 33.

$$MR = \frac{w}{s} \quad (33)$$

Where s is the spring deflection and w the wheel vertical displacement.

Related to the motion ratio is the concept of wheel rate, which is represented by a virtual spring acting on the wheel plane, meaning it is the actual effective stiffness of the suspension given by the real springs and amplified by the suspension mechanism. It is expressed in Equation 34.

$$K = k MR^{-2} \quad (34)$$

Where K is the wheel rate and k the stiffness of the real suspension spring. This means that the relation between spring deflection and wheel travel alters the

stiffness of the suspension. For motion ratios lesser than one, the deflection of the springs is greater than the wheel displacements, meaning their actuation is enhanced, increasing the wheel rate (MILLIKEN; MILLIKEN, 1995).

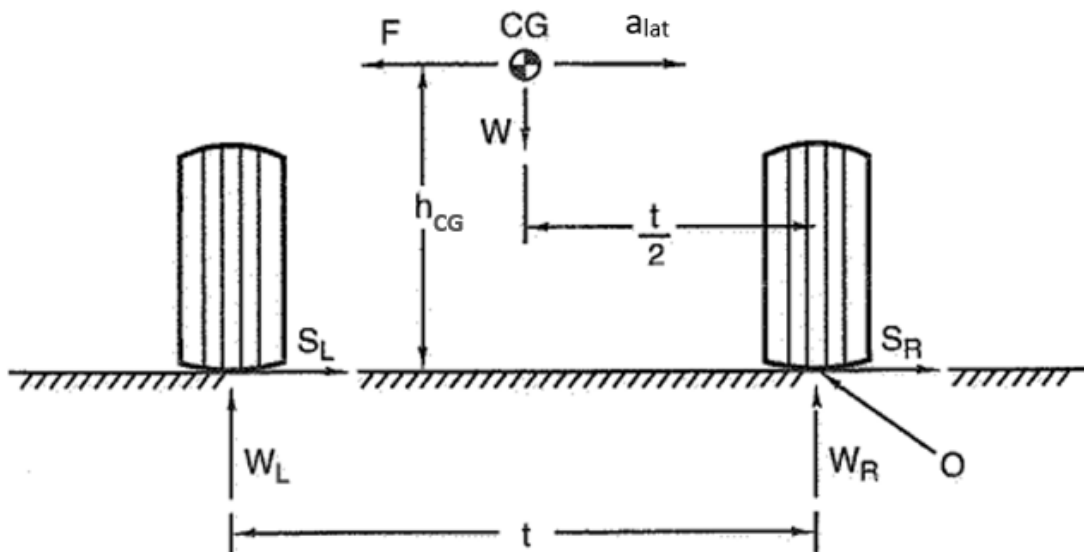
2.5 LOAD TRANSFER

The phenomenon of load transfer causes the vertical loads acting on each tire to vary, increasing for some wheels, and reducing for others, but keeping the sum of all loads equal to the vehicle's total weight. It may occur due to ground inclination, aerodynamic influences and the vehicle's lateral and longitudinal accelerations. This section focuses on the latter cause.

Both lateral and longitudinal load transfers present some nuances which differentiates the usual approach for analysis and calculation of each. However, they are ultimately the same physical phenomenon, occurring under different circumstances.

The sum of lateral forces produced by the tires create an acceleration a_{lat} , which is reacted on the vehicle's center of gravity by an inertial force F , as shown in Figure 25.

Figure 25 – Total lateral load transfer



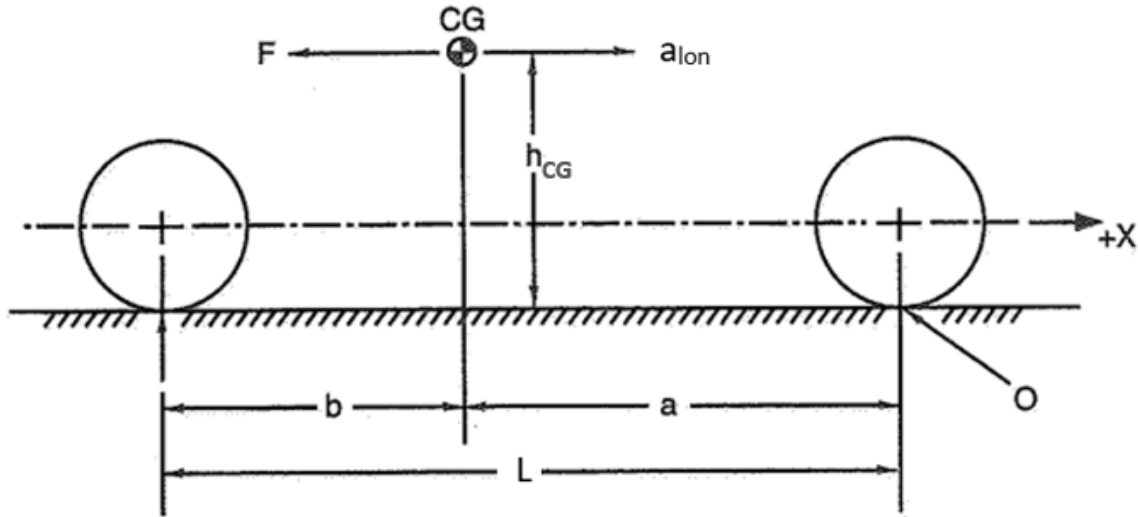
Source: adapted from MILLIKEN; MILLIKEN, 1995.

Where W is the total weight of the vehicle, h_{CG} is the height of the center of gravity and t is the track. The inertial force F creates a moment about ground level, which is reacted by the wheels' vertical loads W_L and W_R as lateral load transfer ΔW_{lat} , which is given by Equation 35.

$$\Delta W_{lat} = \frac{W a_{lat} h_{CG}}{t} \quad (35)$$

Similarly, longitudinal load transfer ΔW_{lon} follows the same principles, as shown in Figure 26.

Figure 26 – Longitudinal load transfer



Source: adapted from MILLIKEN & MILLIKEN, 1995.

It is caused by longitudinal accelerations a_{lon} under braking or traction and is determined as follows in Equation 36.

$$\Delta W_{lon} = \frac{W a_{lon} h_{CG}}{L} \quad (36)$$

Where L is the wheelbase of the vehicle.

2.5.1 Longitudinal load transfer

The suspension system of a car (usually of sports and racing cars) may be designed with a specific geometry that causes a so-called “anti” effect. This causes the longitudinal load transfer not to be totally absorbed by the suspension springs, which may alter the vehicle’s pitch and wheels’ vertical displacements. However, it

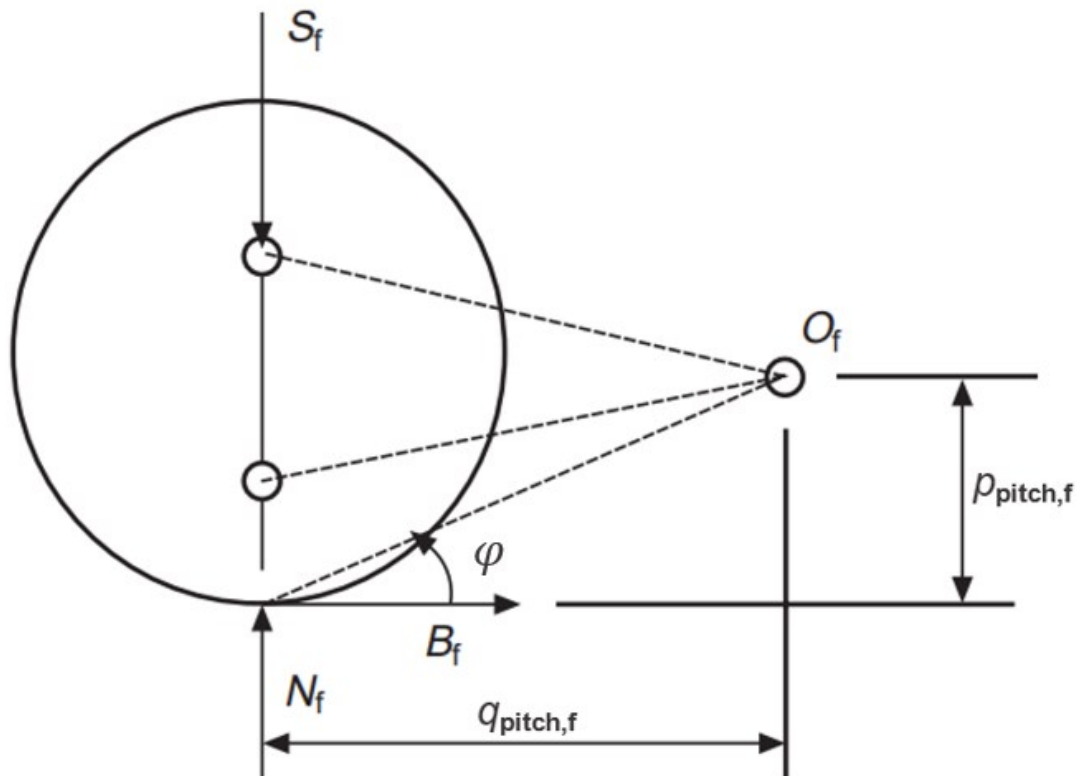
does not alter the total longitudinal load transfer, as it must always satisfy Equation 36 (MILLIKEN; MILLIKEN, 1995).

2.5.1.1 *Braking*

Under braking, the front suspension is said to be designed with anti-dive, and the rear suspension with anti-lift. In essence, both terms refer to the same concept, but are usually analyzed separately.

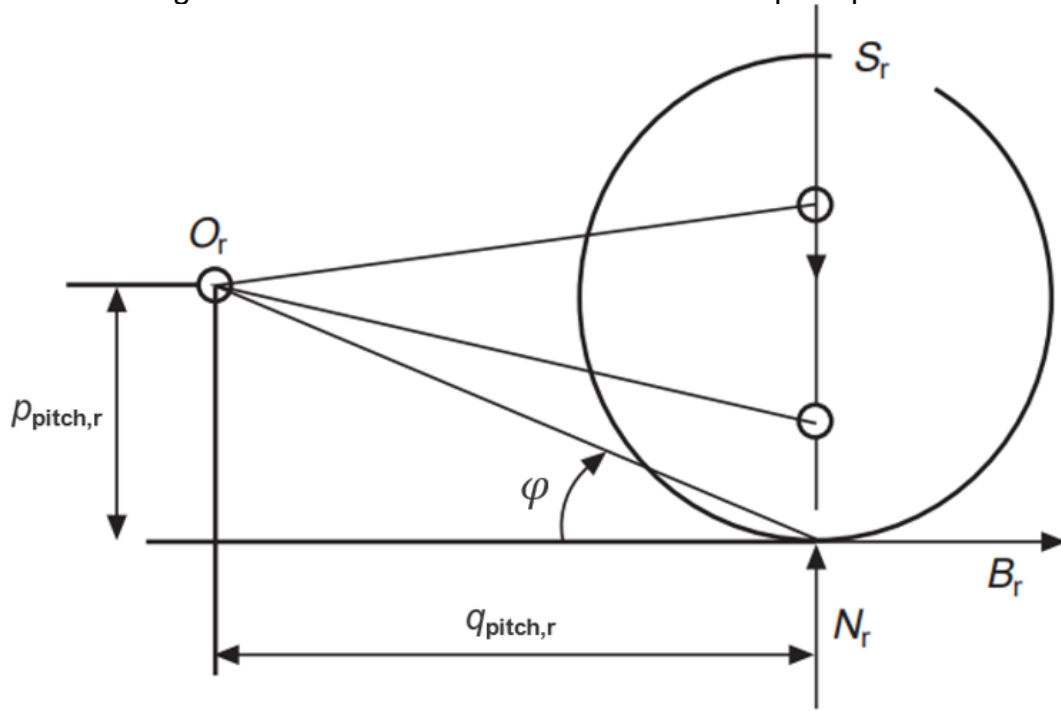
Commonly, the brake systems are mounted directly on the uprights, or in other words, on the unsprung masses of the vehicle. In this scenario, they are classified as outboard brakes. This causes the braking torques on each wheel to be absorbed by the suspension mechanism. As so, the anti-dive and anti-lift geometries are defined as shown in Figures 27 and 28. The “anti” effect is linearly proportional to the angle φ , defined between the side view swing arm and the ground.

Figure 27 – Front wheel forces and effective pivot point



Source: adapted from SMITH, 2002.

Figure 28 – Rear wheel forces and effective pivot point.



Source: adapted from SMITH, 2002.

Where O_f and O_r are respectively the front and rear suspension pivot points, or the instantaneous pitch centers.

Anti-dive is given as a percentage, which is equal to 100% if the vehicle's center of gravity lies within the side view swing arm. Under these circumstances, none of the longitudinal load transfer is absorbed by the front suspension springs, which cause them to not deflect. The actual anti-dive is given by Equation 37.

$$antidive = \frac{p_{pitch,f}}{q_{pitch,f}} \frac{bias L}{h_{CG}} \times 100\% \quad (37)$$

The same is true for anti-lift on the rear suspension, which is given in Equation 38.

$$antilift = \frac{p_{pitch,r}}{q_{pitch,r}} \frac{(1 - bias)L}{h_{CG}} \times 100\% \quad (38)$$

Where *bias* is a fixed braking ratio between front and rear braking forces B_f and B_r , determined in Equation 39.

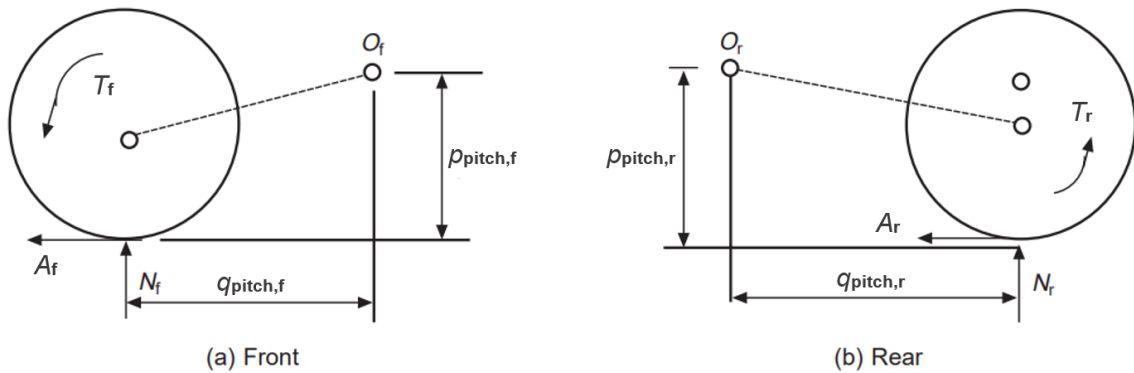
$$bias = \frac{B_f}{B_f + B_r} \quad (39)$$

2.5.1.2 Traction

Rather than anti-dive, under traction conditions the vehicle is said to be designed with anti-squat on the rear suspension and anti-lift on the front.

The analysis is the same as for braking if the vehicle is equipped with an outboard mounted differential, as in a solid axle suspension configuration. However, if the car has independent suspension geometry, the norm is for its differential to be mounted directly on the chassis, which classifies it as an inboard configuration. This causes the tractive torques to be absorbed on the sprung masses, rather than on the uprights. Because of this, the “*anti*” geometry is defined differently, as shown in Figure 29.

Figure 29 – Wheel forces and effective pivot locations



Source: adapted from SMITH, 2002.

As so, the anti-squat percentage on the rear suspension is given by Equation 40.

$$antisquat = \frac{p_{pitch,r} - r_r}{q_{pitch,r}} \frac{(1 - bias)L}{h_{CG}} \times 100\% \quad (40)$$

And the anti-lift on the front suspension is as presented in Equation 41.

$$antilift = \frac{p_{pitch,f} - r_f}{q_{pitch,f}} \frac{bias L}{h_{CG}} \times 100\% \quad (41)$$

Where r_f and r_r are the radiuses of the front and rear tires, respectively, and $bias$ is the ratio of tractive forces A_f and A_r , given by Equation 42.

$$bias = \frac{A_f}{A_f + A_r} \quad (42)$$

2.5.2 Lateral load transfer

The lateral accelerations a vehicle is subjected to are reacted as inertial forces acting on the sprung masses center of gravity, which generates a moment about the roll axis, that rotates the chassis causing a roll angle. The proportion of moment absorbed by each axle causes a distribution of load transfer, which in turn influences the handling and behavior of the vehicle. This proportion is defined by many factors, such as springs and ARB stiffnesses, unsprung masses and the type of suspension geometry (NICOLAZZI, 2023).

NICOLAZZI (2023) presents a mathematical model for lateral load transfer for an independent suspension geometry equipped with anti-roll bars. The lateral load transfer on each axle is a sum of four portions, which are caused by:

1. inertial forces absorbed by the suspension springs;
2. sprung masses inertial forces acting on the roll axis;
3. inertial forces absorbed by the anti-roll bars;
4. action of unsprung masses inertia.

2.5.2.1 Suspension springs portion

The first portion of lateral load transfer is caused by the deflection of suspension springs, which is caused by the chassis roll angle Ψ . The variation of load to which the suspension is subjected is written for each axle as:

$$\Delta W_k = \Psi K \frac{t}{2} \quad (43)$$

Where ΔW_K is the first portion of lateral load transfer for each axle and K is the wheel rate associated the suspension springs.

2.5.2.2 *Sprung masses portion*

The inertia caused by the lateral acceleration of the sprung masses is reacted on each axle by its roll center. This causes a moment to appear about ground level, which is reacted by the second portion of lateral load transfer ΔW_s . As so, this portion can be expressed as in Equation 44.

$$\Delta W_s = \frac{m_s}{m} F_{lat} \frac{h_{roll}}{t} \quad (44)$$

Where h_{roll} is the height of the roll center of either axle, F_{lat} is the sum of lateral friction forces generated by the tires of an axle, m_s is the sprung mass and m is the total mass of the vehicle.

2.5.2.3 *Anti-roll bars portion*

This portion is similar to the first, and is equally a function of the chassis roll angle Ψ , being equated as in Equation 45.

$$\Delta W_{arb} = \Psi K_{arb} \frac{t}{2} \quad (45)$$

Where K_{arb} is the wheel rate associated with the axle's ARB.

2.5.2.4 *Unsprung masses portion*

The fourth and final contribution to load transfer is caused by the inertia of unsprung masses. It generates a moment about ground level, which is reacted by a vertical force acting on the roll instantaneous center. As so, the fourth contribution to load transfer ΔW_u on each axle is determined as given in Equation 46.

$$\Delta W_u = \frac{m_u}{m} F_{lat} \frac{h_{CG,u}}{q_{roll}} \quad (46)$$

Where m_u is the value of unsprung masses of the vehicle, $h_{CG,u}$ is equal to the height of the unsprung masses center of gravity and q_{roll} the lateral distance from the roll instantaneous center to the wheel plane.

2.5.2.5 Chassis roll angle

As a means to determine the roll angle, a compatibility Equation is presented. It describes an equilibrium of moments about the roll axis, as follows in Equation 47.

$$\sum roll\ moments = \sum reaction\ moments \quad (47)$$

Being roll moments the ones caused by inertia of the vehicle's masses and reaction moments generated by deflection of the suspension springs and ARBs.

Part of the roll moments are caused by the fourth portion of load transfer acting on the roll instantaneous center, being expressed as in Equation 48.

$$M_u = \Delta W_u \left(q - \frac{t}{2} \right) \quad (48)$$

Where M_u is the contribution of the unsprung masses to the roll moments.

Additionally, the inertia forces acting on the vehicle's center of gravity also contributes with a moment about the roll axis M_s , which can be determined as in Equation 49.

$$M_s = F_{lat} h_o \quad (49)$$

Where h_o is the vertical distance between the center of gravity and the roll axis.

Equations 50 and 51 give the reaction moments, which can be derived from the first and third portions of load transfer as the sum of moments caused by the suspension springs and ARBs, as follows:

$$M_k = \Psi K \frac{t^2}{2} \quad (50)$$

$$M_{arb} = \Psi K_{arb} \frac{t^2}{2} \quad (51)$$

Where M_k is the reaction moment caused by deflection of suspension springs of each axle and M_{arb} is the contribution of the anti-roll bars.

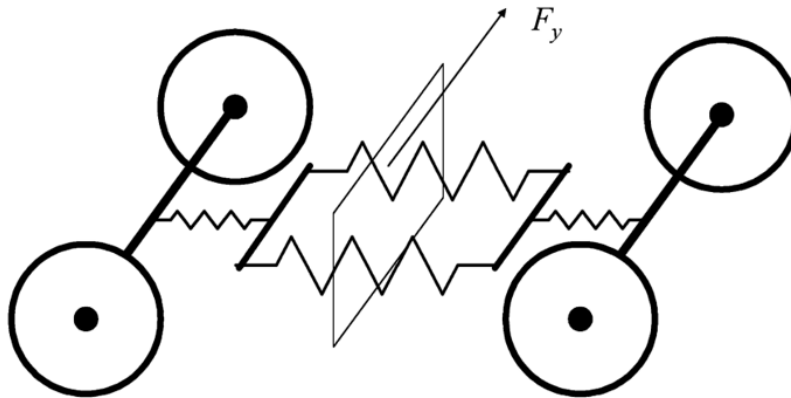
Finally, inserting each contribution into Equation 43 and isolating Ψ , an expression for the chassis roll angle is determined as Equation 52.

$$\Psi = 2 \frac{M_s + M_u}{t_f^2(K_f + K_{arb,f}) + t_r^2(K_r + K_{arb,r})} \quad (52)$$

2.5.3 Chassis torsional stiffness

As the vehicle is subjected to lateral load transfer, the inertial reaction forces developed on its center of gravity create a moment that tends to twist the chassis and are transmitted to the suspension system. Figure 30 shows the representation of the chassis as torsional springs.

Figure 30 – Elementary scheme for modelling the torsional deformation of a vehicle



Source: adapted from GENTA, 2008.

The percentage of this moment transmitted to the front or rear suspensions is determined by the chassis torsional stiffness distribution. Considering the chassis as two torsional springs connected to the front and rear axles, the stiffer element will absorb a larger portion of those loads. This effect alters both the distribution of lateral load transfer to each axle and the roll angle of the chassis.

If the chassis torsional stiffness to the front axle $k_{c,f}$ is determined, then a linear wheel rate for the chassis on the front wheels $K_{c,f}$ can be expressed as Equation 53.

$$K_{c,f} = k_{c,f} t_f^2 \quad (53)$$

The same expression is written for the chassis torsional stiffness to the rear axle $k_{c,r}$ and its wheel rate on the rear wheels $K_{c,r}$, given in Equation 54.

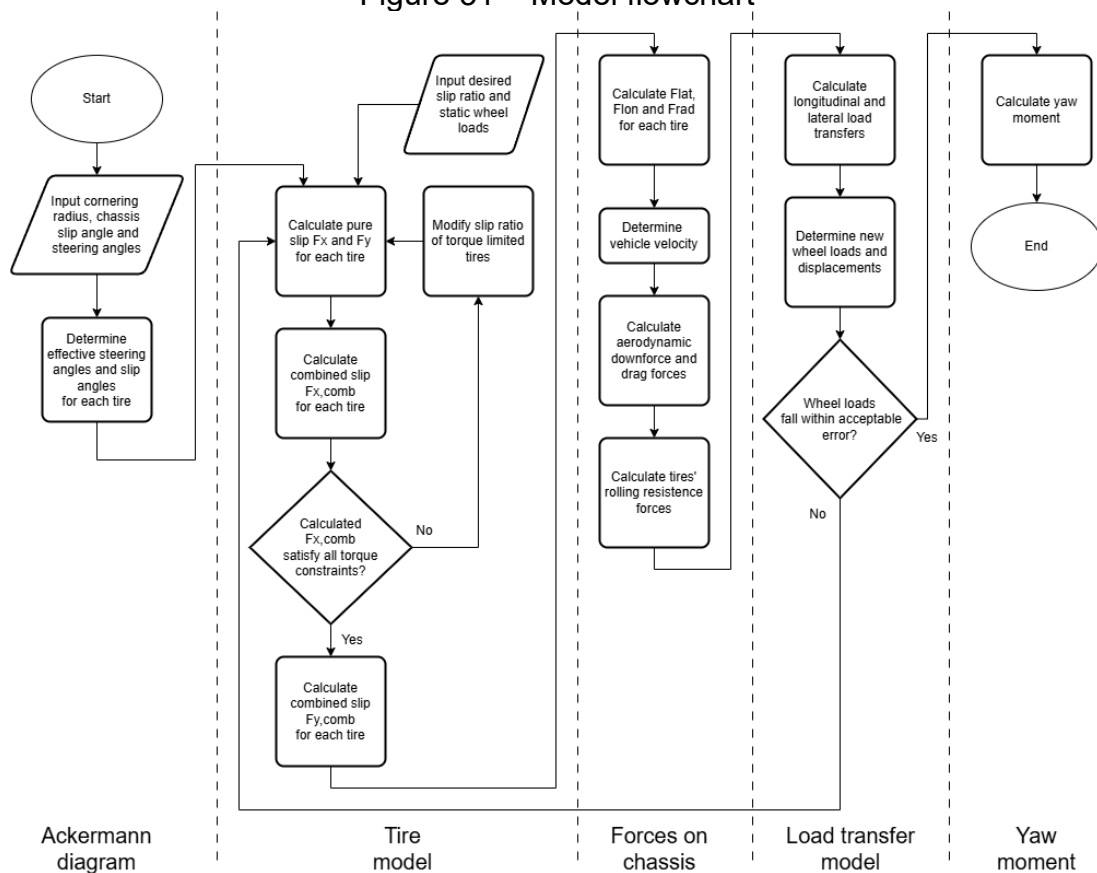
$$K_{c,r} = k_{c,r} t_r^2 \quad (54)$$

3 DEVELOPMENT

This section presents the mathematical development of a quasi-steady state four-wheel model for vehicle dynamics simulation, which uses a Formula SAE prototype as basis, addressing driver input, tire behavior and load transfer, as well as an insight on its computational implementation in C++. The choice for this programming language is based on its superior speed and performance, when compared to others. The model receives input vehicle data, such as the vehicle's size and masses, tire parameters, aerodynamic influences and suspension stiffnesses and geometry, to predict vehicle behavior under certain conditions.

To facilitate the understanding of the model's structure and sequence, a flowchart has been developed, highlighting each major step in the routine, as presented in Figure 31.

Figure 31 – Model flowchart



Source: the author.

The model receives the vehicle's geometric parameters and uses an Ackermann diagram model to determine the slip angle of each wheel. It then applies a root finding algorithm to define each wheel's slip ratios by satisfying physical torque constraints, emulating the driver's control over wheel slip. The tire model uses these parameters to calculate the combined friction forces generated by each tire, which are reoriented to the vehicle coordinate system. Next, the forces imposed on the chassis are inserted in a load transfer model, which determines new vertical loads and vertical displacement of the suspension mechanism to each wheel, both caused by lateral and longitudinal accelerations. These loads are checked against a tolerance error and, if they do not satisfy the ending condition, all calculated information is fed back to the tire model, to reiterate the whole process. Finally, after achieving iterative convergence, the yaw moment about the vehicle's center of gravity may be determined, ending the procedure.

3.1 ACKERMANN DIAGRAM

As mentioned, the proposed model follows a four-wheel approach, which raises the need for a complete Ackermann diagram analysis, as presented in Figure 32.

$$\delta_{ro} = f(w_{ro})$$

The actual cornering radius R_a , perpendicular to the vehicle's midplane (VIEIRA et. al, 2012), can be determined, followed by the longitudinal distances from the cornering center C to each axle S_f and S_r , as given in Equations 56 – 68.

$$R_a = R \cos(\beta) \quad (56)$$

$$S_f = a + R \sin(\beta) \quad (57)$$

$$S_r = L - S_f \quad (58)$$

As for each wheel's effective steering angle θ_{ab} , perpendicular to the corner center radius for said wheel, are given by Equations 59.

$$\begin{aligned} \theta_{fi} &= \text{atan} \left(\frac{S_f}{R_a - \frac{t_f}{2}} \right) \\ \theta_{fo} &= \text{atan} \left(\frac{S_f}{R_a + \frac{t_f}{2}} \right) \\ \theta_{ri} &= \text{atan} \left(\frac{S_r}{R_a - \frac{t_r}{2}} \right) \\ \theta_{ro} &= \text{atan} \left(\frac{S_r}{R_a + \frac{t_r}{2}} \right) \end{aligned} \quad (59)$$

The next step is to define the slip angle α_{ab} for each wheel as the difference between its steering angle δ_{ab} and effective steering angle θ_{ab} for the front wheels and as their sum for the rear wheels, as follows in Equations 60:

$$\begin{aligned} \alpha_{fi} &= \delta_{fi} - \theta_{fi} \\ \alpha_{fo} &= \delta_{fo} - \theta_{fo} \\ \alpha_{ri} &= \delta_{ri} + \theta_{ri} \\ \alpha_{ro} &= \delta_{ro} + \theta_{ro} \end{aligned} \quad (60)$$

Additionally, the cornering radius for each wheel can be defined as means to determine each tire's individual longitudinal velocity, which will later be used calculate the tire rolling resistance forces, as given in Equations 61.

$$\begin{aligned}
R_{fi} &= \sqrt{S_f^2 + \left(R_a - \frac{t_f}{2}\right)^2} \\
R_{fo} &= \sqrt{S_f^2 + \left(R_a + \frac{t_f}{2}\right)^2} \\
R_{ri} &= \sqrt{S_r^2 + \left(R_a - \frac{t_r}{2}\right)^2} \\
R_{ro} &= \sqrt{S_r^2 + \left(R_a + \frac{t_r}{2}\right)^2}
\end{aligned} \tag{61}$$

3.2 TIRE MODELLING

In the presented model, each tire is subjected to different conditions, which raises the need for a tire model that covers all steady-state non-linearities of tire behavior. Additionally, it must be capable of representing generation of both lateral and longitudinal loads simultaneously, while satisfying all constraints imposed by the physical nature of the simulated vehicle. For that matter, the chosen tire model is a combination of the *Magic Formula* (PACEJKA, 2012) for pure slip conditions and the MNC (Modified Nicolas-Comstock) model for combined slip (BRACH, 2011).

3.2.1 Pure slip

The *Magic Formula Tire Model* is used to calculate a tire's generation of longitudinal forces $F_{x,ab}$ and lateral forces $F_{y,ab}$, both emulating conditions with slip in one direction only, the former considering only a longitudinal slip ratio κ_{ab} and the latter only lateral slip angle α_{ab} . Both forces are aligned with the tire's plane.

The Equations for the *Magic Formula* are given in section 2. For a given tire, its camber angle γ_{ab} is equal to the static camber angle of its axle, $\gamma_{f,static}$ for the front tires and $\gamma_{r,static}$ for the rear tires added to the influences of steering and wheel displacement, function of both the steering and suspension geometries, as given by Equations 62.

$$\begin{aligned}
\gamma_{fi} &= \gamma_{f,static} + f(w_{fi}) + f(\delta_{fi}) \\
\gamma_{fo} &= \gamma_{f,static} + f(w_{fo}) + f(\delta_{fo}) \\
\gamma_{ri} &= \gamma_{r,static} + f(w_{ri}) \\
\gamma_{ro} &= \gamma_{r,static} + f(w_{ro})
\end{aligned} \tag{62}$$

The radial load F_z acting on each tire is initially set equal to the static wheel load W_{ab} , due to the vehicle's weight W lateral and longitudinal load distribution, as presented in Equations 63.

$$\begin{aligned}
F_{z,fi} &= W_{fi} = W \ x \ y \\
F_{z,fo} &= W_{fo} = W \ x \ (1 - y) \\
F_{z,ri} &= W_{ri} = W \ (1 - x) \ y \\
F_{z,ro} &= W_{ro} = W \ (1 - x) \ (1 - y)
\end{aligned} \tag{63}$$

Being:

$$W = m \ g \tag{64}$$

Where m is the total mass of the vehicle, g is the vertical acceleration due to gravity, x is the percentage of weight on the front axle and y is the percentage of weight on the left side of the vehicle for left hand corners and on its right side for right hand corners.

The slip angles α_{ab} are used as defined in the Ackermann diagram analysis, and the four slip ratios κ_{ab} will be defined by the torque constraints later explained.

3.2.2 Combined slip

To account for non-linearities caused by the combination of simultaneous lateral and longitudinal slips, the MNC tire model is used. It receives the pure slip forces $F_{x,ab}$ and $F_{y,ab}$ obtained with the *Magic Formula*, the slip ratios κ_{ab} and slip angles α_{ab} to calculate the combined longitudinal and lateral forces $F_{x,comb,ab}$ and $F_{y,comb,ab}$, respectively. These forces are also aligned with the tire's plane. Equations for the MNC model are given in section 2.

3.2.3 Torque constraints

The Formula SAE prototype vehicle used as basis for the proposed model possesses inherent properties which impose constraints to the torque generated by each tire both under braking and traction conditions. This section presents the efforts to model those constraints.

The vehicle is set with four-wheel braking and is equipped with independent brake lines for the front and rear axles, which both grants the possibility for a brake bias adjustment, being it, the percentage of the braking torques generated by the front tires relative to the total braking torques and determines that braking torques generated by tires on the same axle must be equal. The prototype also presents a rear-wheel-drive configuration with an open differential, which means the engine torque under acceleration is sent only to the rear tires and is split equally between both wheels.

These considerations create three constraints for the torques on each tire and, ultimately, for the combined longitudinal tire forces $F_{x,comb,ab}$ under any condition. The three constraints are given by Equations 65 – 67:

$$T_{fi} = T_{fo} \quad (65)$$

$$T_{ri} = T_{ro} \quad (66)$$

$$\frac{T_{fi} + T_{fo}}{T_{fi} + T_{fo} + T_{ri} + T_{ro}} = bias \quad (67)$$

Where T_{ab} is the torque generated by a given tire, which is defined as the product of its deflected radius r_{ab} and its combined longitudinal force $F_{x,comb,ab}$, as shown in Equations 68.

$$\begin{aligned} T_{fi} &= r_{fi} F_{x,comb,fi} \\ T_{fo} &= r_{fo} F_{x,comb,fo} \\ T_{ri} &= r_{ri} F_{x,comb,ri} \\ T_{ro} &= r_{ro} F_{x,comb,ro} \end{aligned} \quad (68)$$

And *bias* is defined as the brake bias setup under braking conditions, usually set between 60% and 70%, and as 0% under traction conditions, due to the rear-wheel-drive configuration. It is important to note that the model can simulate vehicles with any brake and drive configurations, which is done by modifying the input value of *bias* for either condition.

These constraints create a problem that can be interpreted as an irreversible and non-linear function having its result known, but not its input variable, such as $F_{x,comb,ab} = f(\kappa_{ab})$, where $F_{x,comb,ab}$ is known by the restrictions described above, but the slip ratio κ_{ab} that satisfies them is unknown, which has to be determined to later be used to compute the combined lateral tire forces $F_{y,comb,ab}$.

To overcome this problem, a Brent's method (VAN LITH, *et al.*, 2018) root finding algorithm has been implemented to find a suitable slip ratio κ_{ab} for each tire which satisfies all constraints cited above. The algorithm receives a target for a given tire's combined longitudinal force $F_{x,comb,ab,tar}$ and modifies its κ_{ab} . To define the targets, one of the tires must be set as the main tire, which will always be subjected to a desired slip ratio κ_{des} , emulating the driver's effort to control slip on that wheel, as the other three will have their slip ratios (and thus their torques) limited by it.

The main tire will always be on one of the inner wheels, as they always limit the torque of the outer ones due to lateral load transfer effects, and will be set by the axle which generates the most torque, being the front axle under braking or the rear axle under traction conditions, due to longitudinal load transfer effects. This selection is made through the value of *bias*, which being set greater than 50% sets the front inner tire as main tire, opposed to being set less than or equal to 50%, defining the rear inner tire as the main tire. This allows each tire's $F_{x,comb,ab,tar}$ to be set for each case as follows:

Equations 69 – 72 give the targets for *bias* > 50%:

$$\kappa_{fi} = \kappa_{des} \quad (69)$$

$$F_{x,comb,fo,tar} = \frac{T_{fi}}{r_{fo}} \quad (70)$$

$$F_{x,comb,ri,tar} = \frac{1 - bias}{bias} \frac{T_{fi} + T_{fo}}{2 r_{ri}} \quad (71)$$

$$F_{x,comb,ro,tar} = \frac{1 - bias}{bias} \frac{T_{fi} + T_{fo}}{2 r_{ro}} \quad (72)$$

Applying the Brent's method algorithm to the front outer tire before defining the rear tires' targets.

Equations 73 – 76 give the targets for $bias \leq 50\%$:

$$\kappa_{ri} = \kappa_{des} \quad (73)$$

$$F_{x,comb,ro,tar} = \frac{T_{ri}}{r_{ro}} \quad (74)$$

$$F_{x,comb,fi,tar} = \frac{bias}{1 - bias} \frac{T_{ri} + T_{ro}}{2 r_{fi}} \quad (75)$$

$$F_{x,comb,fo,tar} = \frac{bias}{1 - bias} \frac{T_{ri} + T_{ro}}{2 r_{fo}} \quad (76)$$

Similarly, applying the root finding algorithm to the rear outer tire before defining the front tires' targets.

Following these steps, each tire's $F_{x,comb,ab}$ should satisfy the torque constraints given, and each κ_{ab} is known, what allows for the computation of the four $F_{y,comb,ab}$ in sequence.

3.3 FORCES ACTING ON THE CHASSIS

To account for effects of load transfer and to later calculate the vehicle's yaw moment, first all the forces acting on the chassis of the vehicle must be determined.

3.3.1 Lateral and longitudinal forces

The forces that will later be used in load transfer calculations must align with the chassis coordinate system. To differ the nomenclature from the variables obtained from the tire model, loads on the Y axis will be named lateral forces $F_{lat,ab}$ and loads on the X axis are longitudinal forces $F_{lon,ab}$.

Lateral forces are the sum of the ones obtained from the tire models and the tire rolling resistance forces acting laterally to the chassis due to the steering angles, as given in Equations 77.

$$\begin{aligned}
F_{lat,fi} &= F_{y,comb,fi} \cos(\delta_{fi}) + (lon_{sign} F_{x,comb,fi} - F_{rr,fi}) \sin(\delta_{fi}) \\
F_{lat,fo} &= F_{y,comb,fo} \cos(\delta_{fo}) + (lon_{sign} F_{x,comb,fo} - F_{rr,fo}) \sin(\delta_{fo}) \\
F_{lat,ri} &= F_{y,comb,ri} \cos(\delta_{ri}) - (lon_{sign} F_{x,comb,ri} - F_{rr,ri}) \sin(\delta_{ri}) \\
F_{lat,ro} &= F_{y,comb,ro} \cos(\delta_{ro}) - (lon_{sign} F_{x,comb,ro} - F_{rr,ro}) \sin(\delta_{ro})
\end{aligned} \tag{77}$$

The same analysis can be made to determine the longitudinal forces of each tire, as presented by Equations 78.

$$\begin{aligned}
F_{lon,fi} &= -F_{y,comb,fi} \sin(\delta_{fi}) + (lon_{sign} F_{x,comb,fi} - F_{rr,fi}) \cos(\delta_{fi}) \\
F_{lon,fo} &= -F_{y,comb,fo} \sin(\delta_{fo}) + (lon_{sign} F_{x,comb,fo} - F_{rr,fo}) \cos(\delta_{fo}) \\
F_{lon,ri} &= F_{y,comb,ri} \sin(\delta_{ri}) + (lon_{sign} F_{x,comb,ri} - F_{rr,ri}) \cos(\delta_{ri}) \\
F_{lon,ro} &= F_{y,comb,ro} \sin(\delta_{ro}) + (lon_{sign} F_{x,comb,ro} - F_{rr,ro}) \cos(\delta_{ro})
\end{aligned} \tag{78}$$

Where $F_{rr,ab}$ is the rolling resistance force for each tire and lon_{sign} is a parameter that assumes value 1 for traction conditions as -1 for braking.

Additionally, the vehicle's accelerations on each direction can be calculated to give a better understanding of what conditions the car is subjected to. They are determined by Equations 79 – 82.

$$F_{lat} = F_{lat,fi} + F_{lat,fo} + F_{lat,ri} + F_{lat,ro} \tag{79}$$

$$F_{lon} = F_{lon,fi} + F_{lon,fo} + F_{lon,ri} + F_{lon,ro} - F_{drag} \tag{80}$$

$$a_{lat} = \frac{F_{lat}}{W} \tag{81}$$

$$a_{lon} = \frac{F_{lon}}{W} \tag{82}$$

Where a_{lat} and a_{lon} are the lateral and longitudinal accelerations of the vehicle, respectively, both in g's, and F_{drag} is the aerodynamic drag force.

3.3.2 Vehicle velocity

Some of the forces needed in the model are function the vehicle velocity, such as each tire's $F_{rr,ab}$ and the F_{drag} cited above and the aerodynamic downforce F_{down} . To determine the velocity V , it is considered that the vehicle's mass follows a circular trajectory of cornering radius R maintained by each tire's centripetal forces $F_{rad,ab}$, which will be called radial forces and are determined via the effective steering angles θ_{ab} , such as shown in Equations 83.

$$\begin{aligned}
 F_{rad,fi} &= F_{lat,fi} \cos(\theta_{fi}) - F_{lon,fi} \sin(\theta_{fi}) \\
 F_{rad,fo} &= F_{lat,fo} \cos(\theta_{fo}) - F_{lon,fo} \sin(\theta_{fo}) \\
 F_{rad,ri} &= F_{lat,ri} \cos(\theta_{ri}) - F_{lon,ri} \sin(\theta_{ri}) \\
 F_{rad,ro} &= F_{lat,ro} \cos(\theta_{ro}) - F_{lon,ro} \sin(\theta_{ro})
 \end{aligned} \tag{83}$$

Being:

$$F_{rad} = F_{rad,fi} + F_{rad,fo} + F_{rad,ri} + F_{rad,ro} - F_{drag} \sin(\beta) \tag{84}$$

$$a_{rad} = \frac{F_{rad}}{W} \tag{85}$$

Where F_{rad} is the total radial force and a_{rad} is the radial acceleration the vehicle is subjected to towards the cornering center in g's. With this value, the maximum velocity the vehicle can carry through a corner with radius R , called V_{skid} , can be determined as in Equation 86.

$$V_{skid} = \sqrt{a_{rad} g R} \tag{86}$$

The actual vehicle velocity V is defined as the minimum between V_{skid} and an user input velocity V_{input} . This is a safety measure to the implementation, as a straight-line simulation results in an infinite cornering radius, which causes V_{skid} to tend to infinity.

The aerodynamic drag and tire rolling resistance forces depend on the vehicle's velocity, which in turn depends on their value. For this reason, those forces are set to zero in the first iteration, and are later calculated with the value of V as described as follows.

3.3.3 Aerodynamic forces

The forces imposed by the aerodynamic devices on the vehicle's body can be considered functions of many variables, such as the velocity, pitch and roll angles and the platform height of the vehicle. For simplicity, in this model both the aerodynamic forces intensities and their pressure center positions will vary only with the value of V , as in Equations 87 – 89.

$$F_{down} = f(V) \quad (87)$$

$$F_{down,x} = f(V) \quad (88)$$

$$F_{down,y} = 50\% \quad (89)$$

Where F_{down} is the total amount of downforce generated by the aerodynamic package, $F_{down,x}$ the longitudinal distribution of those forces and $F_{down,y}$ the lateral distribution, being set as 50% as the asymmetries related to the roll angle and other effects are disconsidered. With these values, the downforce applied to each wheel may be computed, as given by Equations 90.

$$\begin{aligned} F_{down,fi} &= F_{down} F_{down,x} F_{down,y} \\ F_{down,fo} &= F_{down} F_{down,x} (1 - F_{down,y}) \\ F_{down,fi} &= F_{down} (1 - F_{down,x}) F_{down,y} \\ F_{down,fi} &= F_{down} (1 - F_{down,x}) (1 - F_{down,y}) \end{aligned} \quad (90)$$

Similarly, the drag force F_{drag} and its pressure center height $F_{drag,z}$ are defined as functions of V , as its lateral distribution $F_{drag,y}$ is similarly set to 50%, as follows in Equations 91 – 93:

$$F_{drag} = f(V) \quad (91)$$

$$F_{drag,z} = f(V) \quad (92)$$

$$F_{drag,y} = 50\% \quad (93)$$

3.3.4 Tire rolling resistance forces

The model presented by NICOLAZZI (2023) for tire rolling resistances gives forces which are function of the wheel's longitudinal velocity $V_{x,ab}$, instead of the slip ratio, as described in the tire model section. $V_{x,ab}$ can be determined for each wheel with the chassis velocity V , the ratio between the tire cornering radius and the chassis cornering radius and the slip angle α_{ab} , as shown in Equations 94.

$$\begin{aligned}
 V_{x,fi} &= V \frac{R_{fi}}{R} \cos(\alpha_{fi}) \\
 V_{x,fo} &= V \frac{R_{fo}}{R} \cos(\alpha_{fo}) \\
 V_{x,ri} &= V \frac{R_{ri}}{R} \cos(\alpha_{ri}) \\
 V_{x,ro} &= V \frac{R_{ro}}{R} \cos(\alpha_{ro})
 \end{aligned} \tag{94}$$

As so, the rolling resistance force for each tire can be determined using the Equations given in section 2.

3.4 LOAD TRANSFER

As the vehicle is subjected to lateral and longitudinal accelerations, the vertical loads imposed on each wheel will change by a mechanism called load transfer, as explained in section 2. This phenomenon will also cause roll and pitch angles on the chassis, which displaces each wheel differently, causing the suspension geometry to become asymmetrical both longitudinally and laterally, even if its design considers symmetry under static conditions, which in turn alters completely the behavior of the suspension mechanism and of the load transfer phenomenon itself.

For this matter, a load transfer model is developed, based on the NICOLAZZI (2023) model presented in section 2. Its distinction being the disconsideration for any simplifications based on lateral and/or longitudinal symmetry of the vehicle, also accounting for tire and chassis deflection and aerodynamic influence on load transfer. It is a generalized model which may be used in either direction as,

physically, both phenomena are similar in nature, but having their own peculiarities, which are explored below.

This section presents the whole mathematical deduction for the model, following NICOLAZZI's as reference.

3.4.1 Lateral load transfer

For an easier understanding of the model, the lateral load transfer is initially presented, as it is relatively simpler.

The first consideration is that the total load transfer on each wheel is the sum of contributions of geometrical load transfer $\Delta W_{lat,g,ab}$, divided into sprung masses load transfer $\Delta W_{lat,s,ab}$ and unsprung masses load transfer $\Delta W_{lat,u,ab}$, and elastic load transfer $\Delta W_{lat,k,ab}$, which agglutinates NICOLAZZI's springs and anti-roll bars portions.

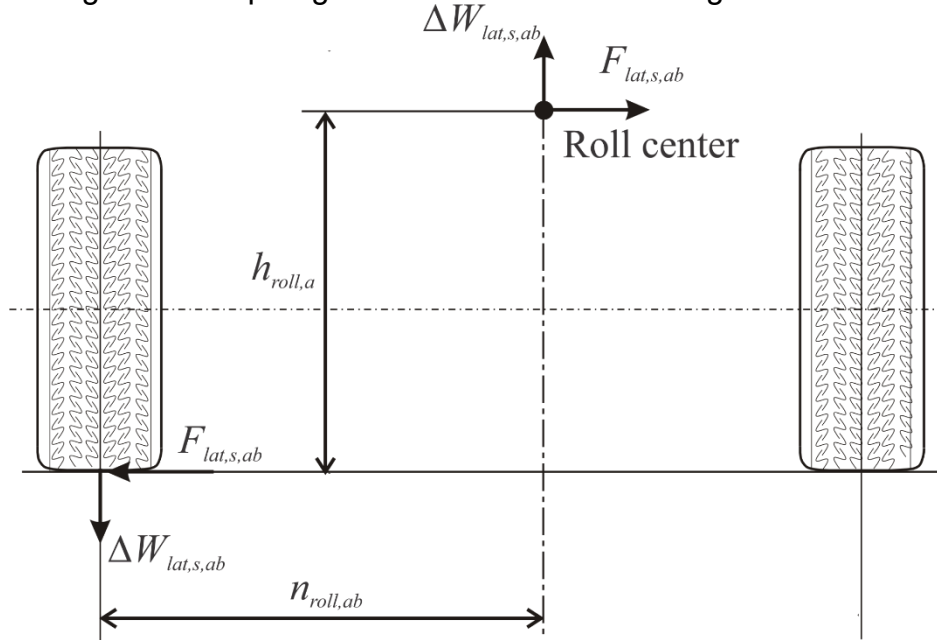
The second is that each axle and each wheel may have different proportions to each of the three contributions due to different stiffnesses and anti-geometries, but when analyzing the vehicle as a whole, the total load transfer must satisfy Equation 35. It is important to note that this Equation is a simplification for when the front and rear tracks are equal in length, for when they differ, the affirmation above loses its veracity.

The third is that both axles are subjected to the same roll angle Ψ , even when considering that the chassis twists differently on each axle.

3.4.1.1 *Sprung masses load transfer*

The first portion of load transfer is caused by the inertia of the sprung masses under lateral acceleration.

Figure 33 – Sprung masses load transfer of a generic wheel



Source: adapted from NICOLAZZI, 2023.

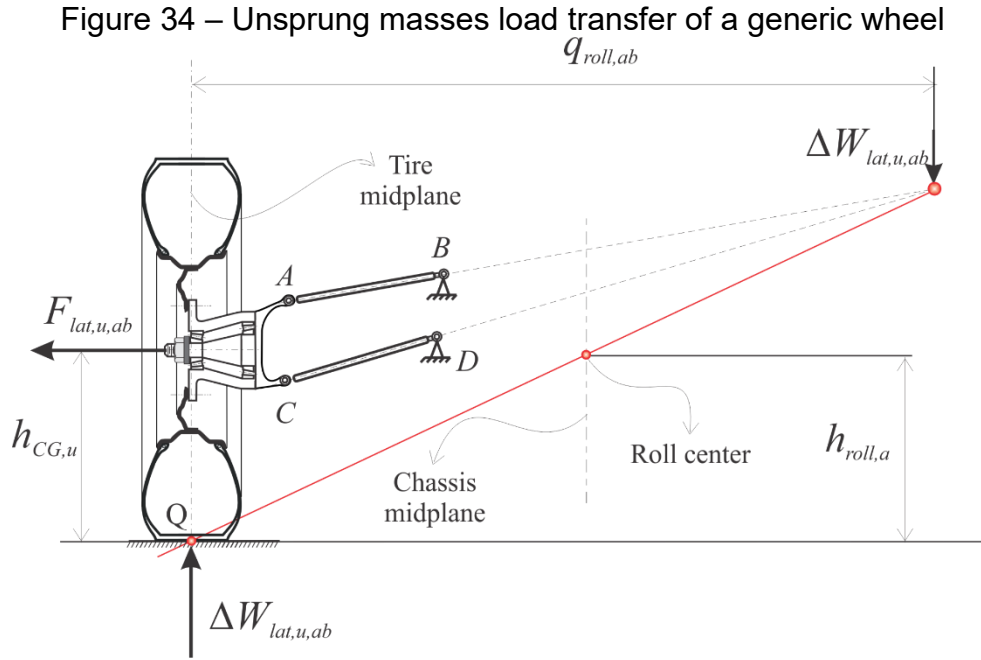
Figure 33 shows that this contribution to load transfer is caused by an equilibrium of moments, which occurs due to the tire lateral force multiplied by the ratio of sprung and total masses on the vehicle, similar to its counterpart from Nicolazzi's model, and is given to each wheel by the Equations 95.

$$\begin{aligned}
 \Delta W_{lat,s,fi} &= \frac{m_s}{m} \frac{F_{lat,fi} h_{roll,f}}{n_{roll,fi}} \\
 \Delta W_{lat,s,fo} &= \frac{m_s}{m} \frac{F_{lat,fo} h_{roll,f}}{n_{roll,fo}} \\
 \Delta W_{lat,s,ri} &= \frac{m_s}{m} \frac{F_{lat,ri} h_{roll,r}}{n_{roll,ri}} \\
 \Delta W_{lat,s,ro} &= \frac{m_s}{m} \frac{F_{lat,ro} h_{roll,r}}{n_{roll,ro}}
 \end{aligned} \tag{95}$$

Where m_s are the vehicle's sprung masses, $h_{roll,f}$ and $h_{roll,r}$ are the front and rear axle's roll center heights, respectively and $n_{roll,ab}$ are the lateral distances from each wheel's medium plane to the roll center.

3.4.1.2 Unsprung masses load transfer

Similarly, the second portion follows Nicolazzi's model as being the result of an equilibrium of moments caused by the inertia of the unsprung masses under lateral acceleration. Instead of acting on the axle's the roll center, this contribution to load transfer occurs around the wheel's instantaneous center, as presented in Figure 34.



Source: adapted from NICOLAZZI, 2023.

Each wheel's unsprung load transfer portion is given by the following Equations 96.

$$\begin{aligned}\Delta W_{lat,u,fi} &= \frac{m_u}{m} \frac{F_{lat,fi} h_{CG,u}}{q_{roll,fi}} \\ \Delta W_{lat,u,fo} &= \frac{m_u}{m} \frac{F_{lat,fo} h_{CG,u}}{q_{roll,fo}} \\ \Delta W_{lat,u,ri} &= \frac{m_u}{m} \frac{F_{lat,ri} h_{CG,u}}{q_{roll,ri}} \\ \Delta W_{lat,u,ro} &= \frac{m_u}{m} \frac{F_{lat,ro} h_{CG,u}}{q_{roll,ro}}\end{aligned}\tag{96}$$

Where m_u is the sum of all unsprung masses, $h_{CG,u}$ is the average height of the unsprung masses center of gravity and $q_{roll,ab}$ are the lateral distances from each

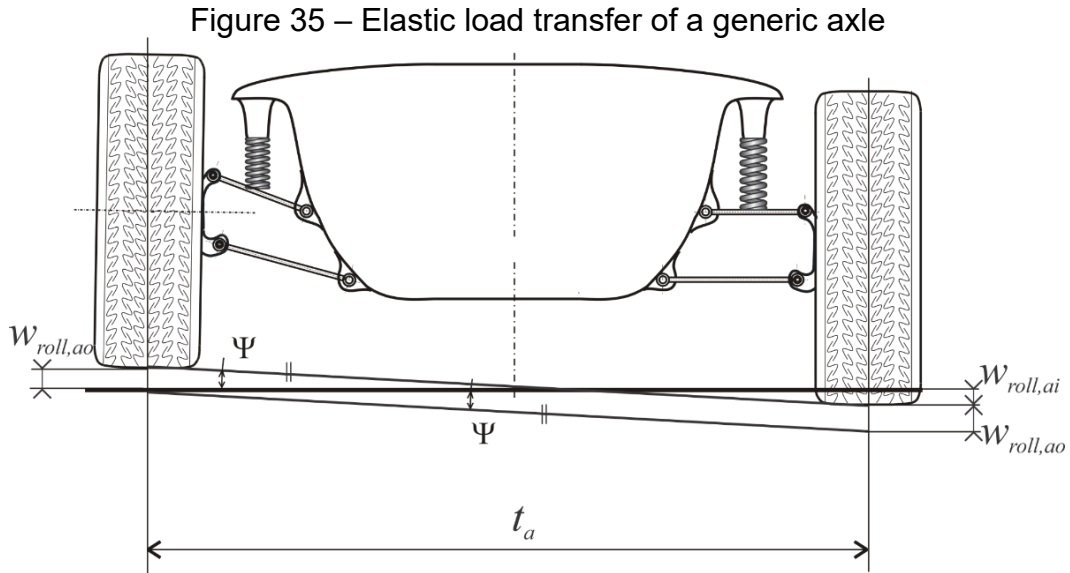
wheel's midplane to its roll instantaneous center. The value of $h_{CG,u}$ is obtained by the Equation 97.

$$h_{CG,u} = \frac{h_{CG,u,fi} m_{u,fi} + h_{CG,u,fo} m_{u,fo} + h_{CG,u,ri} m_{u,ri} + h_{CG,u,ro} m_{u,ro}}{m_u} \quad (97)$$

This comes from the consideration that each wheel has its own unsprung masses value and unsprung masses center of gravity, which height is considered to be equal as its tire's deflected radius, being located at the wheel's center, for simplification.

3.4.1.3 Elastic load transfer

The last portion of load transfer is the one associated with the deflection of suspension springs, ARBs, chassis and tires. As the chassis roll, each spring is subjected to a load variation, which causes them to deflect proportionally to their stiffnesses. Analyzing each axle separately, it is possible to equate its inner and outer wheel displacements as function of Ψ , as shown in Figure 35.



Source: adapted from NICOLAZZI, 2023.

Where:

$$\tan(\Psi) t_a = w_{roll,ai} + w_{roll,ao} \quad (98)$$

When linearizing $\tan(\Psi) = \Psi$ for small roll angles gives Equation 99.

$$\Psi t_a = w_{roll,ai} + w_{roll,ao} \quad (99)$$

And considering each wheel's displacement as the sum of deflections of the suspension springs, tires and chassis, given by Equations 100 and 101:

$$w_{roll,ai} = w_{roll,k,ai} + w_{roll,t,ai} + w_{roll,c,ai} \quad (100)$$

$$w_{roll,ao} = w_{roll,k,ao} + w_{roll,t,ao} + w_{roll,c,ao} \quad (101)$$

Gives Equation 102:

$$\Psi t_a = w_{roll,k,ai} + w_{roll,t,ai} + w_{roll,c,ai} + w_{roll,k,ao} + w_{roll,t,ao} + w_{roll,c,ao} \quad (102)$$

Where $w_{roll,k,ai}$ and $w_{roll,k,ao}$ are the wheel displacements caused by springs and ARBs deflections of the inner and front outer wheels, respectively, $w_{roll,t,ai}$ and $w_{roll,t,ao}$ are the tires' deflections and $w_{roll,c,ai}$ and $w_{roll,c,ao}$ are deflections caused by the torsion of the chassis. These displacements are equal to the loads each spring is subjected to divided by the stiffness associated with it, as follows in Equations 103 – 107:

$$w_{roll,k,ai} = \frac{\Delta W_{lat,k,ai}}{K_{susp,ai} + K_{arb,ai}} \quad (103)$$

$$w_{roll,k,ao} = \frac{\Delta W_{lat,k,ao}}{K_{susp,ao} + K_{arb,ao}} \quad (104)$$

$$w_{roll,t,ai} = \frac{\Delta W_{lat,a}}{K_{t,ai}} \quad (105)$$

$$w_{roll,t,ao} = \frac{\Delta W_{lat,a}}{K_{t,ao}} \quad (106)$$

$$w_{roll,c,ai} = w_{roll,c,ao} = \frac{\Delta W_{lat,a}}{K_{c,a}} \quad (107)$$

Where the suspension springs are subjected only to the elastic portion of lateral load transfer on each wheel $\Delta W_{lat,k,ai}$ and $\Delta W_{lat,k,ao}$, and both the tires and the chassis absorb the axle's lateral load transfer $\Delta W_{lat,a}$ entirely. Additionally, $K_{susp,ai}$ and $K_{susp,ao}$ are the wheel rates associated with the suspension springs, $K_{arb,ai}$ and $K_{arb,ao}$ are the wheel rates of the axle's ARB on each wheel, $K_{t,ai}$ and $K_{t,ao}$ are the

tires' radial stiffnesses and $K_{c,a}$ is the wheel rate of the chassis torsional stiffness to the axle, which is assumed to be equal to the inner and outer wheels. Substituting these values into Equation 99 gives Equation 108:

$$\Psi t_a = \frac{\Delta W_{lat,k,ai}}{K_{susp,ai} + K_{arb,ai}} + \frac{\Delta W_{lat,k,ao}}{K_{susp,ao} + K_{arb,ao}} + \frac{\Delta W_{lat,a}}{K_{t,ai}} + \frac{\Delta W_{lat,a}}{K_{t,ao}} + \frac{2 \Delta W_{lat,a}}{K_{c,a}} \quad (108)$$

But the total load transfer on each axle is equal to the sum of elastic and geometric load transfer for each wheel, as given in Equation 109.

$$\Delta W_{lat,a} = \Delta W_{lat,k,ai} + \Delta W_{lat,g,ai} = \Delta W_{lat,k,ao} + \Delta W_{lat,g,ao} \quad (109)$$

Where:

$$\Delta W_{lat,g,ai} = \Delta W_{lat,s,ai} + \Delta W_{lat,u,ai} \quad (110)$$

$$\Delta W_{lat,g,ao} = \Delta W_{lat,s,ao} + \Delta W_{lat,u,ao} \quad (111)$$

Manipulating and substituting into Equation 108 gives Equation 112.

$$\Psi t_a = \Delta W_{lat,a} \left(\frac{1}{K_{susp,ai} + K_{arb,ai}} + \frac{1}{K_{susp,ao} + K_{arb,ao}} + \frac{1}{K_{t,ai}} + \frac{1}{K_{t,ao}} + \frac{2}{K_{c,a}} \right) - \frac{\Delta W_{lat,g,ai}}{K_{susp,ai} + K_{arb,ai}} - \frac{\Delta W_{lat,g,ao}}{K_{susp,ao} + K_{arb,ao}} \quad (112)$$

Then isolating $\Delta W_{lat,a}$ and substituting the term in parenthesis for an equivalent total roll stiffness for the axle $K_{roll,tot,a}$ gives Equation 113.

$$\Delta W_{lat,a} = \Psi t_a K_{roll,tot,a} + \left(\frac{\Delta W_{lat,g,ai}}{K_{susp,ai} + K_{arb,ai}} + \frac{\Delta W_{lat,g,ao}}{K_{susp,ao} + K_{arb,ao}} \right) K_{roll,tot,a} \quad (113)$$

To simplify the Equation, the term in parenthesis will be substituted by $w_{roll,n,a}$, which can be interpreted as the displacement negated to the suspension springs and ARB due to anti-roll geometry. This gives Equation 114.

$$\Delta W_{lat,a} = \Psi t_a K_{roll,tot,a} + w_{roll,n,a} K_{roll,tot,a} \quad (114)$$

This Equation represents a hyperstatic system for each axle, which will be solved by determining the roll angle Ψ , as shown below.

3.4.1.4 Roll angle

To solve the hyperstatic system problem described above a compatibility Equation is defined as the equilibrium of moments about the roll axis, as it is done in Nicolazzi's model, presented in Equation 115.

$$M_{roll,k} = M_{roll,s} + M_{roll,u} \quad (115)$$

Where $M_{roll,s}$ and $M_{roll,u}$ are the sprung and unsprung masses roll moments caused by their inertia and $M_{roll,k}$ is the elastic reaction roll moment caused by the deflection of springs. Each wheel has its own contribution to each of the three moments, as shown in Equations 116 – 118.

$$M_{roll,k} = M_{roll,k,fi} + M_{roll,k,fo} + M_{roll,k,ri} + M_{roll,k,ro} \quad (116)$$

$$M_{roll,s} = M_{roll,s,fi} + M_{roll,s,fo} + M_{roll,s,ri} + M_{roll,s,ro} \quad (117)$$

$$M_{roll,u} = M_{roll,u,fi} + M_{roll,u,fo} + M_{roll,u,ri} + M_{roll,u,ro} \quad (118)$$

The sprung masses roll moments are caused by inertia forces acting on the sprung masses center of gravity about its axle's roll center. As so, each wheel's sprung masses roll moment is given by Equations 119.

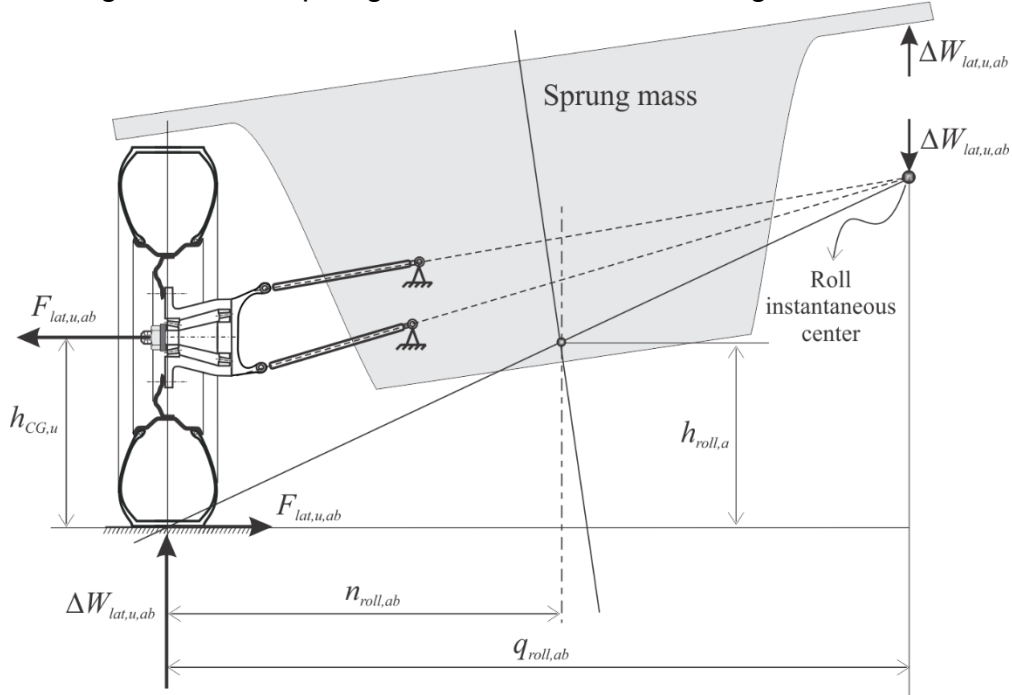
$$\begin{aligned} M_{roll,s,fi} &= \frac{m_s}{m} F_{lat,fi} (h_{CG,s} - h_{roll,f}) \\ M_{roll,s,fo} &= \frac{m_s}{m} F_{lat,fo} (h_{CG,s} - h_{roll,f}) \\ M_{roll,s,ri} &= \frac{m_s}{m} F_{lat,ri} (h_{CG,s} - h_{roll,r}) \\ M_{roll,s,ro} &= \frac{m_s}{m} F_{lat,ro} (h_{CG,s} - h_{roll,r}) \end{aligned} \quad (119)$$

Where $h_{CG,s}$ is the sprung masses center of gravity height, given by the Equation 120.

$$h_{CG,s} = \frac{m h_{CG} - m_u h_{CG,u}}{m_s} \quad (120)$$

The unsprung masses roll moments are caused by the reaction force on the chassis due to the unsprung masses load transfer $\Delta W_{lat,u,ab}$ of each wheel acting on its roll instantaneous center, as shown in Figure 36.

Figure 36 – Unsprung masses roll moment of a generic wheel



Source: adapted from NICOLAZZI, 2023.

As so, each wheel's contribution to the unsprung masses roll moments about the roll axis is given by Equations 121.

$$\begin{aligned}
 M_{roll,u,fi} &= \Delta W_{lat,u,fi} (q_{roll,fi} - n_{roll,fi}) \\
 M_{roll,u,fo} &= \Delta W_{lat,u,fo} (q_{roll,fo} - n_{roll,fo}) \\
 M_{roll,u,ri} &= \Delta W_{lat,u,ri} (q_{roll,ri} - n_{roll,ri}) \\
 M_{roll,u,ro} &= \Delta W_{lat,u,ro} (q_{roll,ro} - n_{roll,ro})
 \end{aligned} \tag{121}$$

The elastic reaction roll moments are caused by each wheel's elastic load transfer $\Delta W_{lat,k,ab}$ acting on its midplane. As so, each wheel's contribution to the elastic reaction roll moments is given by Equations 122.

$$\begin{aligned}
 M_{roll,k,fi} &= \Delta W_{lat,k,fi} n_{roll,fi} \\
 M_{roll,k,fo} &= \Delta W_{lat,k,fo} n_{roll,fo}
 \end{aligned} \tag{122}$$

$$M_{roll,k,ri} = \Delta W_{lat,k,ri} n_{roll,ri}$$

$$M_{roll,k,ro} = \Delta W_{lat,k,ro} n_{roll,ro}$$

Using the definition given in Equation 109 yields Equations 123.

$$\begin{aligned} M_{roll,k,fi} &= (\Delta W_{lat,f} - \Delta W_{lat,g,fi}) n_{roll,fi} \\ M_{roll,k,fo} &= (\Delta W_{lat,f} - \Delta W_{lat,g,fo}) n_{roll,fo} \end{aligned} \quad (123)$$

$$M_{roll,k,ri} = (\Delta W_{lat,r} - \Delta W_{lat,g,ri}) n_{roll,ri}$$

$$M_{roll,k,ro} = (\Delta W_{lat,r} - \Delta W_{lat,g,ro}) n_{roll,ro}$$

Substituting into Equation 115 and manipulating gives Equation 124.

$$\begin{aligned} \Delta W_{lat,f} t_f + \Delta W_{lat,r} t_r &= M_{roll,s} + M_{roll,u} + \Delta W_{lat,g,fi} n_{roll,fi} + \Delta W_{lat,g,fo} n_{roll,fo} \\ &\quad + \Delta W_{lat,g,ri} n_{roll,ri} + \Delta W_{lat,g,ro} n_{roll,ro} \end{aligned} \quad (124)$$

Since:

$$n_{roll,fi} + n_{roll,fo} = t_f \quad (125)$$

$$n_{roll,ri} + n_{roll,ro} = t_r \quad (126)$$

Where each wheel contributes with a $\Delta W_{lat,g,ab} n_{roll,ab}$ term, which can be interpreted as an additional moment $M_{roll,g,ab}$ caused by geometric load transfers as given by Equations 127.

$$M_{roll,g,fi} = \Delta W_{lat,g,fi} n_{roll,fi}$$

$$M_{roll,g,fo} = \Delta W_{lat,g,fo} n_{roll,fo} \quad (127)$$

$$M_{roll,g,ri} = \Delta W_{lat,g,ri} n_{roll,ri}$$

$$M_{roll,g,ro} = \Delta W_{lat,g,ro} n_{roll,ro}$$

As $M_{roll,g}$ is the sum of all four wheels' contributions, as presented in Equation 128:

$$M_{roll,g} = M_{roll,g,fi} + M_{roll,g,fo} + M_{roll,g,ri} + M_{roll,g,ro} \quad (128)$$

Which gives Equation 129.

$$\Delta W_{lat,f} t_f + \Delta W_{lat,r} t_r = M_{roll,s} + M_{roll,u} + M_{roll,g} \quad (129)$$

Substituting the front and rear axle load transfer given by Equation 114, gives Equation 130:

$$\begin{aligned} \Psi t_f^2 K_{roll,tot,f} + w_{roll,n,f} K_{roll,tot,f} t_f + \Psi t_r^2 K_{roll,tot,r} + w_{roll,n,r} K_{roll,tot,r} t_r \\ = M_{roll,s} + M_{roll,u} + M_{roll,g} \end{aligned} \quad (130)$$

Finally, isolating for Ψ gives an expression for the chassis roll angle, described in Equation 131:

$$\Psi = \frac{M_{roll,s} + M_{roll,u} + M_{roll,g} - w_{roll,n,f} K_{roll,tot,f} t_f - w_{roll,n,r} K_{roll,tot,r} t_r}{t_f^2 K_{roll,tot,f} + t_r^2 K_{roll,tot,r}} \quad (131)$$

3.4.1.5 Total load transfer

Inserting the resultant roll angle into Equation 114 gives the total lateral load transfer absorbed by the front and rear axles, $\Delta W_{lat,f}$ and $\Delta W_{lat,r}$ respectively, which in turn can be used to calculate each wheel's elastic load transfer as defined in Equations 132.

$$\begin{aligned} \Delta W_{lat,k,fi} &= \Delta W_{lat,f} - \Delta W_{lat,g,fi} \\ \Delta W_{lat,k,fo} &= \Delta W_{lat,f} - \Delta W_{lat,g,fo} \\ \Delta W_{lat,k,ri} &= \Delta W_{lat,r} - \Delta W_{lat,g,ri} \\ \Delta W_{lat,k,ro} &= \Delta W_{lat,r} - \Delta W_{lat,g,ro} \end{aligned} \quad (132)$$

These values will later be used to determine the displacement of each wheel.

3.4.2 Longitudinal load transfer

As previously explained, lateral and longitudinal load transfer are similar phenomena, differing only by some peculiarities listed below:

1. Longitudinal load transfer is affected by the aerodynamic drag forces;
2. The same analysis made separately for the front and rear axles in lateral load transfer are instead made for the inner and outer sides of the vehicle;
3. The inner and outer wheelbases are considered to be equal in length, as it is unusual to design a vehicle with unequal wheelbases, an affirmation not applicable to the tracks;
4. There is no influence of the anti-roll bars in pitch stiffness;

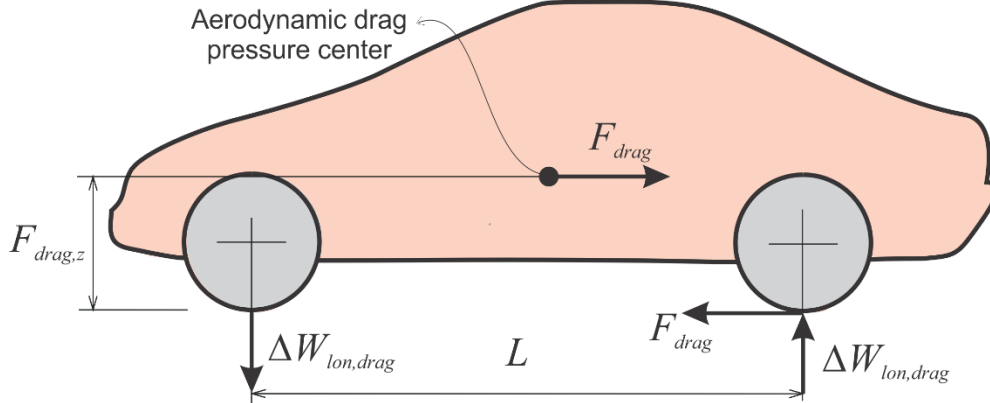
5. The mounting position of brake calipers and differentials alters the phenomenon for geometrical load transfers, whether being inboard or outboard.

Despite these differences, the presented model still must ultimately satisfy the simplest model for longitudinal load transfer, similar to its lateral counterpart, given by Equation 36 (except for load transfers caused by aerodynamic drag).

3.4.2.1 Aerodynamic influences

The aerodynamic drag forces acting on its pressure center cause a disequilibrium of moments, which is reacted by the aerodynamic drag load transfer $\Delta W_{lon,drag}$, as the tractive tires generate longitudinal forces equal in value to the drag force F_{drag} to maintain constant speed, as shown in Figure 37.

Figure 37 – Aerodynamic drag influence on longitudinal load transfer



Source: adapted from NICOLAZZI, 2023.

This portion of load transfer is equated as in Equation 133.

$$\Delta W_{lon,drag} = \frac{F_{drag} F_{drag,z}}{L} \quad (133)$$

Additionally, the drag force causes a moment $M_{pitch,drag}$ about the pitch axis, which interferes with the pitch angle ϕ and, ultimately, with longitudinal elastic load transfers. This moment is described as in Equation 134.

$$M_{pitch,drag} = F_{drag} h_{drag} \quad (134)$$

Where h_{drag} is the perpendicular distance between the drag pressure center and the pitch axis, which is determined by the following Equation 135:

$$h_{drag} = \frac{F_{drag,z} t - h_{pitch,o} t + (h_{pitch,o} - h_{pitch,i})\left(\frac{t}{2} - F_{drag,y}\right)}{\sqrt{(h_{pitch,o} - h_{pitch,i})^2 + t^2}} \quad (135)$$

Where $h_{pitch,i}$ and $h_{pitch,o}$ are the inner and outer pitch center heights, respectively and t is the average between the front and rear tracks.

3.4.2.2 Outboard brakes

As previously explained, the mounting position for the force actuators that cause longitudinal accelerations alters the load transfer phenomenon. As the prototype vehicle used as basis for the model is equipped with a four-wheel outboard mounted braking system, the calculations will follow the Equations below for braking conditions, which are similar to the ones described for lateral load transfer, differing only by the first four statements cited above.

This section presents the adaptation of the lateral load transfer model to a longitudinal one as it differentiates and names the most pertinent of variables used in the Equations.

Similarly to its lateral counterpart, the longitudinal load transfer ΔW_{lon} is the sum of geometric load transfers $\Delta W_{lon,g,ab}$, which are divided into sprung masses load transfers $\Delta W_{lon,s,ab}$ and unsprung masses load transfers $\Delta W_{lon,u,ab}$, and elastic load transfers $\Delta W_{lon,k,ab}$.

The sprung masses load transfer for each wheel is determined as in Equations 136.

$$\begin{aligned} \Delta W_{lon,s,fi} &= \frac{m_s}{m} \frac{F_{lon,fi} h_{pitch,i}}{n_{pitch,fi}} \\ \Delta W_{lon,s,fo} &= \frac{m_s}{m} \frac{F_{lon,fo} h_{pitch,o}}{n_{pitch,fo}} \\ \Delta W_{lon,s,ri} &= \frac{m_s}{m} \frac{F_{lon,ri} h_{pitch,i}}{n_{pitch,ri}} \end{aligned} \quad (136)$$

$$\Delta W_{lon,s,ro} = \frac{m_s}{m} \frac{F_{lon,ro} h_{pitch,o}}{n_{pitch,ro}}$$

Where $n_{pitch,ab}$ are the longitudinal distances of each wheel's centerline to its side's pitch center.

As for unsprung masses load transfer, each wheel's contribution is described as follows in Equations 137:

$$\begin{aligned} \Delta W_{lon,u,fi} &= \frac{m_u}{m} \frac{F_{lon,fi} h_{CG,u}}{q_{pitch,fi}} \\ \Delta W_{lon,u,fo} &= \frac{m_u}{m} \frac{F_{lon,fo} h_{CG,u}}{q_{pitch,fo}} \\ \Delta W_{lon,u,ri} &= \frac{m_u}{m} \frac{F_{lon,ri} h_{CG,u}}{q_{pitch,ri}} \\ \Delta W_{lon,u,ro} &= \frac{m_u}{m} \frac{F_{lon,ro} h_{CG,u}}{q_{pitch,ro}} \end{aligned} \quad (137)$$

Where $q_{pitch,ro}$ is the longitudinal distances of each wheel's centerline to its pitch instantaneous center.

Next, the total of transferred load each side of the vehicle absorbs is equated as in Equation 138.

$$\Delta W_{lon,b} = \Psi L K_{pitch,tot,b} + w_{pitch,n,b} K_{pitch,tot,b} \quad (138)$$

Where $w_{pitch,n,b}$ are the negated displacements for the suspension springs on each side of the vehicle, described in Equation 139.

$$w_{pitch,n,b} = \frac{\Delta W_{lon,g,fb}}{K_{susp,fb}} + \frac{\Delta W_{lat,g,rb}}{K_{susp,rb}} \quad (139)$$

And $K_{pitch,tot,b}$ is the total pitch stiffnesses for the inner and outer sides, given as in Equation 140.

$$K_{pitch,tot,b} = \frac{1}{K_{susp,ai}} + \frac{1}{K_{susp,ao}} + \frac{1}{K_{t,ai}} + \frac{1}{K_{t,ao}} + \frac{2}{K_{c,b}} \quad (140)$$

Being $K_{c,b}$ the wheel rate of the chassis longitudinal torsional stiffness to the inner and outer sides.

The expression for the pitch angle ϕ , which carries the influence of aerodynamic drag as previously described, is given below by Equation 141:

$$\phi = \frac{M_{pitch,drag} + M_{pitch,s} + M_{pitch,u} + M_{pitch,g} - L (w_{pitch,n,f} K_{pitch,tot,f} + w_{pitch,n,r} K_{pitch,tot,r})}{L^2 (K_{roll,tot,f} + K_{roll,tot,r})} \quad (141)$$

Where $M_{pitch,s}$, $M_{pitch,u}$ and $M_{pitch,g}$ are the sprung and unsprung masses and the additional geometric moments, respectively, given by Equations 142 – 144.

$$M_{pitch,s} = M_{pitch,s,fi} + M_{pitch,s,fo} + M_{pitch,s,ri} + M_{pitch,s,ro} \quad (142)$$

$$M_{pitch,u} = M_{pitch,u,fi} + M_{pitch,u,fo} + M_{pitch,u,ri} + M_{pitch,u,ro} \quad (143)$$

$$M_{pitch,g} = M_{pitch,g,fi} + M_{pitch,g,fo} + M_{pitch,g,ri} + M_{pitch,g,ro} \quad (144)$$

And each wheel's contribution to the sprung masses' moment is equated as in Equations 145.

$$\begin{aligned} M_{pitch,s,fi} &= \frac{m_s}{m} F_{lon,fi} (h_{CG,s} - h_{pitch,i}) \\ M_{pitch,s,fo} &= \frac{m_s}{m} F_{lon,fo} (h_{CG,s} - h_{pitch,o}) \\ M_{pitch,s,ri} &= \frac{m_s}{m} F_{lon,ri} (h_{CG,s} - h_{pitch,i}) \\ M_{pitch,s,ro} &= \frac{m_s}{m} F_{lon,ro} (h_{CG,s} - h_{pitch,o}) \end{aligned} \quad (145)$$

As for the unsprung masses moments are given by in Equation 146.

$$\begin{aligned} M_{pitch,u,fi} &= \Delta W_{lon,u,fi} (q_{pitch,fi} - n_{pitch,fi}) \\ M_{pitch,u,fo} &= \Delta W_{lon,u,fo} (q_{pitch,fo} - n_{pitch,fo}) \\ M_{pitch,u,ri} &= \Delta W_{lon,u,ri} (q_{pitch,ri} - n_{pitch,ri}) \\ M_{pitch,u,ro} &= \Delta W_{lon,u,ro} (q_{pitch,ro} - n_{pitch,ro}) \end{aligned} \quad (146)$$

And the additional geometric moments are given by Equations 147.

$$\begin{aligned} M_{pitch,g,fi} &= \Delta W_{lon,g,fi} n_{pitch,fi} \\ M_{pitch,g,fo} &= \Delta W_{lon,g,fo} n_{pitch,fo} \\ M_{pitch,g,ri} &= \Delta W_{lon,g,ri} n_{pitch,ri} \\ M_{pitch,g,ro} &= \Delta W_{lon,g,ro} n_{pitch,ro} \end{aligned} \quad (147)$$

And finally, each wheel's elastic longitudinal load transfer is determined by subtracting its geometrical load transfer from its side's total load transfer, as shown below in Equations 148.

$$\begin{aligned}
 \Delta W_{lon,k,fi} &= \Delta W_{lon,i} - \Delta W_{lon,g,fi} \\
 \Delta W_{lon,k,fo} &= \Delta W_{lon,o} - \Delta W_{lon,g,fo} \\
 \Delta W_{lon,k,ri} &= \Delta W_{lon,i} - \Delta W_{lon,g,ri} \\
 \Delta W_{lon,k,ro} &= \Delta W_{lon,o} - \Delta W_{lon,g,ro}
 \end{aligned} \tag{148}$$

3.4.2.3 Inboard differential

Under traction conditions, the longitudinal load transfer model will suffer a slight modification due to the inboard mounting position of the differential in the vehicle. This causes each wheel's torque to be absorbed on the sprung masses, instead of on the unsprung masses as in braking conditions.

This section presents the inboard variation of the longitudinal load transfer model, which alters the expressions for the sprung and unsprung masses load transfers $\Delta W_{lon,s,ab}$ and $\Delta W_{lon,u,ab}$ as the one for the pitch angle ϕ . These modifications may alter the values of elastic and geometric load transfers $\Delta W_{lon,k,ab}$ and $\Delta W_{lon,g,ab}$, but should not change the total longitudinal load transfer ΔW_{lon} .

For both the sprung and unsprung masses load transfers, each tire's longitudinal force $F_{lon,ab}$ acts on its wheel center point. This causes the sprung masses load transfers to be equated as in Equations 149.

$$\begin{aligned}
 \Delta W_{lon,s,fi} &= \frac{m_s}{m} \frac{F_{lon,fi} h_{pitch,i} - T_{fi}}{n_{pitch,fi}} \\
 \Delta W_{lon,s,fo} &= \frac{m_s}{m} \frac{F_{lon,fo} h_{pitch,o} - T_{fo}}{n_{pitch,fo}} \\
 \Delta W_{lon,s,ri} &= \frac{m_s}{m} \frac{F_{lon,ri} h_{pitch,i} - T_{ri}}{n_{pitch,ri}} \\
 \Delta W_{lon,s,ro} &= \frac{m_s}{m} \frac{F_{lon,ro} h_{pitch,o} - T_{ro}}{n_{pitch,ro}}
 \end{aligned} \tag{149}$$

And the unsprung masses load transfers are as follows in Equations 150:

$$\begin{aligned}
\Delta W_{lon,u,fi} &= \frac{m_u}{m} \frac{F_{lon,fi} h_{CG,u} - T_{fi}}{q_{pitch,fi}} \\
\Delta W_{lon,u,fo} &= \frac{m_u}{m} \frac{F_{lon,fo} h_{CG,u} - T_{fo}}{q_{pitch,fo}} \\
\Delta W_{lon,u,ri} &= \frac{m_u}{m} \frac{F_{lon,ri} h_{CG,u} - T_{ri}}{q_{pitch,ri}} \\
\Delta W_{lon,u,ro} &= \frac{m_u}{m} \frac{F_{lon,ro} h_{CG,u} - T_{ro}}{q_{pitch,ro}}
\end{aligned} \tag{150}$$

Additionally, the torques will be added to the moments acting on the pitch axis, influencing the pitch angle. The new expression is shown in Equation 151:

$$\phi = \frac{T + M_{pitch,drag} + M_{pitch,s} + M_{pitch,u} + M_{pitch,g} - L (w_{pitch,n,f} K_{pitch,tot,f} + w_{pitch,n,r} K_{pitch,tot,r})}{L^2 (K_{roll,tot,f} + K_{roll,tot,r})} \tag{151}$$

Where T is the sum of the four wheel torques, given as Equation 152.

$$T = T_{fi} + T_{fo} + T_{ri} + T_{ro} \tag{152}$$

Other expressions are kept identical to the outboard model presented above.

It is important to note that, as the Formula SAE prototype used as basis has a rear-wheel-drive configuration, the torques should be generated only by the rear tires. However, the model is left open to simulate vehicles with any brake and drive configurations, as well as equipped with any combination of inboard and outboard brakes and/or differentials on both the front and rear.

3.4.3 Wheel loads and displacements

Having determined the values for longitudinal and lateral load transfer, it is possible to calculate the new vertical loads on each wheel and their displacements.

The new vertical load on each wheel is determined considering the inner wheels lose load equal to the lateral load transfer of their axle, as the outer ones gain the same amount. Similarly, the front wheels gain load equal to their side's longitudinal load transfer, as the rear ones lose it. It is important to note that during

braking conditions longitudinal load transfers assume negative values, causing the last statement to be reversed. Additionally, each wheel's load is influenced by the aerodynamic downforce. The four expressions for vertical loads are given below by Equations 153.

$$\begin{aligned}
 F_{z,fi} &= W_{fi} - \Delta W_{lat,f} - \Delta W_{lon,i} + F_{down,fi} - \frac{\Delta W_{lon,drag}}{2} \\
 F_{z,fo} &= W_{fo} + \Delta W_{lat,f} - \Delta W_{lon,o} + F_{down,fo} - \frac{\Delta W_{lon,drag}}{2} \\
 F_{z,ri} &= W_{ri} - \Delta W_{lat,r} + \Delta W_{lon,i} + F_{down,ri} + \frac{\Delta W_{lon,drag}}{2} \\
 F_{z,ro} &= W_{ro} + \Delta W_{lat,r} + \Delta W_{lon,o} + F_{down,ro} + \frac{\Delta W_{lon,drag}}{2}
 \end{aligned} \tag{153}$$

As for the displacement of each wheel, it is considered that the lateral load transfer acts upon the springs and ARB wheel rates, while both the longitudinal load transfer and aerodynamic downforce load influence only the spring wheel rates, as follows in Equations 154:

$$\begin{aligned}
 w_{fi} &= \frac{-\Delta W_{lat,f}}{K_{susp,fi} + K_{arb,fi}} + \frac{-\Delta W_{lon,i} + F_{down,fi}}{K_{susp,fi}} \\
 w_{fo} &= \frac{\Delta W_{lat,f}}{K_{susp,fo} + K_{arb,fo}} + \frac{-\Delta W_{lon,o} + F_{down,fo}}{K_{susp,fo}} \\
 w_{ri} &= \frac{-\Delta W_{lat,r}}{K_{susp,ri} + K_{arb,ri}} + \frac{\Delta W_{lon,i} + F_{down,ri}}{K_{susp,ri}} \\
 w_{ro} &= \frac{\Delta W_{lat,r}}{K_{susp,ro} + K_{arb,ro}} + \frac{\Delta W_{lon,o} + F_{down,ro}}{K_{susp,ro}}
 \end{aligned} \tag{154}$$

3.5 ITERATIVE LOOP

Following the steps described above, it is necessary to create an iterative loop in the implementation which feeds back the determined vertical loads to the tire model and recalculates each step until convergence is achieved.

A stopping condition is created for the loop as a means to determine convergence. It compares the difference between the current value of vertical load

$F_{z,ab}$ for each tire and its previous one $F_{z,past,ab}$ with an acceptable tolerance for the error of vertical loads $F_{z,tol}$, which is set to 0.01 N, as presented in Equation 155.

$$F_{z,ab} - F_{z,past,ab} \leq F_{z,tol} \quad (155)$$

If the expression above is true for all four wheels, the main iterative loop is terminated, allowing the final step of the model to be followed.

3.6 YAW MOMENT

Finally, with converged values for vertical, lateral and longitudinal forces on each tire, it is possible to determine the yaw moment M_{yaw} around the vehicle's center of gravity.

Each wheel grants its contribution to the total yaw moment, being their lateral and longitudinal forces multiplied by their perpendicular distance to the center of gravity. The expressions are given below by Equations 156.

$$\begin{aligned} M_{yaw,fi} &= F_{lat,fi} a - F_{lon,fi} \frac{t_f}{2} \\ M_{yaw,fo} &= F_{lat,fo} a + F_{lon,fo} \frac{t_f}{2} \\ M_{yaw,ri} &= -F_{lat,ri} b - F_{lon,ri} \frac{t_r}{2} \\ M_{yaw,ro} &= -F_{lat,ro} b + F_{lon,ro} \frac{t_r}{2} \end{aligned} \quad (156)$$

Being:

$$M_{yaw} = M_{yaw,fi} + M_{yaw,fo} + M_{yaw,ri} + M_{yaw,ro} \quad (157)$$

By convention, the yaw moment is defined as positive to the inner side of the corner, as in causing oversteer to the vehicle, opposed to being negative when causing understeer.

4 RESULTS AND DISCUSSION

This section starts by presenting the source parameters used to run simulations in the proposed model. Then, the results are analytically validated and compared to a commercial vehicle dynamics simulation solution.

4.1 SIMULATION PARAMETERS

All the sources for the parameters that represent the physical characteristics of the TYPE 25/C Formula SAE are presented below.

4.1.1 Tire parameters

The *Magic Formula* parameters used to emulate tire behavior in the model were acquired via the Formula SAE Tire Test Consortium (FSAE TTC). The FSAE TTC is a non-profit organization which aims at providing engineering-quality tire data to the Formula SAE member teams. The test rounds are funded and conducted by the Calspan Tire Research Facility (TIRF), where the tires are subjected to extensive destructive bench testing on specialized machinery, as shown in Figure 38.

Figure 38 – The main testing machine at Calspan TIRF



Source: KASPRZAK, GENTZ, 2006.

The parameters used in the simulations are given by the SES model, made by Stackpole Engineering Services, referring to the Hoosier 18.0 x 7.5 10 R25B tire, with an inflation pressure of 10 psi. This tire was developed exclusively for use in Formula SAE. The data itself will not be presented in this work, as it is protected by the FSAE TTC Licensing Agreement. Alternatively, the curve plots for lateral and longitudinal forces obtained via the *Magic Formula* and MNC models are presented in Figures 42 – 47, in Appendix A. The parameter that defines the maximum friction coefficient has been divided by two, to better represent realistic tire operations on asphalt.

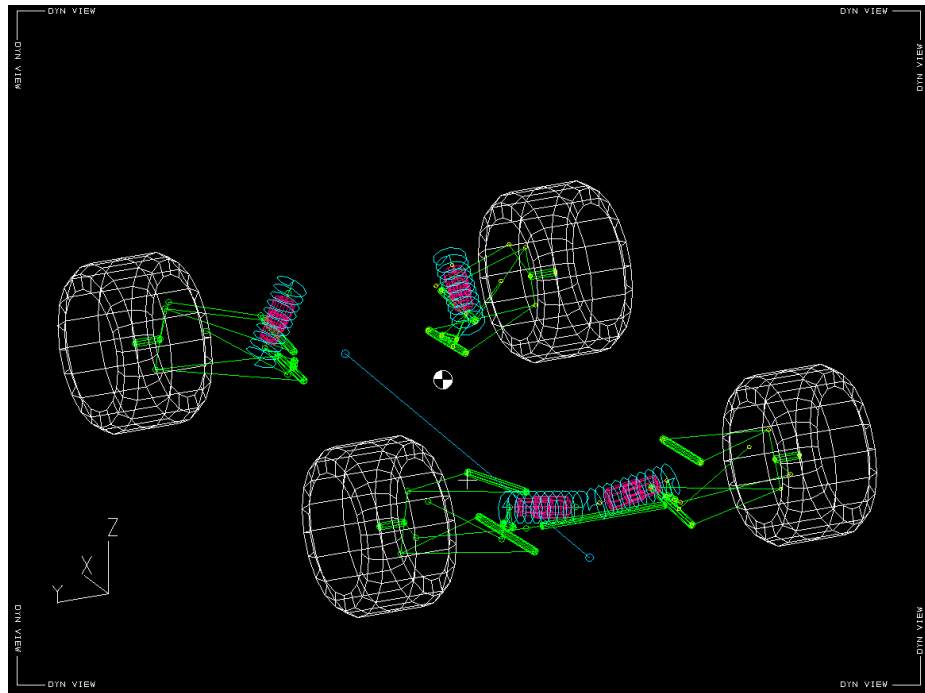
4.1.2 Vehicle parameters

The specifications for the simulated vehicle refer to the TYPE 25/C Formula SAE prototype developed by the Protto UFSC Motorsport team. They consist of static parameters, such as the vehicle mass, tracks, wheelbase, etc. and dynamic ones, which may be associated as a function of either wheel displacement or chassis roll/pitch angles, such as motion ratios and suspension geometric points locations.

The static data was taken from the design of the prototype itself and from bench and on-track validations made by the team. They are presented in Table 10, in Appendix B.

Dynamic parameter functions were defined with the help of the vehicle's CAD and a kinematic simulation software, the Lotus Suspension, shown in Figure 39.

Figure 39 – Suspension geometry of the TYPE 25/C prototype in Lotus Suspension



Source: the author.

The curve plots and functions of dynamic vehicle parameters are presented in Figures 48 – 55, in Appendix C.

4.1.3 Aerodynamic parameters

The influences of aerodynamic drag and downforce used in the simulations also refer to the TYPE 25/C prototype. These parameters are dynamic, being defined as functions of the vehicle velocity V . The functions were determined by simulating the prototype's full aerodynamic package on a CFD software with different air velocity inputs. The curve plots and functions that describe the aerodynamic influences are presented in Figures 56 – 59, in Appendix D.

4.2 ANALYTICAL VALIDATION

In this subsection, the implementation of the proposed model is analytically tested for physical inconsistencies. To achieve this, the five driver input conditions (pure braking, trail braking, pure cornering, unwinding and pure traction) were

simulated and compared to the analytical Equations for lateral and longitudinal load transfers given in section 2.

The aerodynamic drag influences were not considered in these simulations, as they are not represented in the analytic models.

The input variables were defined to emulate the driver's input at each condition. They are given in Table 1.

Table 1 – Input variables for analytic validation of the model

Condition	β	R	κ_{des}	δ_d	$bias$	V
Braking	0°	10 ²⁰ m	8%	0°	75%	60 km/h
Trail braking	1.4°	14 m	5.65%	81°	75%	45 km/h
Pure cornering	2°	10 m	0%	115°	-	40 km/h
Unwinding	1.4°	14 m	5.65%	81°	0%	45 km/h
Traction	0°	10 ²⁰ m	8%	0°	0%	60 km/h

Source: the author.

The results for each simulation are given in Tables 2 – 6.

Table 2 – Braking results for analytic validation

	fi	fo	ri	ro
$F_{z,ab}$	1320.13 N	1320.13 N	462.64 N	462.64 N
$F_{x,comb,ab}$	1632.92 N	1632.92 N	528.76 N	528.76 N
$F_{y,comb,ab}$	24.24 N	-24.24 N	13.99 N	-13.99 N
α_{ab}	0.00°	0.00°	0.00°	0.00°
κ_{ab}	8.00%	8.00%	4.50%	4.50%
F_{lat}	0.00 N			
F_{lon}	-4420.50 N			
ΔW_{lat}	0.00 N			
ΔW_{lon}	-679.63 N			

Source: the author.

Table 3 – Trail braking results for analytic validation

	<i>fi</i>	<i>fo</i>	<i>ri</i>	<i>ro</i>
$F_{z,ab}$	656.60 N	1316.00 N	294.58 N	1036.42 N
$F_{x,comb,ab}$	337.08 N	344.70 N	111.01 N	113.80 N
$F_{y,comb,ab}$	851.28 N	1827.86 N	307.21 N	983.36 N
α_{ab}	9.21°	9.62°	1.64°	1.50°
κ_{ab}	5.56%	2.48%	0.71%	0.09%
F_{lat}	3710.89 N			
F_{lon}	-1623.73 N			
ΔW_{lat}	700.61 N			
ΔW_{lon}	-249.63 N			

Source: the author.

Table 4 – Pure cornering results for analytic validation

	<i>fi</i>	<i>fo</i>	<i>ri</i>	<i>ro</i>
$F_{z,ab}$	545.97 N	1255.92 N	322.09 N	1109.33 N
$F_{x,comb,ab}$	0.00 N	0.00 N	0.00 N	0.00 N
$F_{y,comb,ab}$	722.73 N	1757.97 N	392.54 N	1256.00 N
α_{ab}	12.91°	13.72°	2.37°	2.09°
κ_{ab}	0.00%	0.00%	0.00%	0.00%
F_{lat}	3964.97 N			
F_{lon}	-926.83 N			
ΔW_{lat}	748.58 N			
ΔW_{lon}	-142.49 N			

Source: the author.

Table 5 – Unwinding results for analytic validation

	<i>fi</i>	<i>fo</i>	<i>ri</i>	<i>ro</i>
$F_{z,ab}$	488.96 N	1133.85 N	476.31 N	1204.48 N
$F_{x,comb,ab}$	0.00 N	0.00 N	636.97 N	652.81 N
$F_{y,comb,ab}$	656.48 N	1613.51 N	324.04 N	1120.80 N
α_{ab}	9.21°	9.62°	1.64°	1.50°
κ_{ab}	0.00%	0.00%	5.65%	1.2%
F_{lat}	3636.26 N			
F_{lon}	651.38 N			
ΔW_{lat}	686.52 N			
ΔW_{lon}	100.14 N			

Source: the author.

Table 6 – Traction results for analytic validation

	<i>fi</i>	<i>fo</i>	<i>ri</i>	<i>ro</i>
$F_{z,ab}$	789.12 N	789.12 N	993.64 N	993.64 N
$F_{x,comb,ab}$	0.00 N	0.00 N	1292.07 N	1292.07 N
$F_{y,comb,ab}$	30.96 N	-30.96 N	15.84 N	-15.84 N
α_{ab}	0.00°	0.00°	0.00°	0.00°
κ_{ab}	0.00%	0.00%	8.00%	8.00%
F_{lat}	0.00 N			
F_{lon}	2487.01 N			
ΔW_{lat}	0.00 N			
ΔW_{lon}	382.36 N			

Source: the author.

A preliminary analysis of the results reveals that the total longitudinal forces F_{lon} are greater under pure braking than for the pure traction condition. This is expected, since the brakes are applied on all four wheels, whereas the traction forces are generated only on the rear axle. Additionally, on both of these conditions, the tires generate lateral loads due to their anti-symmetric behavior. As expected, these lateral forces cancel out, resulting in null total lateral load transfer and forces.

On the pure cornering scenario, longitudinal forces and load transfer occur even as there is no braking or traction inputs. This happens because a portion of the lateral forces generated by the front tires are longitudinally oriented on the vehicle's coordinate system.

Finally, the results for the combined accelerations (trail braking and unwinding) show that the torque constraint algorithm is able to maintain the slip ratio on the main tire as specified, while successfully limiting the torque for the others.

Substituting the values of h_{CG} , t , L and the obtained F_{lat} and F_{lon} in Equations 35 and 36 gives the analytical load transfer results, since the expressions given in Equations 158 and 159 are true:

$$F_{lat} = W a_{lat} \quad (158)$$

$$F_{lon} = W a_{lon} \quad (159)$$

The analytical results are shown in Table 7.

Table 7 – Analytical results for longitudinal and lateral load transfers

Condition	ΔW_{lon}	ΔW_{lat}
Braking	-679.63 N	0.00 N
Trail braking	-249.63 N	700.61 N
Pure cornering	-142.49 N	748.58 N
Unwinding	100.14 N	686.52 N
Traction	382.36 N	0.00 N

Source: the author.

All the analytical results match the numerical ones, giving a 0.00% error for all values. This shows that the presented load transfer model is physically consistent by satisfying Equations 35 and 36.

As previously mentioned, the comparison made for lateral load transfer is only possible because the front and rear tracks of the vehicle have the same length.

4.3 BENCHMARKING

As a means to further validate the proposed model, its results were compared to a commercially available solution. The chosen software is the MathWorks Vehicle Dynamics Block Set (VDBS). It is a transient application for vehicle dynamics simulation implemented in Simulink. The VDBS offers a variety of vehicle customization, active controlling parametrization and pre-defined reference scenarios to test the virtual car.

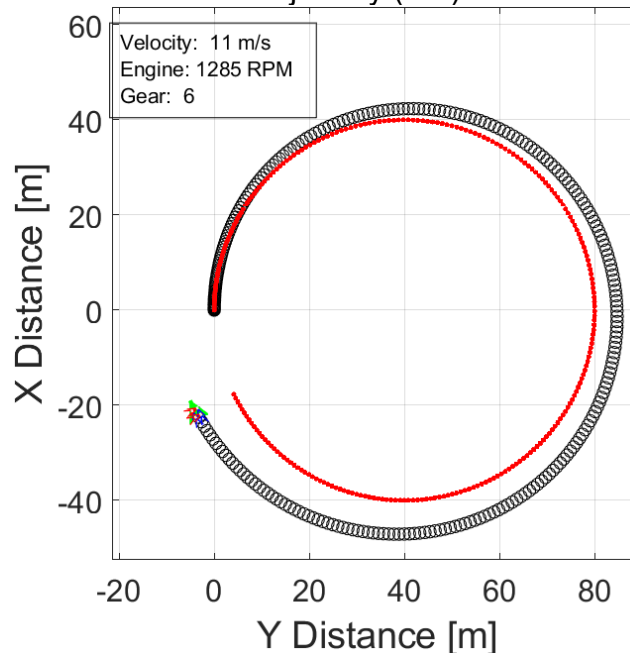
The first step for the VDBS simulations was to set the input parameters in order to emulate the properties of the TYPE 25/C prototype. The following settings were defined in the software for that purpose:

- The 14 degree-of-freedom vehicle model was selected;
- The tire model was set to a four-wheel *Magic Formula* with the same parameters for the Hoosier 18.0 x 7.5 10 R25B tire, with an inflation pressure of 10 psi;
- The static vehicle parameters were set as presented in Appendix B;
- Suspension kinematic parameters were inserted as the interpolation equations shown in Appendix C;

- The steering mechanism was set to a parallel Ackermann configuration;
- Aerodynamic inputs were defined to represent the interpolation equations presented in Appendix D;
- The differential was set to a rear-mounted configuration and the brakes defined as disc-type;
- Every active controlling feature of the vehicle was disabled.

The simulation was run under the predefined constant radius reference application available in the VDBS model. The vehicle follows a circular trajectory of constant radius R of 40 m, shown in Figure 40, with a determined longitudinal velocity V of 40 km/h. The duration of the simulation was set to 30 s. This scenario differs from a pure cornering condition because the driver gives sufficient throttle input to generate tractive forces that are able overcome the aerodynamic drag and rolling resistance forces. This allows the vehicle to maintain a constant longitudinal velocity.

Figure 40 – Constant radius trajectory (red) and vehicle path (black)

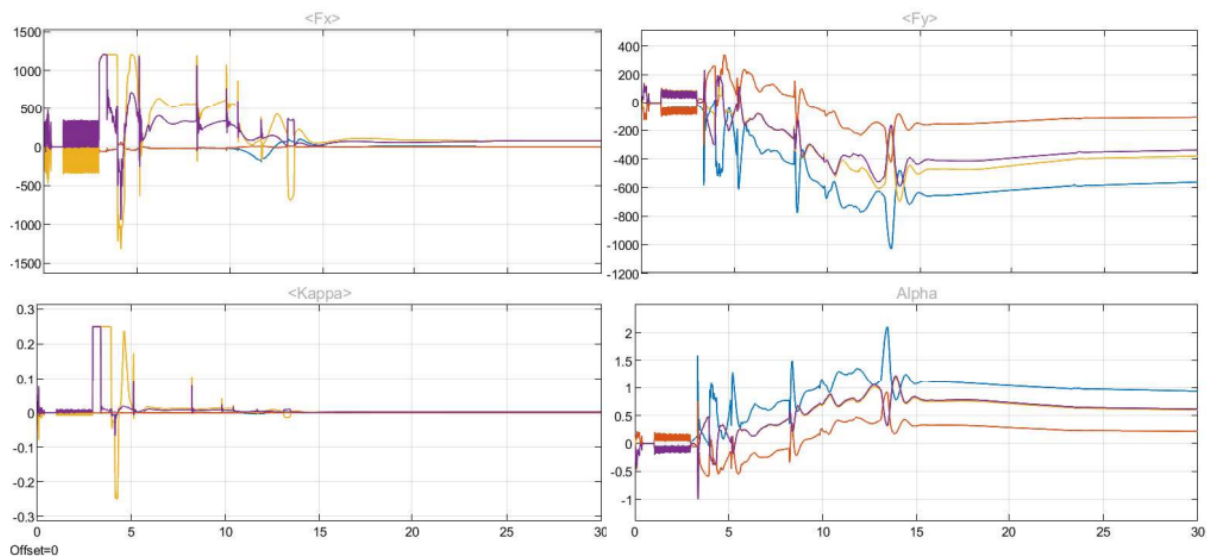


Source: the author.

It is possible to perceive that the vehicle struggles to follow the determined trajectory. Figure 41 shows some of the simulation outputs for each of the four

wheels over time. The top graphs present the longitudinal and lateral forces of each tire, F_x and F_y , respectively, and the bottom ones show the slip ratio κ (kappa) and slip angles α (alpha). By inspecting the loads and slips on each tire, it is clear that the virtual driver constantly tries to correct the path of the vehicle, causing abrupt transient changes, which prevents a correct comparison with the quasi-steady state model. However, eventually the behavior of the vehicle stabilizes, allowing for more accurate analysis of the output values. For this matter, the presented results were taken from $t = 18$ s.

Figure 41 – Longitudinal and lateral tire loads and slip for each wheel on the constant radius reference application



Source: the author.

For both models, the cornering radius was adjusted to 45 m, to compensate for the deviation of trajectory observed in Figure 40.

The result values for both the VDBS and the proposed model simulations are presented in Tables 8 and 9, respectively. Additionally, Table 10 presents the percentual error between the results of both models.

Table 8 – Constant radius results for the VDBS model

	<i>fi</i>	<i>fo</i>	<i>ri</i>	<i>ro</i>
<i>R</i>	45 m			
<i>V</i>	39.27 km/h			
β	0.2658°			
δ_d	11.50°			
$F_{z,ab}$	504.30 N	1006.00 N	671.00 N	1006.00 N
$F_{x,comb,ab}$	-0.49 N	7.79 N	59.38 N	102.50 N
$F_{y,comb,ab}$	151.40 N	639.20 N	413.40 N	472.70 N
α_{ab}	0.32°	1.08°	0.78°	0.76°
κ_{ab}	0.00%	0.00%	1.04%	1.56%
a_{lat}	0.27 g			
a_{lon}	0.00 g			
M_{yaw}	-4.26 Nm			

Source: the author.

Table 9 – Constant radius results for the proposed model

	<i>fi</i>	<i>fo</i>	<i>ri</i>	<i>ro</i>
<i>R</i>	45 m			
<i>V</i>	39.27 km/h			
β	0.2658°			
δ_d	11.50°			
$F_{z,ab}$	696.64 N	835.78 N	764.20 N	926.76 N
$F_{x,comb,ab}$	0.00 N	0.00 N	102.01 N	102.57 N
$F_{y,comb,ab}$	167.02 N	224.89 N	192.93 N	215.79 N
α_{ab}	0.68°	0.72°	0.68°	0.66°
κ_{ab}	0.00%	0.00%	1.80%	1.18%
a_{lat}	0.27 g			
a_{lon}	0.00 g			
M_{yaw}	4.48 Nm			

Source: the author.

Table 10 – Results differences between the VDBS and proposed models

	<i>fi</i>	<i>fo</i>	<i>ri</i>	<i>ro</i>
$F_{z,ab}$	-192.34 N	170.22 N	-93.20 N	79.24 N
$F_{x,comb,ab}$	-0.49 N	7.79 N	-42.63 N	0.00 N
$F_{y,comb,ab}$	15.62 N	414.31 N	220.47 N	256.91 N
α_{ab}	-0.36°	0.36°	0.10°	0.10°
κ_{ab}	0.00 %	0.00 %	-0.76%	0.38%
a_{lat}	0.00 N			
a_{lon}	0.00 N			
M_{yaw}	-8.74 N			

Source: the author.

The overall comparison is satisfactory, as the main output values for the proposed model (a_{lat} , a_{lon} and M_{yaw}) almost perfectly match the VDBS ones. The vehicle on the commercial solution seems to be subjected to greater lateral load transfer, despite the fact that the values for tracks, vehicle mass, lateral acceleration and center of gravity height are equal between models. The longitudinal forces and tire slip also behave differently from the proposed one. This is true because the VDBS uses a limited slip differential (LSD) model, not having an option to implement an open configuration. The errors for R , V , β and δ_d are not considered, as these parameters are input parameters for the proposed model.

5 CONCLUSIONS

This work presented the development of a four-wheel quasi-steady state numerical model for vehicle dynamics simulation, implemented in C++, based on the design of a Formula SAE prototype. Commercially available virtual models are widely used in the automotive industry, in order to reduce design expenses and time. However, these solutions are often too expensive and complex, which limits access to engineering students and smaller companies. This motivated the development of an accessible and intuitive tool.

The first step was to evaluate existing mathematical models used in vehicle simulation. Through a literature review, different tire models were assessed, culminating in the choice of the *Magic Formula* and MNC models to represent the non-linearities of tire behavior, following a quasi-steady state four-wheel vehicle approach. Additionally, suspension geometry and kinematics fundamentals were revised, as a means to explore models to represent physical phenomena, such as lateral and longitudinal load transfer.

As follows, a complete tire model was formulated, based on the forementioned *Magic Formula* and MNC ones, with an integrated algorithm to find the appropriate slip ratios subjecting each tire. The model assesses external conditions imposed on each tire, such as vertical load and slip angles, and determines each slip ratio by following the torque constraints, imposed by the vehicle's traction and braking systems physical natures.

A generic load transfer model was proposed, following the physical principles established by the literature. The whole mathematical deduction was presented, highlighting the nuances that differ lateral and longitudinal load transfer, from a physical standpoint. Its proposal is to represent general asymmetries of the vehicle geometry. This is a relevant characteristic to the model, since the kinematics of the suspension and steering mechanisms determine that geometric parameters may be dynamic in nature, varying as the chassis pitches, rolls and heaves.

Finally, the model's results were analytically validated and compared to a commercial solution. Simulations were run using the tires, suspension and aerodynamic properties of a Formula SAE prototype vehicle. The proposed load transfer model was compared to a simpler analytical one, which matching results proved its physical consistency. The MathWorks Vehicle Dynamics Block Set was

used to create a benchmarking. Both models ran simulations following a constant radius trajectory with a constant longitudinal velocity. The comparison was satisfactory, as the main output values for horizontal accelerations and yaw moment marginally differed. Additionally, the overall results showed a correct behavior of the slip ratio finding algorithm.

5.1 FUTURE DEVELOPMENTS

As a means to further improve the study of vehicle dynamics and enhance the physical representation of the model, future development examples are suggested.

The implementation of additional physical phenomena such as jacking forces, rotational inertia effects, overturning moments and self-aligning tire moments may result on better structural optimization of the suspension components, as the generated loads would be more realistically represented. Additionally, a powertrain model would allow for the simulation of traction conditions where the vehicle's performance is not limited by the tires' grip.

By integrating a suspension and steering kinematics model, the dynamic geometric parameters of said systems could be more accurately determined, without the need to input them as equations. Additionally, this would give the user a better understanding of how these characteristics interfere in the overall vehicle performance. The kinematics and dynamics models could iteratively interact; the former determining the geometric location of suspension points for the latter to calculate load transfer and suspension displacement, which in turn feeds back to the first.

The conversion of the proposed model to a semi or full transient application would allow for the representation of damping effects and proper damper sizing. This is a critical concept in motorsport and for vehicle dynamics in general, as it significantly affects vehicle performance, safety and comfort. Furthermore, the integration with track and virtual driver models could result in an extremely powerful lap time simulation tool, or even a driving simulator. However, these additions would substantially increase the level of complexity, decreasing the didactic aspect of the proposed model.

Lastly, an experimental validation of the model on a real Formula SAE prototype would be of great value. Bench and on-track tests with proper data acquisition on the vehicle could give insights to actual usefulness of the model and where it should be enhanced.

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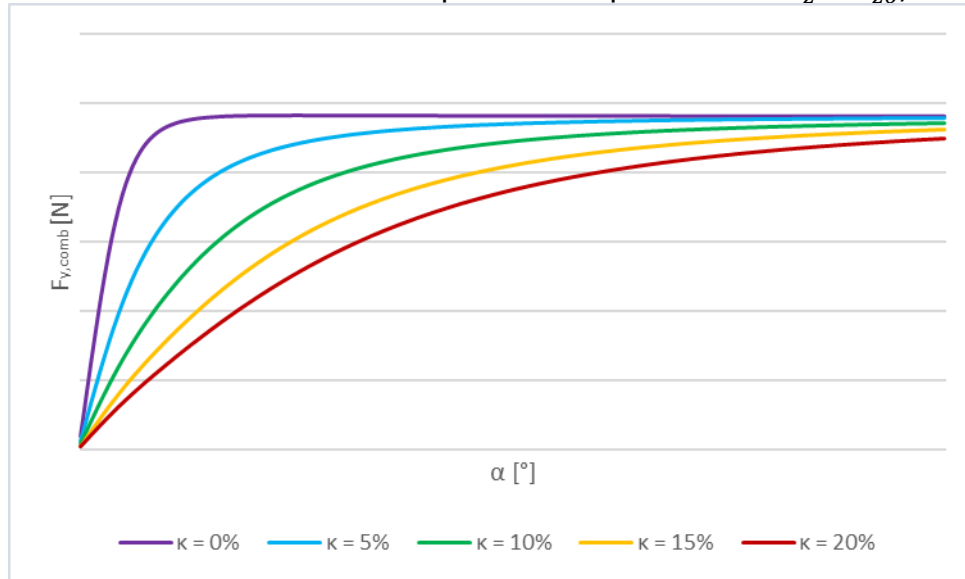
VAN LITH, B. S.; BOONKKAMP J. H.M. ten T.; IJZERMAN, W. L. **Full linear multistep methods as root-finders**. Applied Mathematics and Computation, Volume 320, 2018, Pages 190-201, ISSN 0096-3003, <https://doi.org/10.1016/j.amc.2017.09.003>.

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APPENDIX A – TIRE CURVE PLOTS

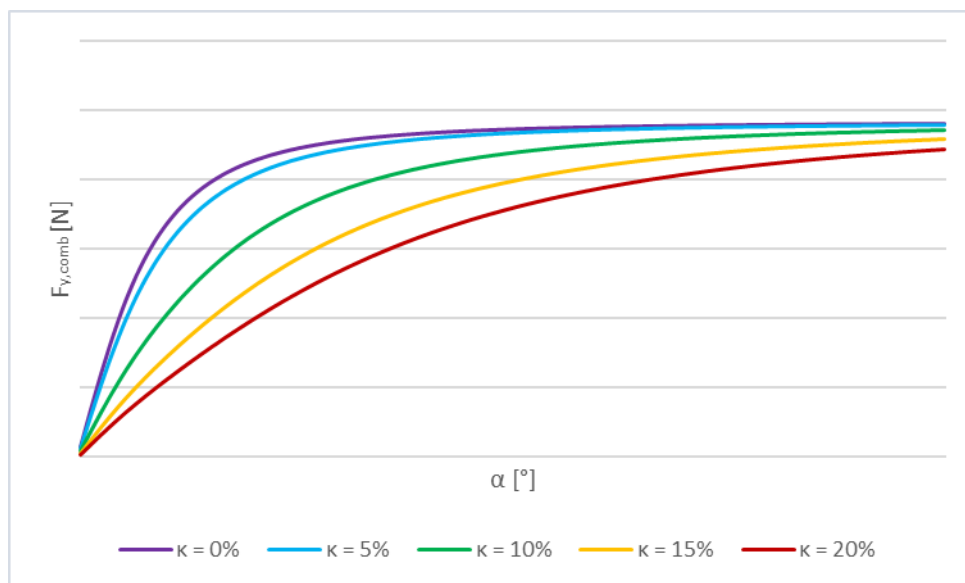
The scales of the graphs presented below are not shown, since the data used to plot them is protected by a licensing agreement, as explained in Chapter 4.

Figure 42 – *Magic Formula* and MNC curve plot of combined lateral forces for the Hoosier 18.0 x 7.5 10 R25B tire at 10 psi inflation pressure for $F_z = F_{z0}/2$ and $\gamma = 0^\circ$



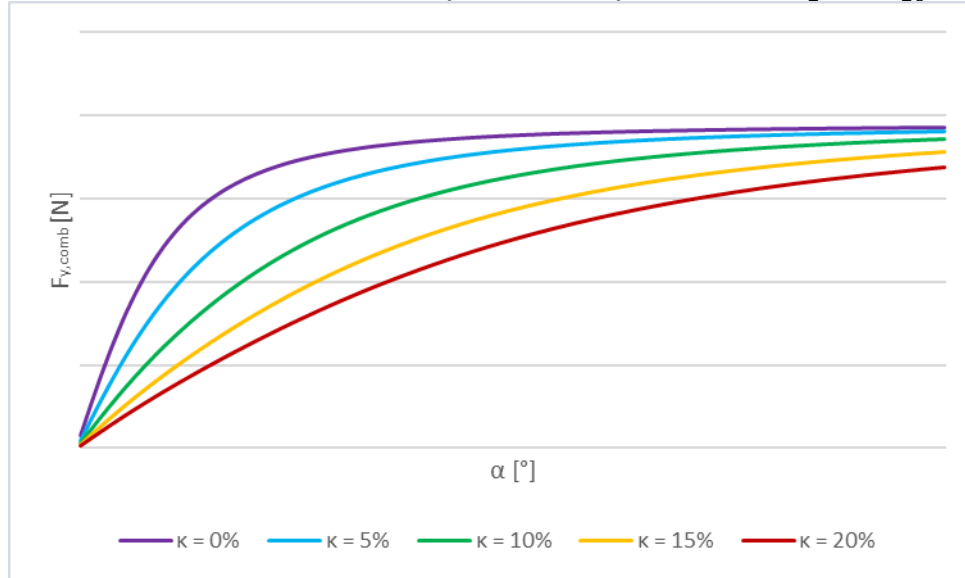
Source: the author.

Figure 43 – *Magic Formula* and MNC curve plot of combined lateral forces for the Hoosier 18.0 x 7.5 10 R25B tire at 10 psi inflation pressure for $F_z = F_{z0}$ and $\gamma = 0^\circ$



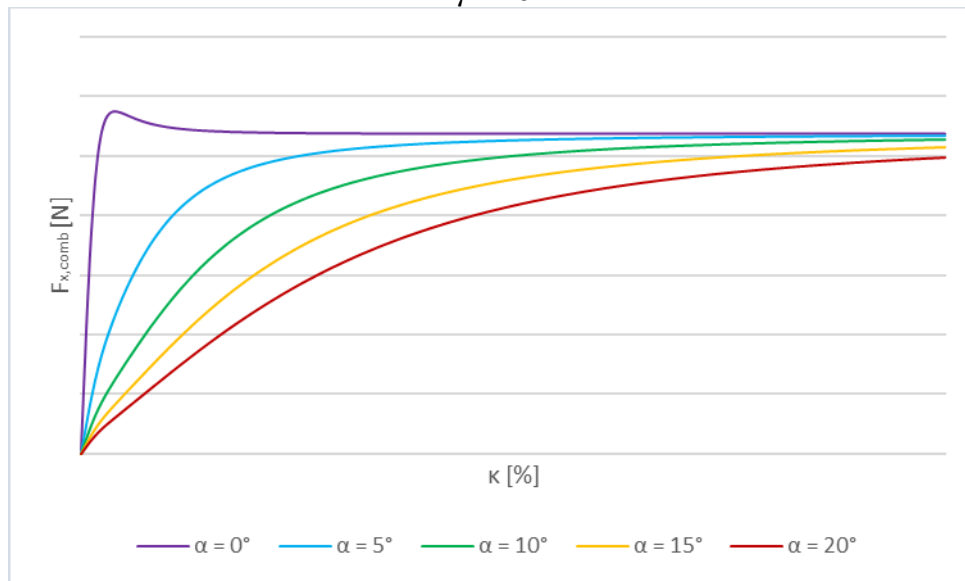
Source: the author.

Figure 44 – *Magic Formula* and MNC curve plot of combined lateral forces for the Hoosier 18.0 x 7.5 10 R25B tire at 10 psi inflation pressure for $F_z = 2 F_{z0}$ and $\gamma = 0^\circ$



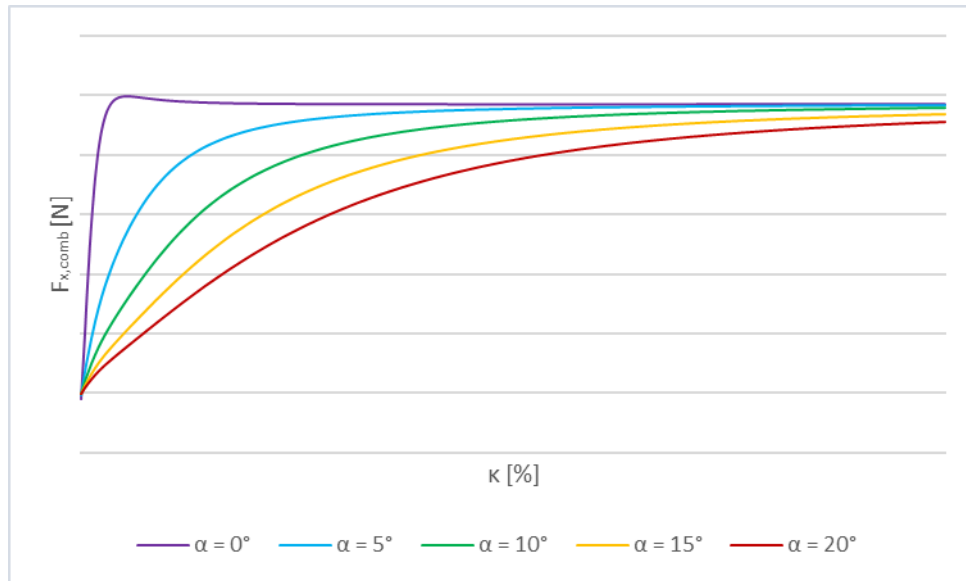
Source: the author.

Figure 45 – *Magic Formula* and MNC curve plot of combined longitudinal forces for the Hoosier 18.0 x 7.5 10 R25B tire at 10 psi inflation pressure for $F_z = F_{z0}/2$ and $\gamma = 0^\circ$



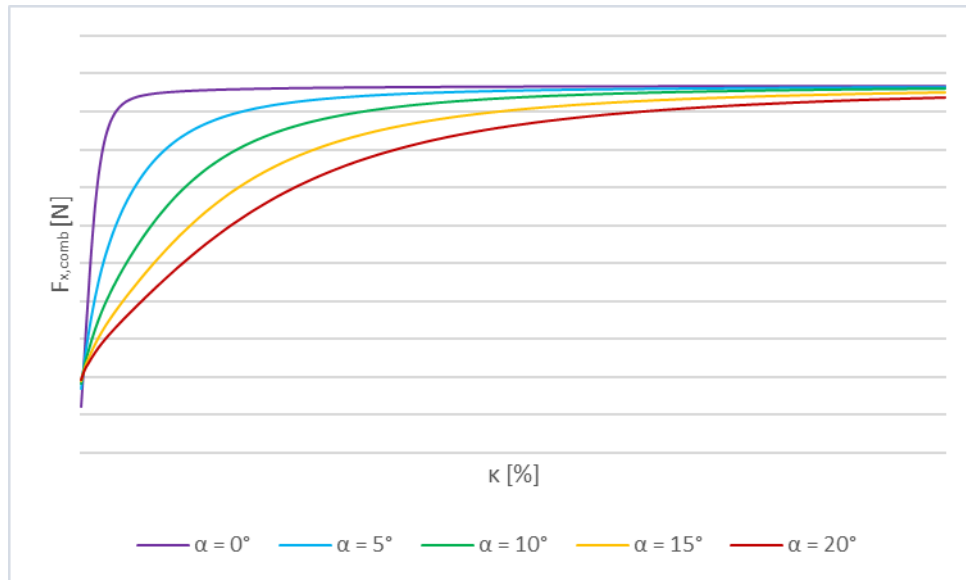
Source: the author.

Figure 46 – *Magic Formula* and MNC curve plot of combined longitudinal forces for the Hoosier 18.0 x 7.5 10 R25B tire at 10 psi inflation pressure for $F_z = F_{z0}$ and $\gamma = 0^\circ$



Source: the author.

Figure 47 – *Magic Formula* and MNC curve plot of combined longitudinal forces for the Hoosier 18.0 x 7.5 10 R25B tire at 10 psi inflation pressure for $F_z = 2 F_{z0}$ and $\gamma = 0^\circ$



Source: the author.

APPENDIX B – STATIC VEHICLE PARAMETERS

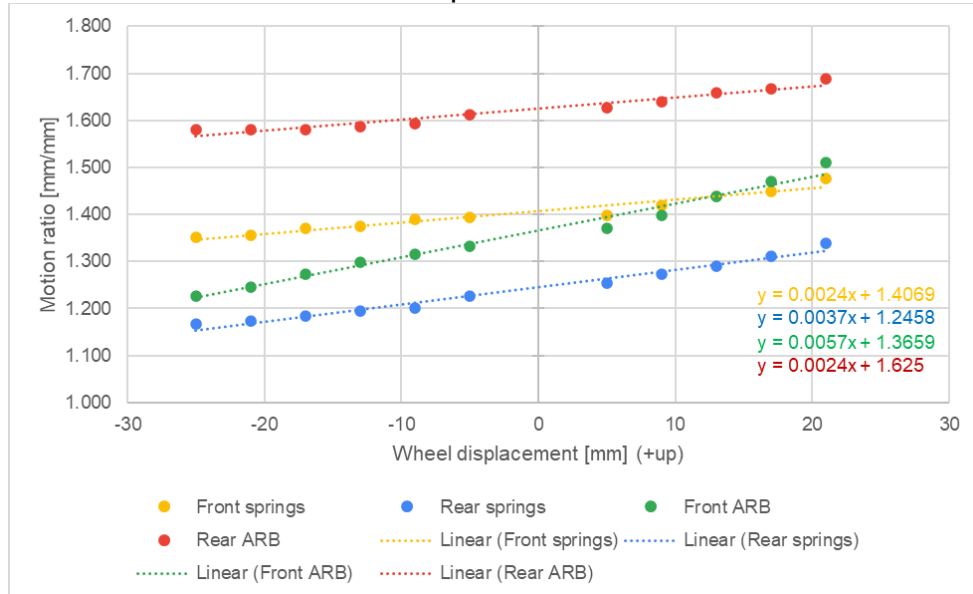
Table 11 – Static vehicle parameters for the TYPE 25/C

Parameter name	Symbol	Value
Wheelbase	L	1.535 m
Front track	t_f	1.25 m
Rear track	t_r	1.25 m
Total mass	m	303.5 kg
Longitudinal weight distribution	x	48%
Lateral weight distribution	y	50%
Front inner unsprung masses	$m_{u,fi}$	11 kg
Front outer unsprung masses	$m_{u,fo}$	11 kg
Rear inner unsprung masses	$m_{u,ri}$	11 kg
Rear outer unsprung masses	$m_{u,ro}$	11 kg
Height of center of gravity	h_{CG}	0.236 m
Chassis front torsional stiffness	$k_{c,f}$	800 Nm/°
Chassis rear torsional stiffness	$k_{c,r}$	800 Nm/°
Front suspension spring stiffness	$k_{susp,f}$	42710 N/m
Rear suspension spring stiffness	$k_{susp,r}$	42710 N/m
Front ARB stiffness	$k_{arb,f}$	95373 N/m
Rear ARB stiffness	$k_{arb,r}$	206581 N/m
Inner pitch center height	$h_{pitch,i}$	0.08 m
Outer pitch center height	$h_{pitch,o}$	0.08 m
Longitudinal distance of inner pitch center to front axle	$n_{pitch,fi}$	0.65598 m
Longitudinal distance of outer pitch center to front axle	$n_{pitch,fo}$	0.65598 m
Longitudinal distance of inner pitch center to rear axle	$n_{pitch,ri}$	0.87902 m
Longitudinal distance of outer pitch center to rear axle	$n_{pitch,ro}$	0.87902 m
Height of the front inner pitch instantaneous center	$p_{pitch,fi}$	0.14774 m
Height of the front outer pitch instantaneous center	$p_{pitch,fo}$	0.14774 m
Height of the rear inner pitch instantaneous center	$p_{pitch,ri}$	0.14742 m
Height of the rear outer pitch instantaneous center	$p_{pitch,ro}$	0.14742 m

Source: the author.

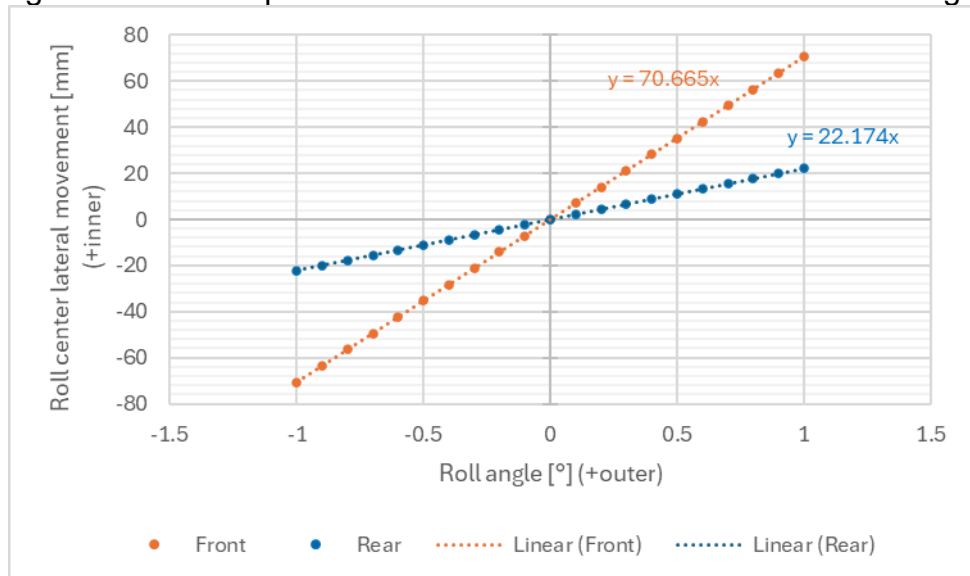
APPENDIX C – DYNAMIC VEHICLE PARAMETERS

Figure 48 – Curve plot for suspension springs and ARBs motion ratios with wheel displacement



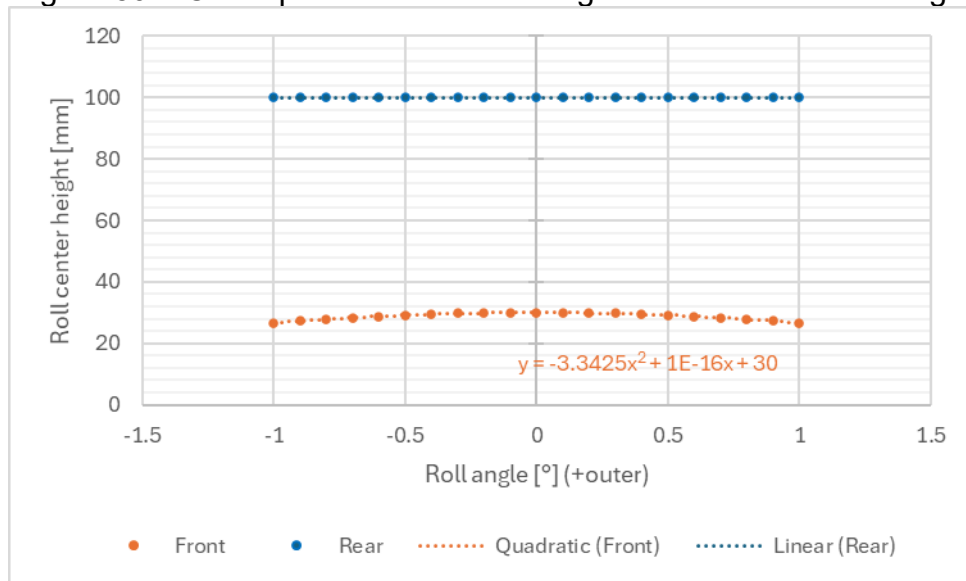
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Figure 49 – Curve plot for roll centers lateral movement with roll angle



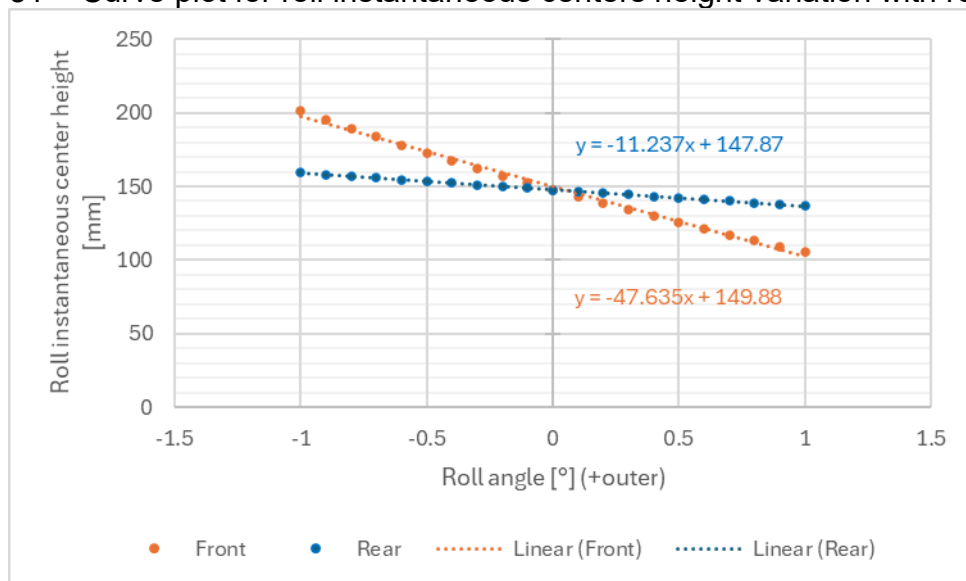
Source: the author.

Figure 50 – Curve plot for roll center heights variation with roll angle



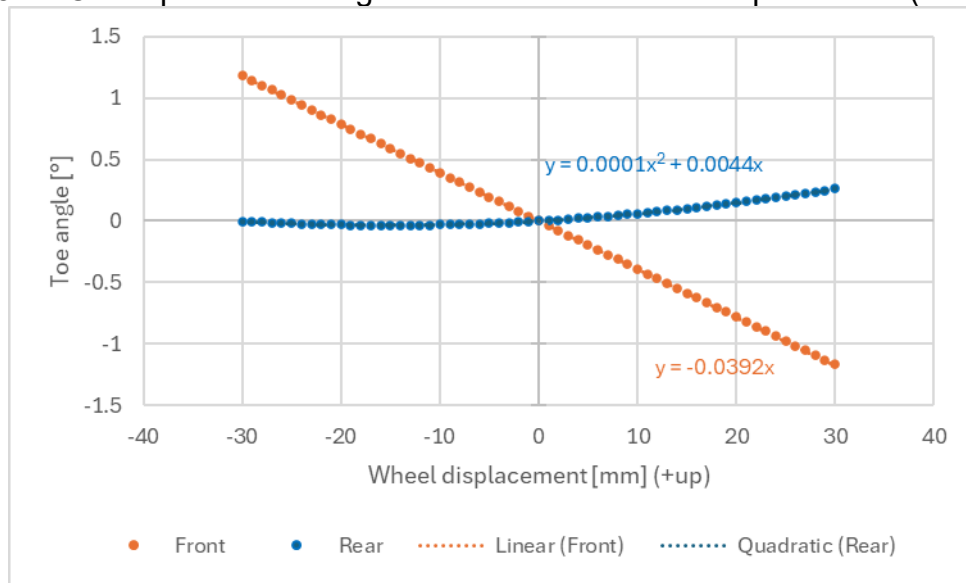
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Figure 51 – Curve plot for roll instantaneous centers height variation with roll angle



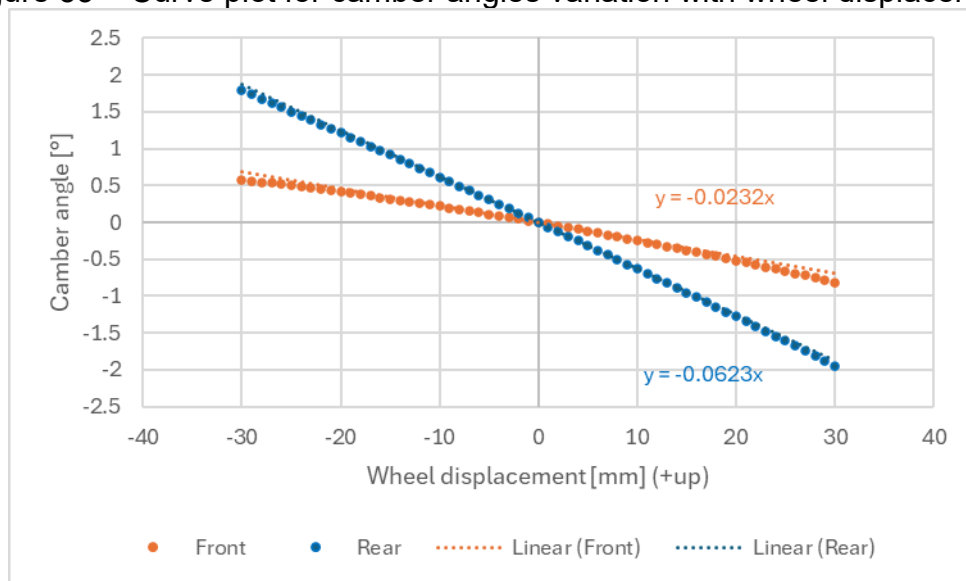
Source: the author.

Figure 52 – Curve plot for toe angles variation with wheel displacement (bump steer)



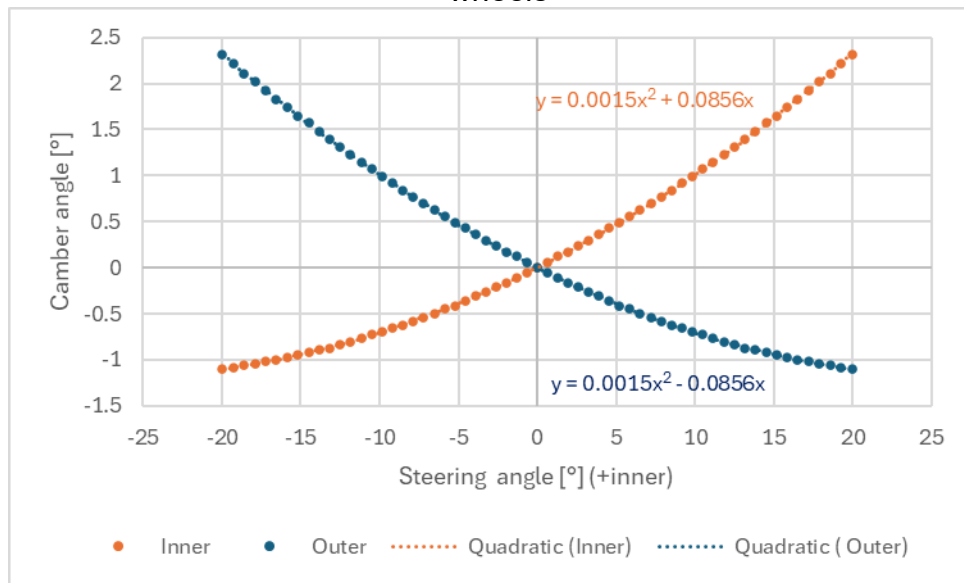
Source: the author.

Figure 53 – Curve plot for camber angles variation with wheel displacement



Source: the author.

Figure 54 – Curve plot for camber angles variation with steering angle on front wheels



Source: the author.

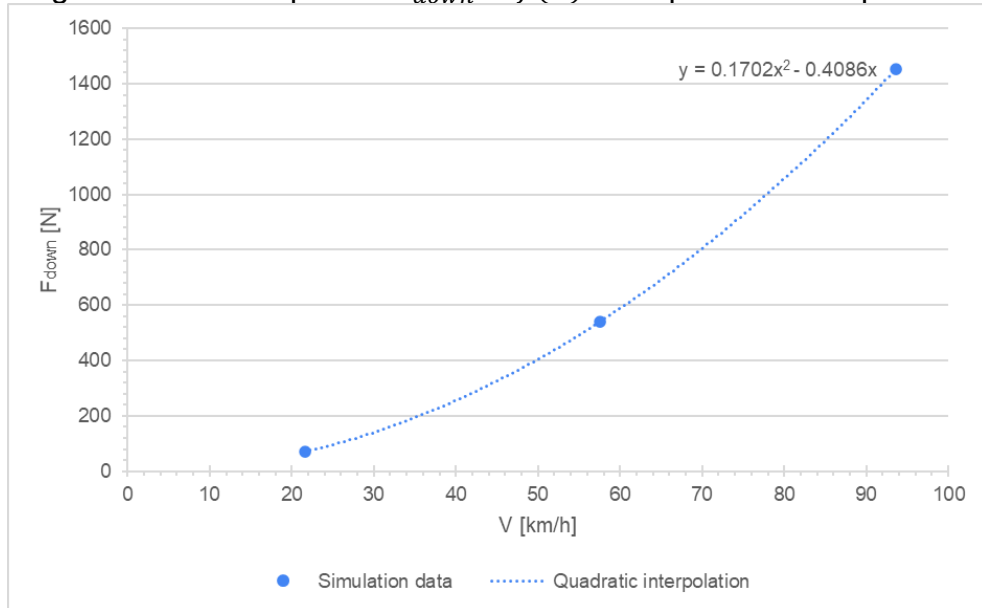
Figure 55 – Curve plot for steering angles variation with steering wheel input angle (Ackermann)



Source: the author.

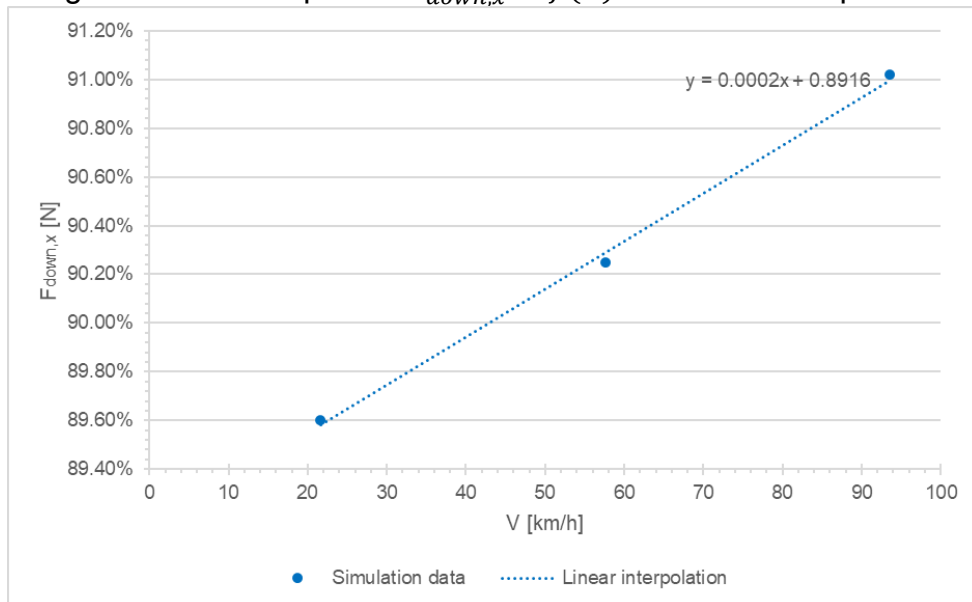
APPENDIX D – AERODYNAMIC PARAMETERS

Figure 56 – Curve plot for $F_{down} = f(V)$ with quadratic interpolation

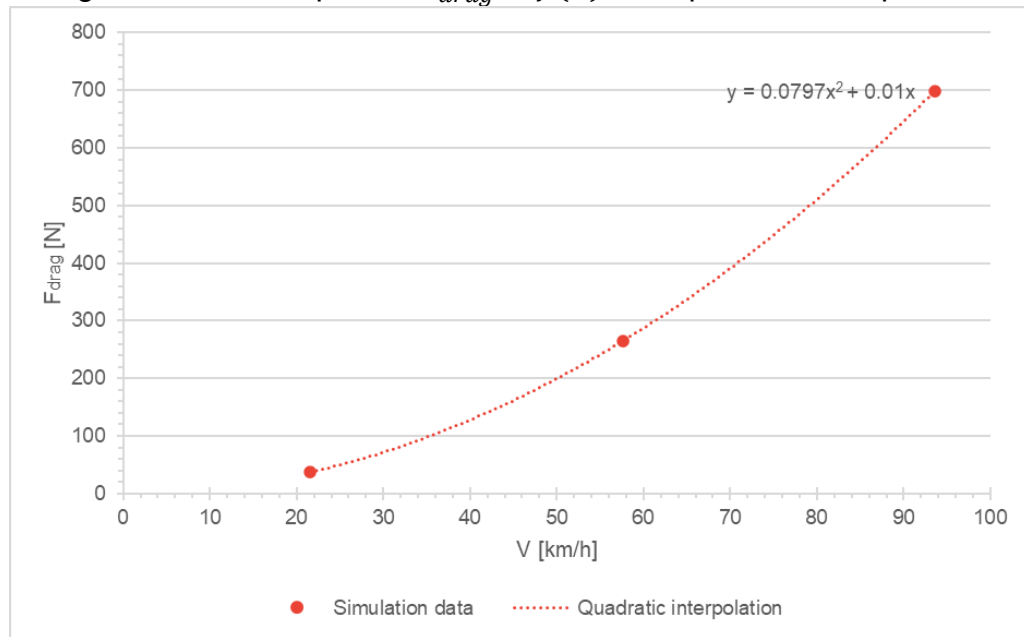


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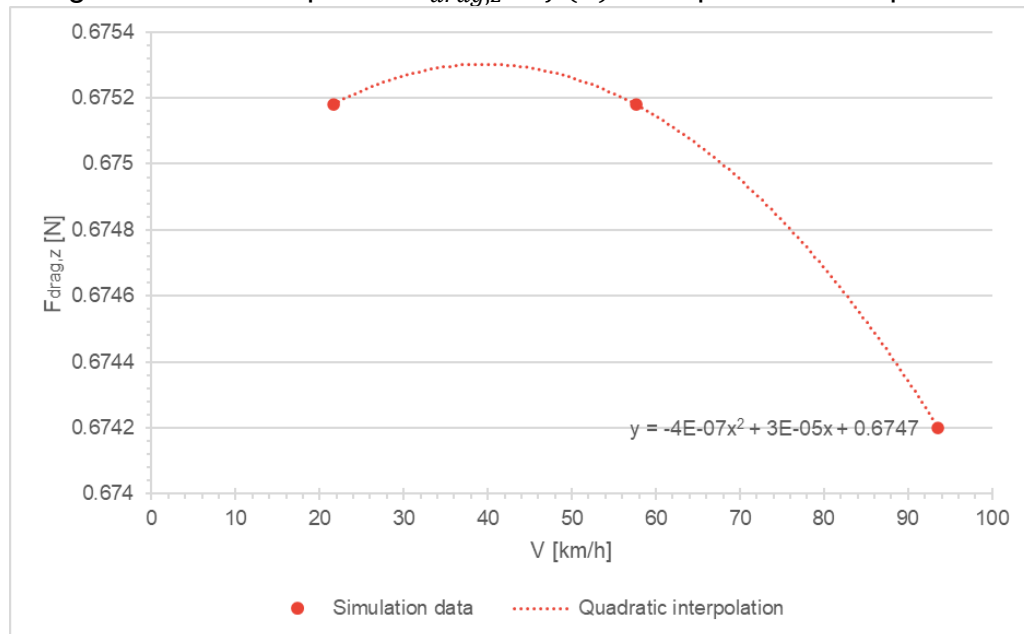
Figure 57 – Curve plot for $F_{down,x} = f(V)$ with linear interpolation



Source: the author.

Figure 58 – Curve plot for $F_{drag} = f(V)$ with quadratic interpolation

Source: the author.

Figure 59 – Curve plot for $F_{drag,z} = f(V)$ with quadratic interpolation

Source: the author.